

Filter Summary Report: CG,TIA,Automated,CROSS,COUPLED,some,parasitic,Z1,Z4,ZL

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Contents

1 Examined $H(z)$ for CG TIA Automated CROSS COUPLED some parasitic Z1 Z4 ZL: $\frac{Z_1 Z_4 Z_L g_m r_o + Z_1 Z_4 Z_L}{Z_1 Z_4 g_m r_o + Z_1 Z_4 + 2Z_1 Z_L g_m r_o + 2Z_1 Z_L + Z_4 Z_L + Z_4 r_o + 2Z_L r_o}$

$$H(z) = \frac{Z_1 Z_4 Z_L g_m r_o + Z_1 Z_4 Z_L}{Z_1 Z_4 g_m r_o + Z_1 Z_4 + 2Z_1 Z_L g_m r_o + 2Z_1 Z_L + Z_4 Z_L + Z_4 r_o + 2Z_L r_o}$$

2 AP

3 BP

3.1 BP-1 $Z(s) = \left(R_1, \infty, \infty, R_4, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$

$$H(s) = \frac{s (L_L R_1 R_4 g_m r_o + L_L R_1 R_4)}{R_1 R_4 g_m r_o + R_1 R_4 + R_4 r_o + s^2 (C_L L_L R_1 R_4 g_m r_o + C_L L_L R_1 R_4 + C_L L_L R_4 r_o) + s (2L_L R_1 g_m r_o + 2L_L R_1 + L_L R_4 + 2L_L r_o)}$$

Parameters:

Q: $\frac{\sqrt{C_L} R_1 R_4 g_m r_o + \sqrt{C_L} R_1 R_4 + \sqrt{C_L} R_4 r_o}{2\sqrt{L_L} R_1 g_m r_o + 2\sqrt{L_L} R_1 + \sqrt{L_L} R_4 + 2\sqrt{L_L} r_o}$
 wo: $\frac{1}{\sqrt{C_L} \sqrt{L_L}}$
 bandwidth: $\frac{2\sqrt{L_L} R_1 g_m r_o + 2\sqrt{L_L} R_1 + \sqrt{L_L} R_4 + 2\sqrt{L_L} r_o}{\sqrt{C_L} \sqrt{L_L} (\sqrt{C_L} R_1 R_4 g_m r_o + \sqrt{C_L} R_1 R_4 + \sqrt{C_L} R_4 r_o)}$
 K-LP: 0
 K-HP: 0
 K-BP: $\frac{R_1 R_4 g_m r_o + R_1 R_4}{2R_1 g_m r_o + 2R_1 + R_4 + 2r_o}$
 Qz: None
 Wz: None

3.2 BP-2 $Z(s) = \left(R_1, \infty, \infty, \frac{1}{C_4 s}, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$

$$H(s) = \frac{s (L_L R_1 g_m r_o + L_L R_1)}{L_L s + R_1 g_m r_o + R_1 + r_o + s^2 (2C_4 L_L R_1 g_m r_o + 2C_4 L_L R_1 + 2C_4 L_L r_o + C_L L_L R_1 g_m r_o + C_L L_L R_1 + C_L L_L r_o)}$$

Parameters:

Q: $\frac{2C_4 R_1 g_m r_o + 2C_4 R_1 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o}{\sqrt{L_L} \sqrt{2C_4 + C_L}}$
 wo: $\frac{1}{\sqrt{2C_4 L_L + C_L L_L}}$
 bandwidth: $\frac{\sqrt{L_L} \sqrt{2C_4 + C_L}}{\sqrt{2C_4 L_L + C_L L_L} (2C_4 R_1 g_m r_o + 2C_4 R_1 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o)}$
 K-LP: 0
 K-HP: 0
 K-BP: $R_1 g_m r_o + R_1$
 Qz: None
 Wz: None

3.3 BP-3 $Z(s) = \left(R_1, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$

$$H(s) = \frac{s (L_L R_1 R_4 g_m r_o + L_L R_1 R_4)}{R_1 R_4 g_m r_o + R_1 R_4 + R_4 r_o + s^2 (2C_4 L_L R_1 R_4 g_m r_o + 2C_4 L_L R_1 R_4 + 2C_4 L_L R_4 r_o + C_L L_L R_1 R_4 g_m r_o + C_L L_L R_1 R_4 + C_L L_L R_4 r_o) + s (2L_L R_1 g_m r_o + 2L_L R_1 + L_L R_4 + 2L_L r_o)}$$

Parameters:

Q: $\frac{2C_4 R_1 R_4 g_m r_o + 2C_4 R_1 R_4 + 2C_4 R_4 r_o + C_L R_1 R_4 g_m r_o + C_L R_1 R_4 + C_L R_4 r_o}{2\sqrt{L_L} R_1 g_m r_o \sqrt{2C_4 + C_L} + 2\sqrt{L_L} R_1 \sqrt{2C_4 + C_L} + \sqrt{L_L} R_4 \sqrt{2C_4 + C_L} + 2\sqrt{L_L} r_o \sqrt{2C_4 + C_L}}$
 wo: $\frac{1}{\sqrt{2C_4 L_L + C_L L_L}}$
 bandwidth: $\frac{2\sqrt{L_L} R_1 g_m r_o \sqrt{2C_4 + C_L} + 2\sqrt{L_L} R_1 \sqrt{2C_4 + C_L} + \sqrt{L_L} R_4 \sqrt{2C_4 + C_L} + 2\sqrt{L_L} r_o \sqrt{2C_4 + C_L}}{\sqrt{2C_4 L_L + C_L L_L} (2C_4 R_1 R_4 g_m r_o + 2C_4 R_1 R_4 + 2C_4 R_4 r_o + C_L R_1 R_4 g_m r_o + C_L R_1 R_4 + C_L R_4 r_o)}$
 K-LP: 0
 K-HP: 0
 K-BP: $\frac{R_1 R_4 g_m r_o + R_1 R_4}{2R_1 g_m r_o + 2R_1 + R_4 + 2r_o}$
 Qz: None
 Wz: None

3.4 BP-4 $Z(s) = \left(R_1, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, R_L \right)$

$$H(s) = \frac{s(L_4 R_1 R_L g_m r_o + L_4 R_1 R_L)}{2R_1 R_L g_m r_o + 2R_1 R_L + 2R_L r_o + s^2(2C_4 L_4 R_1 R_L g_m r_o + 2C_4 L_4 R_1 R_L + 2C_4 L_4 R_L r_o) + s(L_4 R_1 g_m r_o + L_4 R_1 + L_4 R_L + L_4 r_o)}$$

Parameters:

Q: $\frac{2\sqrt{C_4}R_1R_Lg_mr_o+2\sqrt{C_4}R_1R_L+2\sqrt{C_4}R_Lr_o}{\sqrt{L_4}R_1g_mr_o+\sqrt{L_4}R_1+\sqrt{L_4}R_L+\sqrt{L_4}r_o}$
 wo: $\frac{1}{\sqrt{C_4}\sqrt{L_4}}$
 bandwidth: $\frac{\sqrt{L_4}R_1g_mr_o+\sqrt{L_4}R_1+\sqrt{L_4}R_L+\sqrt{L_4}r_o}{\sqrt{C_4}\sqrt{L_4}(2\sqrt{C_4}R_1R_Lg_mr_o+2\sqrt{C_4}R_1R_L+2\sqrt{C_4}R_Lr_o)}$
 K-LP: 0
 K-HP: 0
 K-BP: $\frac{R_1R_Lg_mr_o+R_1R_L}{R_1g_mr_o+R_1+R_L+r_o}$
 Qz: None
 Wz: None

3.5 BP-5 $Z(s) = \left(R_1, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{s(L_4 R_1 g_m r_o + L_4 R_1)}{L_4 s + 2R_1 g_m r_o + 2R_1 + 2r_o + s^2(2C_4 L_4 R_1 g_m r_o + 2C_4 L_4 R_1 + 2C_4 L_4 r_o + C_L L_4 R_1 g_m r_o + C_L L_4 R_1 + C_L L_4 r_o)}$$

Parameters:

Q: $\frac{2\sqrt{2}C_4R_1g_mr_o+2\sqrt{2}C_4R_1+2\sqrt{2}C_4r_o+\sqrt{2}C_LR_1g_mr_o+\sqrt{2}C_LR_1+\sqrt{2}C_Lr_o}{\sqrt{L_4}\sqrt{2C_4+C_L}}$
 wo: $\frac{\sqrt{2}}{\sqrt{2C_4L_4+C_LL_4}}$
 bandwidth: $\frac{\sqrt{2}\sqrt{L_4}\sqrt{2C_4+C_L}}{\sqrt{2C_4L_4+C_LL_4}(2\sqrt{2}C_4R_1g_mr_o+2\sqrt{2}C_4R_1+2\sqrt{2}C_4r_o+\sqrt{2}C_LR_1g_mr_o+\sqrt{2}C_LR_1+\sqrt{2}C_Lr_o)}$
 K-LP: 0
 K-HP: 0
 K-BP: $R_1g_mr_o + R_1$
 Qz: None
 Wz: None

3.6 BP-6 $Z(s) = \left(R_1, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{s(L_4 R_1 R_L g_m r_o + L_4 R_1 R_L)}{2R_1 R_L g_m r_o + 2R_1 R_L + 2R_L r_o + s^2(2C_4 L_4 R_1 R_L g_m r_o + 2C_4 L_4 R_1 R_L + 2C_4 L_4 R_L r_o + C_L L_4 R_1 R_L g_m r_o + C_L L_4 R_1 R_L + C_L L_4 R_L r_o) + s(L_4 R_1 g_m r_o + L_4 R_1 + L_4 R_L + L_4 r_o)}$$

Parameters:

Q: $\frac{2\sqrt{2}C_4R_1R_Lg_mr_o+2\sqrt{2}C_4R_1R_L+2\sqrt{2}C_4R_Lr_o+\sqrt{2}C_LR_1R_Lg_mr_o+\sqrt{2}C_LR_1R_L+\sqrt{2}C_LR_Lr_o}{\sqrt{L_4}R_1g_mr_o\sqrt{2C_4+C_L}+\sqrt{L_4}R_1\sqrt{2C_4+C_L}+\sqrt{L_4}R_L\sqrt{2C_4+C_L}+\sqrt{L_4}r_o\sqrt{2C_4+C_L}}$
 wo: $\frac{\sqrt{2}}{\sqrt{2C_4L_4+C_LL_4}}$
 bandwidth: $\frac{\sqrt{2}(\sqrt{L_4}R_1g_mr_o\sqrt{2C_4+C_L}+\sqrt{L_4}R_1\sqrt{2C_4+C_L}+\sqrt{L_4}R_L\sqrt{2C_4+C_L}+\sqrt{L_4}r_o\sqrt{2C_4+C_L})}{\sqrt{2C_4L_4+C_LL_4}(2\sqrt{2}C_4R_1R_Lg_mr_o+2\sqrt{2}C_4R_1R_L+2\sqrt{2}C_4R_Lr_o+\sqrt{2}C_LR_1R_Lg_mr_o+\sqrt{2}C_LR_1R_L+\sqrt{2}C_LR_Lr_o)}$
 K-LP: 0
 K-HP: 0
 K-BP: $\frac{R_1R_Lg_mr_o+R_1R_L}{R_1g_mr_o+R_1+R_L+r_o}$
 Qz: None
 Wz: None

3.7 BP-7 $Z(s) = \left(R_1, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$

$$H(s) = \frac{s(L_4 L_L R_1 g_m r_o + L_4 L_L R_1)}{L_4 L_L s + L_4 R_1 g_m r_o + L_4 R_1 + L_4 r_o + 2L_L R_1 g_m r_o + 2L_L R_1 + 2L_L r_o + s^2(2C_4 L_4 L_L R_1 g_m r_o + 2C_4 L_4 L_L R_1 + 2C_4 L_4 L_L r_o + C_L L_4 L_L R_1 g_m r_o + C_L L_4 L_L R_1 + C_L L_4 L_L r_o)}$$

Parameters:

Q: $\frac{2C_4R_1g_mr_o\sqrt{L_4+2L_L}+2C_4R_1\sqrt{L_4+2L_L}+2C_4r_o\sqrt{L_4+2L_L}+C_LR_1g_mr_o\sqrt{L_4+2L_L}+C_LR_1\sqrt{L_4+2L_L}+C_Lr_o\sqrt{L_4+2L_L}}{\sqrt{L_4}\sqrt{L_L}\sqrt{2C_4+C_L}}$
 wo: $\frac{\sqrt{L_4+2L_L}}{\sqrt{2C_4L_4L_L+C_LL_4L_L}}$

bandwidth: $\frac{\sqrt{L_4}\sqrt{L_L}\sqrt{2C_4+C_L}\sqrt{L_4+2L_L}}{\sqrt{2C_4L_4L_L+C_LL_4L_L}\left(2C_4R_1g_mr_o\sqrt{L_4+2L_L}+2C_4R_1\sqrt{L_4+2L_L}+2C_4r_o\sqrt{L_4+2L_L}+C_LR_1g_mr_o\sqrt{L_4+2L_L}+C_LR_1\sqrt{L_4+2L_L}+C_Lr_o\sqrt{L_4+2L_L}\right)}$
K-LP: 0
K-HP: 0
K-BP: $R_1g_mr_o + R_1$
Qz: None
Wz: None

3.8 BP-8 $Z(s) = \left(\frac{L_1s}{C_1L_1s^2+1}, \infty, \infty, R_4, \infty, R_L \right)$

$$H(s) = \frac{s(L_1R_4R_Lg_mr_o + L_1R_4R_L)}{R_4R_L + R_4r_o + 2R_Lr_o + s^2(C_1L_1R_4R_L + C_1L_1R_4r_o + 2C_1L_1R_Lr_o) + s(L_1R_4g_mr_o + L_1R_4 + 2L_1R_Lg_mr_o + 2L_1R_L)}$$

Parameters:

Q: $\frac{\sqrt{C_1}R_4R_L+\sqrt{C_1}R_4r_o+2\sqrt{C_1}R_Lr_o}{\sqrt{L_1}R_4g_mr_o+\sqrt{L_1}R_4+2\sqrt{L_1}R_Lg_mr_o+2\sqrt{L_1}R_L}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{\sqrt{L_1}R_4g_mr_o+\sqrt{L_1}R_4+2\sqrt{L_1}R_Lg_mr_o+2\sqrt{L_1}R_L}{\sqrt{C_1}\sqrt{L_1}\left(\sqrt{C_1}R_4R_L+\sqrt{C_1}R_4r_o+2\sqrt{C_1}R_Lr_o\right)}$
K-LP: 0
K-HP: 0
K-BP: $\frac{R_4R_L}{R_4+2R_L}$
Qz: None
Wz: None

4 BS

4.1 BS-1 $Z(s) = \left(R_1, \infty, \infty, R_4, \infty, L_Ls + \frac{1}{C_Ls} \right)$

$$H(s) = \frac{R_1R_4g_mr_o + R_1R_4 + s^2(C_LL_LR_1R_4g_mr_o + C_LL_LR_1R_4)}{2R_1g_mr_o + 2R_1 + R_4 + 2r_o + s^2(2C_LL_LR_1g_mr_o + 2C_LL_LR_1 + C_LL_LR_4 + 2C_LL_Lr_o) + s(C_LR_1R_4g_mr_o + C_LR_1R_4 + C_LR_4r_o)}$$

Parameters:

Q: $\frac{2\sqrt{L_L}R_1g_mr_o+2\sqrt{L_L}R_1+\sqrt{L_L}R_4+2\sqrt{L_L}r_o}{\sqrt{C_L}R_1R_4g_mr_o+\sqrt{C_L}R_1R_4+\sqrt{C_L}R_4r_o}$
wo: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$
bandwidth: $\frac{\sqrt{C_L}R_1R_4g_mr_o+\sqrt{C_L}R_1R_4+\sqrt{C_L}R_4r_o}{\sqrt{C_L}\sqrt{L_L}\left(2\sqrt{L_L}R_1g_mr_o+2\sqrt{L_L}R_1+\sqrt{L_L}R_4+2\sqrt{L_L}r_o\right)}$
K-LP: $\frac{R_1R_4g_mr_o+R_1R_4}{2R_1g_mr_o+2R_1+R_4+2r_o}$
K-HP: $\frac{R_1R_4g_mr_o+R_1R_4}{2R_1g_mr_o+2R_1+R_4+2r_o}$
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$

4.2 BS-2 $Z(s) = \left(R_1, \infty, \infty, L_4s + \frac{1}{C_4s}, \infty, R_L \right)$

$$H(s) = \frac{R_1R_Lg_mr_o + R_1R_L + s^2(C_4L_4R_1R_Lg_mr_o + C_4L_4R_1R_L)}{R_1g_mr_o + R_1 + R_L + r_o + s^2(C_4L_4R_1g_mr_o + C_4L_4R_1 + C_4L_4R_L + C_4L_4r_o) + s(2C_4R_1R_Lg_mr_o + 2C_4R_1R_L + 2C_4R_Lr_o)}$$

Parameters:

Q: $\frac{\sqrt{L_4}R_1g_mr_o+\sqrt{L_4}R_1+\sqrt{L_4}R_L+\sqrt{L_4}r_o}{2\sqrt{C_4}R_1R_Lg_mr_o+2\sqrt{C_4}R_1R_L+2\sqrt{C_4}R_Lr_o}$
wo: $\frac{1}{\sqrt{C_4}\sqrt{L_4}}$
bandwidth: $\frac{2\sqrt{C_4}R_1R_Lg_mr_o+2\sqrt{C_4}R_1R_L+2\sqrt{C_4}R_Lr_o}{\sqrt{C_4}\sqrt{L_4}\left(\sqrt{L_4}R_1g_mr_o+\sqrt{L_4}R_1+\sqrt{L_4}R_L+\sqrt{L_4}r_o\right)}$
K-LP: $\frac{R_1R_Lg_mr_o+R_1R_L}{R_1g_mr_o+R_1+R_L+r_o}$
K-HP: $\frac{R_1R_Lg_mr_o+R_1R_L}{R_1g_mr_o+R_1+R_L+r_o}$
K-BP: 0
Qz: None

$$\text{Wz: } \frac{1}{\sqrt{C_4}\sqrt{L_4}}$$

$$\mathbf{4.3 \quad BS-3} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \infty, \quad R_L \right)$$

$$H(s) = \frac{R_4 R_L g_m r_o + R_4 R_L + s^2 (C_1 L_1 R_4 R_L g_m r_o + C_1 L_1 R_4 R_L)}{R_4 g_m r_o + R_4 + 2 R_L g_m r_o + 2 R_L + s^2 (C_1 L_1 R_4 g_m r_o + C_1 L_1 R_4 + 2 C_1 L_1 R_L g_m r_o + 2 C_1 L_1 R_L) + s (C_1 R_4 R_L + C_1 R_4 r_o + 2 C_1 R_L r_o)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{L_1} R_4 g_m r_o + \sqrt{L_1} R_4 + 2 \sqrt{L_1} R_L g_m r_o + 2 \sqrt{L_1} R_L}{\sqrt{C_1} R_4 R_L + \sqrt{C_1} R_4 r_o + 2 \sqrt{C_1} R_L r_o} \\ \text{wo: } & \frac{1}{\sqrt{C_1} \sqrt{L_1}} \\ \text{bandwidth: } & \frac{\sqrt{C_1} R_4 R_L + \sqrt{C_1} R_4 r_o + 2 \sqrt{C_1} R_L r_o}{\sqrt{C_1} \sqrt{L_1} (\sqrt{L_1} R_4 g_m r_o + \sqrt{L_1} R_4 + 2 \sqrt{L_1} R_L g_m r_o + 2 \sqrt{L_1} R_L)} \\ \text{K-LP: } & \frac{R_4 R_L}{R_4 + 2 R_L} \\ \text{K-HP: } & \frac{R_4 R_L}{R_4 + 2 R_L} \\ \text{K-BP: } & 0 \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \frac{1}{\sqrt{C_1} \sqrt{L_1}} \end{aligned}$$

5 GE

6 HP

7 LP

$$\mathbf{7.1 \quad LP-1} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \infty, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_4 g_m r_o + R_4}{C_1 C_L R_4 r_o s^2 + 2 g_m r_o + s (C_1 R_4 + 2 C_1 r_o + C_L R_4 g_m r_o + C_L R_4) + 2}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2} \sqrt{C_1} \sqrt{C_L} \sqrt{R_4} r_o \sqrt{g_m + \frac{1}{r_o}}}{C_1 R_4 + 2 C_1 r_o + C_L R_4 g_m r_o + C_L R_4} \\ \text{wo: } & \frac{\sqrt{\frac{2 g_m r_o + 2}{r_o}}}{\sqrt{C_1} \sqrt{C_L} \sqrt{R_4}} \\ \text{bandwidth: } & \frac{\sqrt{2} \sqrt{\frac{2 g_m r_o + 2}{r_o}} (C_1 R_4 + 2 C_1 r_o + C_L R_4 g_m r_o + C_L R_4)}{2 C_1 C_L R_4 r_o \sqrt{g_m + \frac{1}{r_o}}} \\ \text{K-LP: } & \frac{R_4}{2} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & 0 \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

$$\mathbf{7.2 \quad LP-2} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \infty, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{R_4 R_L g_m r_o + R_4 R_L}{C_1 C_L R_4 R_L r_o s^2 + R_4 g_m r_o + R_4 + 2 R_L g_m r_o + 2 R_L + s (C_1 R_4 R_L + C_1 R_4 r_o + 2 C_1 R_L r_o + C_L R_4 R_L g_m r_o + C_L R_4 R_L)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_1} \sqrt{C_L} \sqrt{R_4} \sqrt{R_L} r_o \sqrt{R_4 g_m + \frac{R_4}{r_o} + 2 R_L g_m + \frac{2 R_L}{r_o}}}{C_1 R_4 R_L + C_1 R_4 r_o + 2 C_1 R_L r_o + C_L R_4 R_L g_m r_o + C_L R_4 R_L} \\ \text{wo: } & \frac{\sqrt{\frac{R_4 g_m r_o + R_4 + 2 R_L g_m r_o + 2 R_L}{r_o}}}{\sqrt{C_1} \sqrt{C_L} \sqrt{R_4} \sqrt{R_L}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{R_4 g_m r_o + R_4 + 2 R_L g_m r_o + 2 R_L}{r_o}} (C_1 R_4 R_L + C_1 R_4 r_o + 2 C_1 R_L r_o + C_L R_4 R_L g_m r_o + C_L R_4 R_L)}{C_1 C_L R_4 R_L r_o \sqrt{R_4 g_m + \frac{R_4}{r_o} + 2 R_L g_m + \frac{2 R_L}{r_o}}} \end{aligned}$$

K-LP: $\frac{R_4 R_L}{R_4 + 2R_L}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.3 LP-3 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{1}{C_4 s}, \infty, R_L \right)$

$$H(s) = \frac{R_L g_m r_o + R_L}{2C_1 C_4 R_L r_o s^2 + g_m r_o + s(C_1 R_L + C_1 r_o + 2C_4 R_L g_m r_o + 2C_4 R_L) + 1}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_L}r_o\sqrt{g_m+\frac{1}{r_o}}}{C_1 R_L + C_1 r_o + 2C_4 R_L g_m r_o + 2C_4 R_L}$
wo: $\frac{\sqrt{2}\sqrt{\frac{g_m r_o + 1}{r_o}}}{2\sqrt{C_1}\sqrt{C_4}\sqrt{R_L}}$
bandwidth: $\frac{\sqrt{\frac{g_m r_o + 1}{r_o}}(C_1 R_L + C_1 r_o + 2C_4 R_L g_m r_o + 2C_4 R_L)}{2C_1 C_4 R_L r_o \sqrt{g_m + \frac{1}{r_o}}}$
K-LP: R_L
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.4 LP-4 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{1}{C_4 s}, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_L g_m r_o + R_L}{g_m r_o + s^2(2C_1 C_4 R_L r_o + C_1 C_L R_L r_o) + s(C_1 R_L + C_1 r_o + 2C_4 R_L g_m r_o + 2C_4 R_L + C_L R_L g_m r_o + C_L R_L) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{R_L}r_o\sqrt{2C_4+C_L}\sqrt{g_m+\frac{1}{r_o}}}{C_1 R_L + C_1 r_o + 2C_4 R_L g_m r_o + 2C_4 R_L + C_L R_L g_m r_o + C_L R_L}$
wo: $\sqrt{\frac{g_m r_o + 1}{2C_1 C_4 R_L r_o + C_1 C_L R_L r_o}}$
bandwidth: $\frac{\sqrt{\frac{g_m r_o + 1}{2C_1 C_4 R_L r_o + C_1 C_L R_L r_o}}(C_1 R_L + C_1 r_o + 2C_4 R_L g_m r_o + 2C_4 R_L + C_L R_L g_m r_o + C_L R_L)}{\sqrt{C_1}\sqrt{R_L}r_o\sqrt{2C_4+C_L}\sqrt{g_m+\frac{1}{r_o}}}$
K-LP: R_L
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.5 LP-5 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, R_L \right)$

$$H(s) = \frac{R_4 R_L g_m r_o + R_4 R_L}{2C_1 C_4 R_4 R_L r_o s^2 + R_4 g_m r_o + R_4 + 2R_L g_m r_o + 2R_L + s(C_1 R_4 R_L + C_1 R_4 r_o + 2C_1 R_L r_o + 2C_4 R_4 R_L g_m r_o + 2C_4 R_4 R_L)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_4}\sqrt{R_L}r_o\sqrt{R_4 g_m + \frac{R_4}{r_o} + 2R_L g_m + \frac{2R_L}{r_o}}}{C_1 R_4 R_L + C_1 R_4 r_o + 2C_1 R_L r_o + 2C_4 R_4 R_L g_m r_o + 2C_4 R_4 R_L}$
wo: $\frac{\sqrt{2}\sqrt{\frac{R_4 g_m r_o + R_4 + 2R_L g_m r_o + 2R_L}{r_o}}}{2\sqrt{C_1}\sqrt{C_4}\sqrt{R_4}\sqrt{R_L}}$
bandwidth: $\frac{\sqrt{\frac{R_4 g_m r_o + R_4 + 2R_L g_m r_o + 2R_L}{r_o}}(C_1 R_4 R_L + C_1 R_4 r_o + 2C_1 R_L r_o + 2C_4 R_4 R_L g_m r_o + 2C_4 R_4 R_L)}{2C_1 C_4 R_4 R_L r_o \sqrt{R_4 g_m + \frac{R_4}{r_o} + 2R_L g_m + \frac{2R_L}{r_o}}}$
K-LP: $\frac{R_4 R_L}{R_4 + 2R_L}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.6 LP-6 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_4 g_m r_o + R_4}{2g_m r_o + s^2 (2C_1 C_4 R_4 r_o + C_1 C_L R_4 r_o) + s (C_1 R_4 + 2C_1 r_o + 2C_4 R_4 g_m r_o + 2C_4 R_4 + C_L R_4 g_m r_o + C_L R_4) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{R_4}r_o\sqrt{2C_4+C_L}\sqrt{g_m+\frac{1}{r_o}}}{C_1 R_4 + 2C_1 r_o + 2C_4 R_4 g_m r_o + 2C_4 R_4 + C_L R_4 g_m r_o + C_L R_4}$
 wo: $\sqrt{\frac{2g_m r_o + 2}{2C_1 C_4 R_4 r_o + C_1 C_L R_4 r_o}}$
 bandwidth: $\frac{\sqrt{2}\sqrt{\frac{2g_m r_o + 2}{2C_1 C_4 R_4 r_o + C_1 C_L R_4 r_o}}(C_1 R_4 + 2C_1 r_o + 2C_4 R_4 g_m r_o + 2C_4 R_4 + C_L R_4 g_m r_o + C_L R_4)}{2\sqrt{C_1}\sqrt{R_4}r_o\sqrt{2C_4+C_L}\sqrt{g_m+\frac{1}{r_o}}}$
 K-LP: $\frac{R_4}{2}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

7.7 LP-7 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_4 R_L g_m r_o + R_4 R_L}{R_4 g_m r_o + R_4 + 2R_L g_m r_o + 2R_L + s^2 (2C_1 C_4 R_4 R_L r_o + C_1 C_L R_4 R_L r_o) + s (C_1 R_4 R_L + C_1 R_4 r_o + 2C_1 R_L r_o + 2C_4 R_4 R_L g_m r_o + 2C_4 R_4 R_L + C_L R_4 R_L g_m r_o + C_L R_4 R_L)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{R_4}\sqrt{R_L}r_o\sqrt{2C_4+C_L}\sqrt{R_4 g_m + \frac{R_4}{r_o} + 2R_L g_m + \frac{2R_L}{r_o}}}{C_1 R_4 R_L + C_1 R_4 r_o + 2C_1 R_L r_o + 2C_4 R_4 R_L g_m r_o + 2C_4 R_4 R_L + C_L R_4 R_L g_m r_o + C_L R_4 R_L}$
 wo: $\sqrt{\frac{R_4 g_m r_o + R_4 + 2R_L g_m r_o + 2R_L}{2C_1 C_4 R_4 R_L r_o + C_1 C_L R_4 R_L r_o}}$
 bandwidth: $\frac{\sqrt{\frac{R_4 g_m r_o + R_4 + 2R_L g_m r_o + 2R_L}{2C_1 C_4 R_4 R_L r_o + C_1 C_L R_4 R_L r_o}}(C_1 R_4 R_L + C_1 R_4 r_o + 2C_1 R_L r_o + 2C_4 R_4 R_L g_m r_o + 2C_4 R_4 R_L + C_L R_4 R_L g_m r_o + C_L R_4 R_L)}{\sqrt{C_1}\sqrt{R_4}\sqrt{R_L}r_o\sqrt{2C_4+C_L}\sqrt{R_4 g_m + \frac{R_4}{r_o} + 2R_L g_m + \frac{2R_L}{r_o}}}$
 K-LP: $\frac{R_4 R_L}{R_4 + 2R_L}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

7.8 LP-8 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 R_4 g_m r_o + R_1 R_4}{C_1 C_L R_1 R_4 r_o s^2 + 2R_1 g_m r_o + 2R_1 + R_4 + 2r_o + s (C_1 R_1 R_4 + 2C_1 R_1 r_o + C_L R_1 R_4 g_m r_o + C_L R_1 R_4 + C_L R_4 r_o)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_L}\sqrt{R_1}\sqrt{R_4}r_o\sqrt{2R_1 g_m + \frac{2R_1}{r_o} + \frac{R_4}{r_o} + 2}}{C_1 R_1 R_4 + 2C_1 R_1 r_o + C_L R_1 R_4 g_m r_o + C_L R_1 R_4 + C_L R_4 r_o}$
 wo: $\sqrt{\frac{\frac{2R_1 g_m r_o + 2R_1 + R_4 + 2r_o}{r_o}}{\sqrt{C_1}\sqrt{C_L}\sqrt{R_1}\sqrt{R_4}}}$
 bandwidth: $\frac{\sqrt{\frac{2R_1 g_m r_o + 2R_1 + R_4 + 2r_o}{r_o}}(C_1 R_1 R_4 + 2C_1 R_1 r_o + C_L R_1 R_4 g_m r_o + C_L R_1 R_4 + C_L R_4 r_o)}{C_1 C_L R_1 R_4 r_o \sqrt{2R_1 g_m + \frac{2R_1}{r_o} + \frac{R_4}{r_o} + 2}}$
 K-LP: $\frac{R_1 R_4 g_m r_o + R_1 R_4}{2R_1 g_m r_o + 2R_1 + R_4 + 2r_o}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

7.9 LP-9 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_1 R_4 R_L g_m r_o + R_1 R_4 R_L}{C_1 C_L R_1 R_4 R_L r_o s^2 + R_1 R_4 g_m r_o + R_1 R_4 + 2 R_1 R_L g_m r_o + 2 R_1 R_L + R_4 R_L + R_4 r_o + 2 R_L r_o + s (C_1 R_1 R_4 R_L + C_1 R_1 R_4 r_o + 2 C_1 R_1 R_L r_o + C_L R_1 R_4 R_L g_m r_o + C_L R_1 R_4 R_L + C_L R_4 R_L r_o)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} \sqrt{R_4} \sqrt{R_L} r_o \sqrt{R_1 R_4 g_m + \frac{R_1 R_4}{r_o} + 2 R_1 R_L g_m + \frac{2 R_1 R_L}{r_o} + \frac{R_4 R_L}{r_o} + R_4 + 2 R_L}}{C_1 R_1 R_4 R_L + C_1 R_1 R_4 r_o + 2 C_1 R_1 R_L r_o + C_L R_1 R_4 R_L g_m r_o + C_L R_1 R_4 R_L + C_L R_4 R_L r_o}$

wo: $\frac{\sqrt{\frac{R_1 R_4 g_m r_o + R_1 R_4 + 2 R_1 R_L g_m r_o + 2 R_1 R_L + R_4 R_L + R_4 r_o + 2 R_L r_o}{r_o}}}{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} \sqrt{R_4} \sqrt{R_L}}$

bandwidth: $\frac{\sqrt{\frac{R_1 R_4 g_m r_o + R_1 R_4 + 2 R_1 R_L g_m r_o + 2 R_1 R_L + R_4 R_L + R_4 r_o + 2 R_L r_o}{r_o}} (C_1 R_1 R_4 R_L + C_1 R_1 R_4 r_o + 2 C_1 R_1 R_L r_o + C_L R_1 R_4 R_L g_m r_o + C_L R_1 R_4 R_L + C_L R_4 R_L r_o)}{C_1 C_L R_1 R_4 R_L r_o \sqrt{R_1 R_4 g_m + \frac{R_1 R_4}{r_o} + 2 R_1 R_L g_m + \frac{2 R_1 R_L}{r_o} + \frac{R_4 R_L}{r_o} + R_4 + 2 R_L}}$

K-LP: $\frac{R_1 R_4 R_L g_m r_o + R_1 R_4 R_L}{R_1 R_4 g_m r_o + R_1 R_4 + 2 R_1 R_L g_m r_o + 2 R_1 R_L + R_4 R_L + R_4 r_o + 2 R_L r_o}$

K-HP: 0

K-BP: 0

Qz: None

Wz: None

7.10 LP-10 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{1}{C_4 s}, \infty, R_L \right)$

$$H(s) = \frac{R_1 R_L g_m r_o + R_1 R_L}{2 C_1 C_4 R_1 R_L r_o s^2 + R_1 g_m r_o + R_1 + R_L + r_o + s (C_1 R_1 R_L + C_1 R_1 r_o + 2 C_4 R_1 R_L g_m r_o + 2 C_4 R_1 R_L + 2 C_4 R_L r_o)}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_1} \sqrt{C_4} \sqrt{R_1} \sqrt{R_L} r_o \sqrt{R_1 g_m + \frac{R_1}{r_o} + \frac{R_L}{r_o} + 1}}{C_1 R_1 R_L + C_1 R_1 r_o + 2 C_4 R_1 R_L g_m r_o + 2 C_4 R_1 R_L + 2 C_4 R_L r_o}$

wo: $\frac{\sqrt{2} \sqrt{\frac{R_1 g_m r_o + R_1 + R_L + r_o}{r_o}}}{2 \sqrt{C_1} \sqrt{C_4} \sqrt{R_1} \sqrt{R_L}}$

bandwidth: $\frac{\sqrt{\frac{R_1 g_m r_o + R_1 + R_L + r_o}{r_o}} (C_1 R_1 R_L + C_1 R_1 r_o + 2 C_4 R_1 R_L g_m r_o + 2 C_4 R_1 R_L + 2 C_4 R_L r_o)}{2 C_1 C_4 R_1 R_L r_o \sqrt{R_1 g_m + \frac{R_1}{r_o} + \frac{R_L}{r_o} + 1}}$

K-LP: $\frac{R_1 R_L g_m r_o + R_1 R_L}{R_1 g_m r_o + R_1 + R_L + r_o}$

K-HP: 0

K-BP: 0

Qz: None

Wz: None

7.11 LP-11 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{1}{C_4 s}, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 g_m r_o + R_1}{s^2 (2 C_1 C_4 R_1 r_o + C_1 C_L R_1 r_o) + s (C_1 R_1 + 2 C_4 R_1 g_m r_o + 2 C_4 R_1 + 2 C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{R_1} r_o \sqrt{2 C_4 + C_L} \sqrt{\frac{1}{r_o}}}{C_1 R_1 + 2 C_4 R_1 g_m r_o + 2 C_4 R_1 + 2 C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o}$

wo: $\sqrt{\frac{1}{2 C_1 C_4 R_1 r_o + C_1 C_L R_1 r_o}}$

bandwidth: $\frac{(C_1 R_1 + 2 C_4 R_1 g_m r_o + 2 C_4 R_1 + 2 C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o) \sqrt{\frac{1}{2 C_1 C_4 R_1 r_o + C_1 C_L R_1 r_o}}}{\sqrt{C_1} \sqrt{R_1} r_o \sqrt{2 C_4 + C_L} \sqrt{\frac{1}{r_o}}}$

K-LP: $R_1 g_m r_o + R_1$

K-HP: 0

K-BP: 0

Qz: None

Wz: None

7.12 LP-12 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{1}{C_4 s}, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_1 R_L g_m r_o + R_1 R_L}{R_1 g_m r_o + R_1 + R_L + r_o + s^2 (2C_1 C_4 R_1 R_L r_o + C_1 C_L R_1 R_L r_o) + s (C_1 R_1 R_L + C_1 R_1 r_o + 2C_4 R_1 R_L g_m r_o + 2C_4 R_1 R_L + 2C_4 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{R_1} \sqrt{R_L} r_o \sqrt{2C_4 + C_L} \sqrt{R_1 g_m + \frac{R_1}{r_o} + \frac{R_L}{r_o} + 1}}{C_1 R_1 R_L + C_1 R_1 r_o + 2C_4 R_1 R_L g_m r_o + 2C_4 R_1 R_L + 2C_4 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o}$

wo: $\sqrt{\frac{R_1 g_m r_o + R_1 + R_L + r_o}{2C_1 C_4 R_1 R_L r_o + C_1 C_L R_1 R_L r_o}}$

bandwidth: $\frac{\sqrt{\frac{R_1 g_m r_o + R_1 + R_L + r_o}{2C_1 C_4 R_1 R_L r_o + C_1 C_L R_1 R_L r_o}} (C_1 R_1 R_L + C_1 R_1 r_o + 2C_4 R_1 R_L g_m r_o + 2C_4 R_1 R_L + 2C_4 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o)}{\sqrt{C_1} \sqrt{R_1} \sqrt{R_L} r_o \sqrt{2C_4 + C_L} \sqrt{R_1 g_m + \frac{R_1}{r_o} + \frac{R_L}{r_o} + 1}}$

K-LP: $\frac{R_1 R_L g_m r_o + R_1 R_L}{R_1 g_m r_o + R_1 + R_L + r_o}$

K-HP: 0

K-BP: 0

Qz: None

Wz: None

7.13 LP-13 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, R_L \right)$

$$H(s) = \frac{R_1 R_4 R_L g_m r_o + R_1 R_4 R_L}{2C_1 C_4 R_1 R_4 R_L r_o s^2 + R_1 R_4 g_m r_o + R_1 R_4 + 2R_1 R_L g_m r_o + 2R_1 R_L + R_4 R_L + R_4 r_o + 2R_L r_o + s (C_1 R_1 R_4 R_L + C_1 R_1 R_4 r_o + 2C_1 R_1 R_L r_o + 2C_4 R_1 R_4 R_L g_m r_o + 2C_4 R_1 R_4 R_L + 2C_4 R_4 R_L r_o)}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_1} \sqrt{C_4} \sqrt{R_1} \sqrt{R_4} \sqrt{R_L} r_o \sqrt{R_1 R_4 g_m + \frac{R_1 R_4}{r_o} + 2R_1 R_L g_m + \frac{2R_1 R_L}{r_o} + \frac{R_4 R_L}{r_o} + R_4 + 2R_L}}{C_1 R_1 R_4 R_L + C_1 R_1 R_4 r_o + 2C_1 R_1 R_L r_o + 2C_4 R_1 R_4 R_L g_m r_o + 2C_4 R_1 R_4 R_L + 2C_4 R_4 R_L r_o}$

wo: $\frac{\sqrt{2} \sqrt{\frac{R_1 R_4 g_m r_o + R_1 R_4 + 2R_1 R_L g_m r_o + 2R_1 R_L + R_4 R_L + R_4 r_o + 2R_L r_o}{r_o}}}{2\sqrt{C_1} \sqrt{C_4} \sqrt{R_1} \sqrt{R_4} \sqrt{R_L}}$

bandwidth: $\frac{\sqrt{\frac{R_1 R_4 g_m r_o + R_1 R_4 + 2R_1 R_L g_m r_o + 2R_1 R_L + R_4 R_L + R_4 r_o + 2R_L r_o}{r_o}} (C_1 R_1 R_4 R_L + C_1 R_1 R_4 r_o + 2C_1 R_1 R_L r_o + 2C_4 R_1 R_4 R_L g_m r_o + 2C_4 R_1 R_4 R_L + 2C_4 R_4 R_L r_o)}{2C_1 C_4 R_1 R_4 R_L r_o \sqrt{R_1 R_4 g_m + \frac{R_1 R_4}{r_o} + 2R_1 R_L g_m + \frac{2R_1 R_L}{r_o} + \frac{R_4 R_L}{r_o} + R_4 + 2R_L}}$

K-LP: $\frac{R_1 R_4 R_L g_m r_o + R_1 R_4 R_L}{R_1 R_4 g_m r_o + R_1 R_4 + 2R_1 R_L g_m r_o + 2R_1 R_L + R_4 R_L + R_4 r_o + 2R_L r_o}$

K-HP: 0

K-BP: 0

Qz: None

Wz: None

7.14 LP-14 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 R_4 g_m r_o + R_1 R_4}{2R_1 g_m r_o + 2R_1 + R_4 + 2r_o + s^2 (2C_1 C_4 R_1 R_4 r_o + C_1 C_L R_1 R_4 r_o) + s (C_1 R_1 R_4 + 2C_1 R_1 r_o + 2C_4 R_1 R_4 g_m r_o + 2C_4 R_1 R_4 + 2C_4 R_4 r_o + C_L R_1 R_4 g_m r_o + C_L R_1 R_4 + C_L R_4 r_o)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{R_1} \sqrt{R_4} r_o \sqrt{2C_4 + C_L} \sqrt{2R_1 g_m + \frac{2R_1}{r_o} + \frac{R_4}{r_o} + 2}}{C_1 R_1 R_4 + 2C_1 R_1 r_o + 2C_4 R_1 R_4 g_m r_o + 2C_4 R_1 R_4 + 2C_4 R_4 r_o + C_L R_1 R_4 g_m r_o + C_L R_1 R_4 + C_L R_4 r_o}$

wo: $\sqrt{\frac{2R_1 g_m r_o + 2R_1 + R_4 + 2r_o}{2C_1 C_4 R_1 R_4 r_o + C_1 C_L R_1 R_4 r_o}}$

bandwidth: $\frac{\sqrt{\frac{2R_1 g_m r_o + 2R_1 + R_4 + 2r_o}{2C_1 C_4 R_1 R_4 r_o + C_1 C_L R_1 R_4 r_o}} (C_1 R_1 R_4 + 2C_1 R_1 r_o + 2C_4 R_1 R_4 g_m r_o + 2C_4 R_1 R_4 + 2C_4 R_4 r_o + C_L R_1 R_4 g_m r_o + C_L R_1 R_4 + C_L R_4 r_o)}{\sqrt{C_1} \sqrt{R_1} \sqrt{R_4} r_o \sqrt{2C_4 + C_L} \sqrt{2R_1 g_m + \frac{2R_1}{r_o} + \frac{R_4}{r_o} + 2}}$

K-LP: $\frac{R_1 R_4 g_m r_o + R_1 R_4}{2R_1 g_m r_o + 2R_1 + R_4 + 2r_o}$

K-HP: 0

K-BP: 0

Qz: None

Wz: None

7.15 LP-15 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_1 R_4 R_L g_m r_o + R_1 R_4 R_L}{R_1 R_4 g_m r_o + R_1 R_4 + 2R_1 R_L g_m r_o + 2R_1 R_L + R_4 R_L + R_4 r_o + 2R_L r_o + s^2 (2C_1 C_4 R_1 R_4 R_L r_o + C_1 C_L R_1 R_4 R_L r_o) + s (C_1 R_1 R_4 R_L + C_1 R_1 R_4 r_o + 2C_1 R_1 R_L r_o + 2C_4 R_1 R_4 R_L g_m r_o + 2C_4 R_1 R_4 R_L + 2C_4 R_4 R_L r_o + C_L R_1 R_4 R_L g_m r_o + C_L R_1 R_4 R_L + C_L R_4 R_L r_o)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{R_1} \sqrt{R_4} \sqrt{R_L} r_o \sqrt{2C_4 + C_L} \sqrt{R_1 R_4 g_m + \frac{R_1 R_4}{r_o} + 2R_1 R_L g_m + \frac{2R_1 R_L}{r_o} + \frac{R_4 R_L}{r_o} + R_4 + 2R_L}}{C_1 R_1 R_4 R_L + C_1 R_1 R_4 r_o + 2C_1 R_1 R_L r_o + 2C_4 R_1 R_4 R_L g_m r_o + 2C_4 R_1 R_4 R_L + 2C_4 R_4 R_L r_o + C_L R_1 R_4 R_L g_m r_o + C_L R_1 R_4 R_L + C_L R_4 R_L r_o}$

wo: $\sqrt{\frac{R_1 R_4 g_m r_o + R_1 R_4 + 2R_1 R_L g_m r_o + 2R_1 R_L + R_4 R_L + R_4 r_o + 2R_L r_o}{2C_1 C_4 R_1 R_4 R_L r_o + C_1 C_L R_1 R_4 R_L r_o}}$

bandwidth: $\frac{\sqrt{\frac{R_1 R_4 g_m r_o + R_1 R_4 + 2R_1 R_L g_m r_o + 2R_1 R_L + R_4 R_L + R_4 r_o + 2R_L r_o}{2C_1 C_4 R_1 R_4 R_L r_o + C_1 C_L R_1 R_4 R_L r_o}}}{\sqrt{C_1} \sqrt{R_1} \sqrt{R_4} \sqrt{R_L} r_o \sqrt{2C_4 + C_L} \sqrt{R_1 R_4 g_m + \frac{R_1 R_4}{r_o} + 2R_1 R_L g_m + \frac{2R_1 R_L}{r_o} + \frac{R_4 R_L}{r_o} + R_4 + 2R_L}} (C_1 R_1 R_4 R_L + C_1 R_1 R_4 r_o + 2C_1 R_1 R_L r_o + 2C_4 R_1 R_4 R_L g_m r_o + 2C_4 R_1 R_4 R_L + 2C_4 R_4 R_L r_o + C_L R_1 R_4 R_L g_m r_o + C_L R_1 R_4 R_L + C_L R_4 R_L r_o)$

K-LP: $\frac{R_1 R_4 R_L g_m r_o + R_1 R_4 R_L}{R_1 R_4 g_m r_o + R_1 R_4 + 2R_1 R_L g_m r_o + 2R_1 R_L + R_4 R_L + R_4 r_o + 2R_L r_o}$

K-HP: 0

K-BP: 0

Qz: None

Wz: None

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(R_1, \infty, \infty, \frac{1}{C_4 s}, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s (C_L R_1 R_L g_m r_o + C_L R_1 R_L)}{s^2 (2C_4 C_L R_1 R_L g_m r_o + 2C_4 C_L R_1 R_L + 2C_4 C_L R_L r_o) + s (2C_4 R_1 g_m r_o + 2C_4 R_1 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L R_L + C_L r_o) + 1}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_4} \sqrt{C_L} R_1 \sqrt{R_L} g_m r_o \sqrt{\frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{2} \sqrt{C_4} \sqrt{C_L} R_1 \sqrt{R_L} \sqrt{\frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{2} \sqrt{C_4} \sqrt{C_L} \sqrt{R_L} r_o \sqrt{\frac{1}{R_1 g_m r_o + R_1 + r_o}}}{2C_4 R_1 g_m r_o + 2C_4 R_1 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L R_L + C_L r_o}$

wo: $\sqrt{\frac{1}{2C_4 C_L R_1 R_L g_m r_o + 2C_4 C_L R_1 R_L + 2C_4 C_L R_L r_o}}$

bandwidth: $\frac{(2C_4 R_1 g_m r_o + 2C_4 R_1 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L R_L + C_L r_o) \sqrt{\frac{1}{2C_4 C_L R_1 R_L g_m r_o + 2C_4 C_L R_1 R_L + 2C_4 C_L R_L r_o}}}{\sqrt{2} \sqrt{C_4} \sqrt{C_L} R_1 \sqrt{R_L} g_m r_o \sqrt{\frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{2} \sqrt{C_4} \sqrt{C_L} R_1 \sqrt{R_L} \sqrt{\frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{2} \sqrt{C_4} \sqrt{C_L} \sqrt{R_L} r_o \sqrt{\frac{1}{R_1 g_m r_o + R_1 + r_o}}}$

K-LP: $R_1 g_m r_o + R_1$

K-HP: 0

K-BP: $\frac{C_L R_1 R_L g_m r_o + C_L R_1 R_L}{2C_4 R_1 g_m r_o + 2C_4 R_1 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L R_L + C_L r_o}$

Qz: None

Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(R_1, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 R_4 g_m r_o + R_1 R_4 + s (C_L R_1 R_4 R_L g_m r_o + C_L R_1 R_4 R_L)}{2R_1 g_m r_o + 2R_1 + R_4 + 2r_o + s^2 (2C_4 C_L R_1 R_4 R_L g_m r_o + 2C_4 C_L R_1 R_4 R_L + 2C_4 C_L R_4 R_L r_o) + s (2C_4 R_1 R_4 g_m r_o + 2C_4 R_1 R_4 + 2C_4 R_4 r_o + C_L R_1 R_4 g_m r_o + C_L R_1 R_4 + 2C_L R_1 R_L g_m r_o + 2C_L R_1 R_L + C_L R_4 R_L + C_L R_4 r_o + 2C_L R_L r_o)}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_4} \sqrt{C_L} R_1 \sqrt{R_4} \sqrt{R_L} g_m r_o \sqrt{\frac{2R_1 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_1}{R_1 g_m r_o + R_1 + r_o} + \frac{R_4}{R_1 g_m r_o + R_1 + r_o} + \frac{2r_o}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{2} \sqrt{C_4} \sqrt{C_L} R_1 \sqrt{R_4} \sqrt{R_L} \sqrt{\frac{2R_1 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_1}{R_1 g_m r_o + R_1 + r_o} + \frac{R_4}{R_1 g_m r_o + R_1 + r_o} + \frac{2r_o}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{2} \sqrt{C_4} \sqrt{C_L} \sqrt{R_4} \sqrt{R_L} r_o \sqrt{\frac{2R_1 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_1}{R_1 g_m r_o + R_1 + r_o} + \frac{R_4}{R_1 g_m r_o + R_1 + r_o} + \frac{2r_o}{R_1 g_m r_o + R_1 + r_o}}}{2C_4 R_1 R_4 g_m r_o + 2C_4 R_1 R_4 + 2C_4 R_4 r_o + C_L R_1 R_4 g_m r_o + C_L R_1 R_4 + 2C_L R_1 R_L g_m r_o + 2C_L R_1 R_L + C_L R_4 R_L + C_L R_4 r_o + 2C_L R_L r_o}$

wo: $\sqrt{\frac{2R_1 g_m r_o + 2R_1 + R_4 + 2r_o}{2C_4 C_L R_1 R_4 R_L g_m r_o + 2C_4 C_L R_1 R_4 R_L + 2C_4 C_L R_4 R_L r_o}}$

bandwidth: $\frac{\sqrt{\frac{2R_1 g_m r_o + 2R_1 + R_4 + 2r_o}{2C_4 C_L R_1 R_4 R_L g_m r_o + 2C_4 C_L R_1 R_4 R_L + 2C_4 C_L R_4 R_L r_o}}}{\sqrt{2} \sqrt{C_4} \sqrt{C_L} R_1 \sqrt{R_4} \sqrt{R_L} g_m r_o \sqrt{\frac{2R_1 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_1}{R_1 g_m r_o + R_1 + r_o} + \frac{R_4}{R_1 g_m r_o + R_1 + r_o} + \frac{2r_o}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{2} \sqrt{C_4} \sqrt{C_L} R_1 \sqrt{R_4} \sqrt{R_L} \sqrt{\frac{2R_1 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_1}{R_1 g_m r_o + R_1 + r_o} + \frac{R_4}{R_1 g_m r_o + R_1 + r_o} + \frac{2r_o}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{2} \sqrt{C_4} \sqrt{C_L} \sqrt{R_4} \sqrt{R_L} r_o \sqrt{\frac{2R_1 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_1}{R_1 g_m r_o + R_1 + r_o} + \frac{R_4}{R_1 g_m r_o + R_1 + r_o} + \frac{2r_o}{R_1 g_m r_o + R_1 + r_o}}}$

K-LP: $\frac{R_1 R_4 g_m r_o + R_1 R_4}{2R_1 g_m r_o + 2R_1 + R_4 + 2r_o}$

K-HP: 0

K-BP: $\frac{C_L R_1 R_4 R_L g_m r_o + C_L R_1 R_4 R_L}{2C_4 R_1 R_4 g_m r_o + 2C_4 R_1 R_4 + 2C_4 R_4 r_o + C_L R_1 R_4 g_m r_o + C_L R_1 R_4 + 2C_L R_1 R_L g_m r_o + 2C_L R_1 R_L + C_L R_4 R_L + C_L R_4 r_o + 2C_L R_L r_o}$

Qz: None

Wz: None

8.3 X-INVALID-NUMER-3 $Z(s) = \left(R_1, \infty, \infty, R_4 + \frac{1}{C_4 s}, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s(C_4 R_1 R_4 g_m r_o + C_4 R_1 R_4)}{s^2(C_4 C_L R_1 R_4 g_m r_o + C_4 C_L R_1 R_4 + C_4 C_L R_4 r_o) + s(2C_4 R_1 g_m r_o + 2C_4 R_1 + C_4 R_4 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_4} \sqrt{C_L} R_1 \sqrt{R_4} g_m r_o \sqrt{\frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_4} \sqrt{C_L} R_1 \sqrt{R_4} \sqrt{\frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_4} \sqrt{C_L} \sqrt{R_4} r_o \sqrt{\frac{1}{R_1 g_m r_o + R_1 + r_o}}}{2C_4 R_1 g_m r_o + 2C_4 R_1 + C_4 R_4 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o}$

wo: $\sqrt{\frac{1}{C_4 C_L R_1 R_4 g_m r_o + C_4 C_L R_1 R_4 + C_4 C_L R_4 r_o}}$

bandwidth: $\frac{(2C_4 R_1 g_m r_o + 2C_4 R_1 + C_4 R_4 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o) \sqrt{\frac{1}{C_4 C_L R_1 R_4 g_m r_o + C_4 C_L R_1 R_4 + C_4 C_L R_4 r_o}}}{\sqrt{C_4} \sqrt{C_L} R_1 \sqrt{R_4} g_m r_o \sqrt{\frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_4} \sqrt{C_L} R_1 \sqrt{R_4} \sqrt{\frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_4} \sqrt{C_L} \sqrt{R_4} r_o \sqrt{\frac{1}{R_1 g_m r_o + R_1 + r_o}}}$

K-LP: $R_1 g_m r_o + R_1$

K-HP: 0

K-BP: $\frac{C_4 R_1 R_4 g_m r_o + C_4 R_1 R_4}{2C_4 R_1 g_m r_o + 2C_4 R_1 + C_4 R_4 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o}$

Qz: None

Wz: None

8.4 X-INVALID-NUMER-4 $Z(s) = \left(R_1, \infty, \infty, R_4 + \frac{1}{C_4 s}, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_1 R_L g_m r_o + R_1 R_L + s(C_4 R_1 R_4 R_L g_m r_o + C_4 R_1 R_4 R_L)}{R_1 g_m r_o + R_1 + R_L + r_o + s^2(C_4 C_L R_1 R_4 R_L g_m r_o + C_4 C_L R_1 R_4 R_L + C_4 C_L R_4 R_L r_o) + s(C_4 R_1 R_4 g_m r_o + C_4 R_1 R_4 + 2C_4 R_1 R_L g_m r_o + 2C_4 R_1 R_L + C_4 R_4 R_L + C_4 R_4 r_o + 2C_4 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o)}$$

Parameters:

Q: $\frac{\sqrt{C_4} \sqrt{C_L} R_1 \sqrt{R_4} \sqrt{R_L} g_m r_o \sqrt{\frac{R_1 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_1}{R_1 g_m r_o + R_1 + r_o} + \frac{R_L}{R_1 g_m r_o + R_1 + r_o} + \frac{r_o}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_4} \sqrt{C_L} R_1 \sqrt{R_4} \sqrt{R_L} \sqrt{\frac{R_1 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_1}{R_1 g_m r_o + R_1 + r_o} + \frac{R_L}{R_1 g_m r_o + R_1 + r_o} + \frac{r_o}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_4} \sqrt{C_L} \sqrt{R_4} \sqrt{R_L} r_o \sqrt{\frac{R_1 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_1}{R_1 g_m r_o + R_1 + r_o} + \frac{R_L}{R_1 g_m r_o + R_1 + r_o} + \frac{r_o}{R_1 g_m r_o + R_1 + r_o}}}{C_4 R_1 R_4 g_m r_o + C_4 R_1 R_4 + 2C_4 R_1 R_L g_m r_o + 2C_4 R_1 R_L + C_4 R_4 R_L + C_4 R_4 r_o + 2C_4 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o}$

wo: $\sqrt{\frac{R_1 g_m r_o + R_1 + R_L + r_o}{C_4 C_L R_1 R_4 R_L g_m r_o + C_4 C_L R_1 R_4 R_L + C_4 C_L R_4 R_L r_o}}$

bandwidth: $\frac{\sqrt{\frac{R_1 g_m r_o + R_1 + R_L + r_o}{C_4 C_L R_1 R_4 R_L g_m r_o + C_4 C_L R_1 R_4 R_L + C_4 C_L R_4 R_L r_o}} (C_4 R_1 R_4 g_m r_o + C_4 R_1 R_4 + 2C_4 R_1 R_L g_m r_o + 2C_4 R_1 R_L + C_4 R_4 R_L + C_4 R_4 r_o + 2C_4 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o)}{\sqrt{C_4} \sqrt{C_L} R_1 \sqrt{R_4} \sqrt{R_L} g_m r_o \sqrt{\frac{R_1 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_1}{R_1 g_m r_o + R_1 + r_o} + \frac{R_L}{R_1 g_m r_o + R_1 + r_o} + \frac{r_o}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_4} \sqrt{C_L} R_1 \sqrt{R_4} \sqrt{R_L} \sqrt{\frac{R_1 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_1}{R_1 g_m r_o + R_1 + r_o} + \frac{R_L}{R_1 g_m r_o + R_1 + r_o} + \frac{r_o}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_4} \sqrt{C_L} \sqrt{R_4} \sqrt{R_L} r_o \sqrt{\frac{R_1 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_1}{R_1 g_m r_o + R_1 + r_o} + \frac{R_L}{R_1 g_m r_o + R_1 + r_o} + \frac{r_o}{R_1 g_m r_o + R_1 + r_o}}}$

K-LP: $\frac{R_1 R_L g_m r_o + R_1 R_L}{R_1 g_m r_o + R_1 + R_L + r_o}$

K-HP: 0

K-BP: $\frac{C_4 R_1 R_4 R_L g_m r_o + C_4 R_1 R_4 R_L}{C_4 R_1 R_4 g_m r_o + C_4 R_1 R_4 + 2C_4 R_1 R_L g_m r_o + 2C_4 R_1 R_L + C_4 R_4 R_L + C_4 R_4 r_o + 2C_4 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o}$

Qz: None

Wz: None

8.5 X-INVALID-NUMER-5 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, R_4, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_4 g_m r_o + R_4 + s(C_L R_4 R_L g_m r_o + C_L R_4 R_L)}{2g_m r_o + s^2(C_1 C_L R_4 R_L + C_1 C_L R_4 r_o + 2C_1 C_L R_L r_o) + s(C_1 R_4 + 2C_1 r_o + C_L R_4 g_m r_o + C_L R_4 + 2C_L R_L g_m r_o + 2C_L R_L) + 2}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_1} \sqrt{C_L} R_4 R_L \sqrt{\frac{g_m r_o}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{1}{R_4 R_L + R_4 r_o + 2R_L r_o}} + \sqrt{2} \sqrt{C_1} \sqrt{C_L} R_4 r_o \sqrt{\frac{g_m r_o}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{1}{R_4 R_L + R_4 r_o + 2R_L r_o}} + 2\sqrt{2} \sqrt{C_1} \sqrt{C_L} R_L r_o \sqrt{\frac{g_m r_o}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{1}{R_4 R_L + R_4 r_o + 2R_L r_o}}}{C_1 R_4 + 2C_1 r_o + C_L R_4 g_m r_o + C_L R_4 + 2C_L R_L g_m r_o + 2C_L R_L}$

wo: $\sqrt{\frac{2g_m r_o + 2}{C_1 C_L R_4 R_L + C_1 C_L R_4 r_o + 2C_1 C_L R_L r_o}}$

bandwidth: $\frac{\sqrt{\frac{2g_m r_o + 2}{C_1 C_L R_4 R_L + C_1 C_L R_4 r_o + 2C_1 C_L R_L r_o}} (C_1 R_4 + 2C_1 r_o + C_L R_4 g_m r_o + C_L R_4 + 2C_L R_L g_m r_o + 2C_L R_L)}{\sqrt{2} \sqrt{C_1} \sqrt{C_L} R_4 R_L \sqrt{\frac{g_m r_o}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{1}{R_4 R_L + R_4 r_o + 2R_L r_o}} + \sqrt{2} \sqrt{C_1} \sqrt{C_L} R_4 r_o \sqrt{\frac{g_m r_o}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{1}{R_4 R_L + R_4 r_o + 2R_L r_o}} + 2\sqrt{2} \sqrt{C_1} \sqrt{C_L} R_L r_o \sqrt{\frac{g_m r_o}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{1}{R_4 R_L + R_4 r_o + 2R_L r_o}}}$

K-LP: $\frac{R_4}{2}$

K-HP: 0

K-BP: $\frac{C_L R_4 R_L g_m r_o + C_L R_4 R_L}{C_1 R_4 + 2C_1 r_o + C_L R_4 g_m r_o + C_L R_4 + 2C_L R_L g_m r_o + 2C_L R_L}$

Qz: None

Wz: None

8.6 X-INVALID-NUMER-6 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \infty, R_L \right)$

$$H(s) = \frac{R_L g_m r_o + R_L + s(C_4 R_4 R_L g_m r_o + C_4 R_4 R_L)}{g_m r_o + s^2(C_1 C_4 R_4 R_L + C_1 C_4 R_4 r_o + 2C_1 C_4 R_L r_o) + s(C_1 R_L + C_1 r_o + C_4 R_4 g_m r_o + C_4 R_4 + 2C_4 R_L g_m r_o + 2C_4 R_L) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_4} R_4 R_L \sqrt{\frac{g_m r_o}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{1}{R_4 R_L + R_4 r_o + 2R_L r_o}} + \sqrt{C_1} \sqrt{C_4} R_4 r_o \sqrt{\frac{g_m r_o}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{1}{R_4 R_L + R_4 r_o + 2R_L r_o}} + 2\sqrt{C_1} \sqrt{C_4} R_L r_o \sqrt{\frac{g_m r_o}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{1}{R_4 R_L + R_4 r_o + 2R_L r_o}}}{C_1 R_L + C_1 r_o + C_4 R_4 g_m r_o + C_4 R_4 + 2C_4 R_L g_m r_o + 2C_4 R_L}$

wo: $\sqrt{\frac{g_m r_o + 1}{C_1 C_4 R_4 R_L + C_1 C_4 R_4 r_o + 2C_1 C_4 R_L r_o}}$

bandwidth: $\frac{\sqrt{\frac{C_1 C_4 R_4 R_L + C_1 C_4 R_4 r_o + 2C_1 C_4 R_L r_o}{C_1 R_L + C_1 r_o + C_4 R_4 g_m r_o + C_4 R_4 + 2C_4 R_L g_m r_o + 2C_4 R_L}}}{\sqrt{C_1} \sqrt{C_4} R_4 R_L \sqrt{\frac{g_m r_o}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{1}{R_4 R_L + R_4 r_o + 2R_L r_o}} + \sqrt{C_1} \sqrt{C_4} R_4 r_o \sqrt{\frac{g_m r_o}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{1}{R_4 R_L + R_4 r_o + 2R_L r_o}} + 2\sqrt{C_1} \sqrt{C_4} R_L r_o \sqrt{\frac{g_m r_o}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{1}{R_4 R_L + R_4 r_o + 2R_L r_o}}}$

K-LP: R_L

K-HP: 0

K-BP: $\frac{C_4 R_4 R_L g_m r_o + C_4 R_4 R_L}{C_1 R_L + C_1 r_o + C_4 R_4 g_m r_o + C_4 R_4 + 2C_4 R_L g_m r_o + 2C_4 R_L}$

Qz: None

Wz: None

8.7 X-INVALID-NUMER-7 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 R_4 g_m r_o + R_1 R_4 + s(C_L R_1 R_4 R_L g_m r_o + C_L R_1 R_4 R_L)}{2R_1 g_m r_o + 2R_1 + R_4 + 2r_o + s^2(C_1 C_L R_1 R_4 R_L + C_1 C_L R_1 R_4 r_o + 2C_1 C_L R_1 R_L r_o) + s(C_1 R_1 R_4 + 2C_1 R_1 r_o + C_L R_1 R_4 g_m r_o + C_L R_1 R_4 + 2C_L R_1 R_L g_m r_o + 2C_L R_1 R_L + C_L R_4 R_L + C_L R_4 r_o + 2C_L R_L r_o)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} R_4 R_L \sqrt{\frac{2R_1 g_m r_o}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{2R_1}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{R_4}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{2r_o}{R_4 R_L + R_4 r_o + 2R_L r_o}} + \sqrt{C_1} \sqrt{C_L} \sqrt{R_1} R_4 r_o \sqrt{\frac{2R_1 g_m r_o}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{2R_1}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{R_4}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{2r_o}{R_4 R_L + R_4 r_o + 2R_L r_o}} + 2\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} R_L r_o \sqrt{\frac{2R_1 g_m r_o}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{2R_1}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{R_4}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{2r_o}{R_4 R_L + R_4 r_o + 2R_L r_o}}}{C_1 R_1 R_4 + 2C_1 R_1 r_o + C_L R_1 R_4 g_m r_o + C_L R_1 R_4 + 2C_L R_1 R_L g_m r_o + 2C_L R_1 R_L + C_L R_4 R_L + C_L R_4 r_o + 2C_L R_L r_o}$

wo: $\sqrt{\frac{2R_1 g_m r_o + 2R_1 + R_4 + 2r_o}{C_1 C_L R_1 R_4 R_L + C_1 C_L R_1 R_4 r_o + 2C_1 C_L R_1 R_L r_o}}$

bandwidth: $\frac{\sqrt{\frac{2R_1 g_m r_o + 2R_1 + R_4 + 2r_o}{C_1 C_L R_1 R_4 R_L + C_1 C_L R_1 R_4 r_o + 2C_1 C_L R_1 R_L r_o}}}{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} R_4 R_L \sqrt{\frac{2R_1 g_m r_o}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{2R_1}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{R_4}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{2r_o}{R_4 R_L + R_4 r_o + 2R_L r_o}} + \sqrt{C_1} \sqrt{C_L} \sqrt{R_1} R_4 r_o \sqrt{\frac{2R_1 g_m r_o}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{2R_1}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{R_4}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{2r_o}{R_4 R_L + R_4 r_o + 2R_L r_o}} + 2\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} R_L r_o \sqrt{\frac{2R_1 g_m r_o}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{2R_1}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{R_4}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{2r_o}{R_4 R_L + R_4 r_o + 2R_L r_o}}}$

K-LP: $\frac{R_1 R_4 g_m r_o + R_1 R_4}{2R_1 g_m r_o + 2R_1 + R_4 + 2r_o}$

K-HP: 0

K-BP: $\frac{C_L R_1 R_4 R_L g_m r_o + C_L R_1 R_4 R_L}{C_1 R_1 R_4 + 2C_1 R_1 r_o + C_L R_1 R_4 g_m r_o + C_L R_1 R_4 + 2C_L R_1 R_L g_m r_o + 2C_L R_1 R_L + C_L R_4 R_L + C_L R_4 r_o + 2C_L R_L r_o}$

Qz: None

Wz: None

8.8 X-INVALID-NUMER-8 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \infty, R_L \right)$

$$H(s) = \frac{R_1 R_L g_m r_o + R_1 R_L + s(C_4 R_1 R_4 R_L g_m r_o + C_4 R_1 R_4 R_L)}{R_1 g_m r_o + R_1 + R_L + r_o + s^2(C_1 C_4 R_1 R_4 R_L + C_1 C_4 R_1 R_4 r_o + 2C_1 C_4 R_1 R_L r_o) + s(C_1 R_1 R_L + C_1 R_1 r_o + C_4 R_1 R_4 g_m r_o + C_4 R_1 R_4 + 2C_4 R_1 R_L g_m r_o + 2C_4 R_1 R_L + C_4 R_4 R_L + C_4 R_4 r_o + 2C_4 R_L r_o)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_4} \sqrt{R_1} R_4 R_L \sqrt{\frac{R_1 g_m r_o}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{R_1}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{R_L}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{r_o}{R_4 R_L + R_4 r_o + 2R_L r_o}} + \sqrt{C_1} \sqrt{C_4} \sqrt{R_1} R_4 r_o \sqrt{\frac{R_1 g_m r_o}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{R_1}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{R_L}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{r_o}{R_4 R_L + R_4 r_o + 2R_L r_o}} + 2\sqrt{C_1} \sqrt{C_4} \sqrt{R_1} R_L r_o \sqrt{\frac{R_1 g_m r_o}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{R_1}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{R_L}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{r_o}{R_4 R_L + R_4 r_o + 2R_L r_o}}}{C_1 R_1 R_L + C_1 R_1 r_o + C_4 R_1 R_4 g_m r_o + C_4 R_1 R_4 + 2C_4 R_1 R_L g_m r_o + 2C_4 R_1 R_L + C_4 R_4 R_L + C_4 R_4 r_o + 2C_4 R_L r_o}$

wo: $\sqrt{\frac{R_1 g_m r_o + R_1 + R_L + r_o}{C_1 C_4 R_1 R_4 R_L + C_1 C_4 R_1 R_4 r_o + 2C_1 C_4 R_1 R_L r_o}}$

bandwidth: $\frac{\sqrt{\frac{R_1 g_m r_o + R_1 + R_L + r_o}{C_1 C_4 R_1 R_4 R_L + C_1 C_4 R_1 R_4 r_o + 2C_1 C_4 R_1 R_L r_o}}}{\sqrt{C_1} \sqrt{C_4} \sqrt{R_1} R_4 R_L \sqrt{\frac{R_1 g_m r_o}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{R_1}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{R_L}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{r_o}{R_4 R_L + R_4 r_o + 2R_L r_o}} + \sqrt{C_1} \sqrt{C_4} \sqrt{R_1} R_4 r_o \sqrt{\frac{R_1 g_m r_o}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{R_1}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{R_L}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{r_o}{R_4 R_L + R_4 r_o + 2R_L r_o}} + 2\sqrt{C_1} \sqrt{C_4} \sqrt{R_1} R_L r_o \sqrt{\frac{R_1 g_m r_o}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{R_1}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{R_L}{R_4 R_L + R_4 r_o + 2R_L r_o} + \frac{r_o}{R_4 R_L + R_4 r_o + 2R_L r_o}}}$

K-LP: $\frac{R_1 R_L g_m r_o + R_1 R_L}{R_1 g_m r_o + R_1 + R_L + r_o}$

K-HP: 0

K-BP: $\frac{C_4 R_1 R_4 R_L g_m r_o + C_4 R_1 R_4 R_L}{C_1 R_1 R_L + C_1 R_1 r_o + C_4 R_1 R_4 g_m r_o + C_4 R_1 R_4 + 2C_4 R_1 R_L g_m r_o + 2C_4 R_1 R_L + C_4 R_4 R_L + C_4 R_4 r_o + 2C_4 R_L r_o}$

Qz: None

Wz: None

8.9 X-INVALID-NUMER-9 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, R_4, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_4 g_m r_o + R_4 + s(C_1 R_1 R_4 g_m r_o + C_1 R_1 R_4)}{2g_m r_o + s^2(C_1 C_L R_1 R_4 g_m r_o + C_1 C_L R_1 R_4 + C_1 C_L R_4 r_o) + s(2C_1 R_1 g_m r_o + 2C_1 R_1 + C_1 R_4 + 2C_1 r_o + C_L R_4 g_m r_o + C_L R_4) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_L}R_1\sqrt{R_4}g_m r_o\sqrt{\frac{g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{2}\sqrt{C_1}\sqrt{C_L}R_1\sqrt{R_4}\sqrt{\frac{g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{2}\sqrt{C_1}\sqrt{C_L}\sqrt{R_4}r_o\sqrt{\frac{g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{1}{R_1 g_m r_o + R_1 + r_o}}}{2C_1 R_1 g_m r_o + 2C_1 R_1 + C_1 R_4 + 2C_1 r_o + C_L R_4 g_m r_o + C_L R_4}$

wo: $\sqrt{\frac{2g_m r_o + 2}{C_1 C_L R_1 R_4 g_m r_o + C_1 C_L R_1 R_4 + C_1 C_L R_4 r_o}}$

bandwidth: $\frac{\sqrt{\frac{2g_m r_o + 2}{C_1 C_L R_1 R_4 g_m r_o + C_1 C_L R_1 R_4 + C_1 C_L R_4 r_o}}(2C_1 R_1 g_m r_o + 2C_1 R_1 + C_1 R_4 + 2C_1 r_o + C_L R_4 g_m r_o + C_L R_4)}{\sqrt{2}\sqrt{C_1}\sqrt{C_L}R_1\sqrt{R_4}g_m r_o\sqrt{\frac{g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{2}\sqrt{C_1}\sqrt{C_L}R_1\sqrt{R_4}\sqrt{\frac{g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{2}\sqrt{C_1}\sqrt{C_L}\sqrt{R_4}r_o\sqrt{\frac{g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{1}{R_1 g_m r_o + R_1 + r_o}}}$

K-LP: $\frac{R_4}{2}$

K-HP: 0

K-BP: $\frac{C_1 R_1 R_4 g_m r_o + C_1 R_1 R_4}{2C_1 R_1 g_m r_o + 2C_1 R_1 + C_1 R_4 + 2C_1 r_o + C_L R_4 g_m r_o + C_L R_4}$

Qz: None

Wz: None

8.10 X-INVALID-NUMER-10 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, R_4, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_4 R_L g_m r_o + R_4 R_L + s(C_1 R_1 R_4 R_L g_m r_o + C_1 R_1 R_4 R_L)}{R_4 g_m r_o + R_4 + 2R_L g_m r_o + 2R_L + s^2(C_1 C_L R_1 R_4 R_L g_m r_o + C_1 C_L R_1 R_4 R_L + C_1 C_L R_4 R_L r_o) + s(C_1 R_1 R_4 g_m r_o + C_1 R_1 R_4 + 2C_1 R_1 R_L g_m r_o + 2C_1 R_1 R_L + C_1 R_4 R_L + C_1 R_4 r_o + 2C_1 R_L r_o + C_L R_4 R_L g_m r_o + C_L R_4 R_L)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_L}R_1\sqrt{R_4}\sqrt{R_L}g_m r_o\sqrt{\frac{R_4 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_4}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_L g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_L}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_1}\sqrt{C_L}R_1\sqrt{R_4}\sqrt{R_L}\sqrt{\frac{R_4 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_4}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_L g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_L}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_1}\sqrt{C_L}\sqrt{R_4}\sqrt{R_L}r_o\sqrt{\frac{R_4 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_4}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_L g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_L}{R_1 g_m r_o + R_1 + r_o}}}{C_1 R_1 R_4 g_m r_o + C_1 R_1 R_4 + 2C_1 R_1 R_L g_m r_o + 2C_1 R_1 R_L + C_1 R_4 R_L + C_1 R_4 r_o + 2C_1 R_L r_o + C_L R_4 R_L g_m r_o + C_L R_4 R_L}$

wo: $\sqrt{\frac{R_4 g_m r_o + R_4 + 2R_L g_m r_o + 2R_L}{C_1 C_L R_1 R_4 R_L g_m r_o + C_1 C_L R_1 R_4 R_L + C_1 C_L R_4 R_L r_o}}$

bandwidth: $\frac{\sqrt{\frac{R_4 g_m r_o + R_4 + 2R_L g_m r_o + 2R_L}{C_1 C_L R_1 R_4 R_L g_m r_o + C_1 C_L R_1 R_4 R_L + C_1 C_L R_4 R_L r_o}}(C_1 R_1 R_4 g_m r_o + C_1 R_1 R_4 + 2C_1 R_1 R_L g_m r_o + 2C_1 R_1 R_L + C_1 R_4 R_L + C_1 R_4 r_o + 2C_1 R_L r_o + C_L R_4 R_L g_m r_o + C_L R_4 R_L)}{\sqrt{C_1}\sqrt{C_L}R_1\sqrt{R_4}\sqrt{R_L}g_m r_o\sqrt{\frac{R_4 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_4}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_L g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_L}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_1}\sqrt{C_L}R_1\sqrt{R_4}\sqrt{R_L}\sqrt{\frac{R_4 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_4}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_L g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_L}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_1}\sqrt{C_L}\sqrt{R_4}\sqrt{R_L}r_o\sqrt{\frac{R_4 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_4}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_L g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_L}{R_1 g_m r_o + R_1 + r_o}}}$

K-LP: $\frac{R_4 R_L}{R_4 + 2R_L}$

K-HP: 0

K-BP: $\frac{C_1 R_1 R_4 R_L g_m r_o + C_1 R_1 R_4 R_L}{C_1 R_1 R_4 g_m r_o + C_1 R_1 R_4 + 2C_1 R_1 R_L g_m r_o + 2C_1 R_1 R_L + C_1 R_4 R_L + C_1 R_4 r_o + 2C_1 R_L r_o + C_L R_4 R_L g_m r_o + C_L R_4 R_L}$

Qz: None

Wz: None

8.11 X-INVALID-NUMER-11 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{1}{C_4 s}, \infty, R_L \right)$

$$H(s) = \frac{R_L g_m r_o + R_L + s(C_1 R_1 R_L g_m r_o + C_1 R_1 R_L)}{g_m r_o + s^2(2C_1 C_4 R_1 R_L g_m r_o + 2C_1 C_4 R_1 R_L + 2C_1 C_4 R_L r_o) + s(C_1 R_1 g_m r_o + C_1 R_1 + C_1 R_L + C_1 r_o + 2C_4 R_L g_m r_o + 2C_4 R_L) + 1}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_L}g_m r_o\sqrt{\frac{g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_L}\sqrt{\frac{g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_L}r_o\sqrt{\frac{g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{1}{R_1 g_m r_o + R_1 + r_o}}}{C_1 R_1 g_m r_o + C_1 R_1 + C_1 R_L + C_1 r_o + 2C_4 R_L g_m r_o + 2C_4 R_L}$

wo: $\sqrt{\frac{g_m r_o + 1}{2C_1 C_4 R_1 R_L g_m r_o + 2C_1 C_4 R_1 R_L + 2C_1 C_4 R_L r_o}}$

bandwidth: $\frac{\sqrt{\frac{g_m r_o + 1}{2C_1 C_4 R_1 R_L g_m r_o + 2C_1 C_4 R_1 R_L + 2C_1 C_4 R_L r_o}}(C_1 R_1 g_m r_o + C_1 R_1 + C_1 R_L + C_1 r_o + 2C_4 R_L g_m r_o + 2C_4 R_L)}{\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_L}g_m r_o\sqrt{\frac{g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_L}\sqrt{\frac{g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_L}r_o\sqrt{\frac{g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{1}{R_1 g_m r_o + R_1 + r_o}}}$

K-LP: R_L

K-HP: 0

K-BP: $\frac{C_1 R_1 R_L g_m r_o + C_1 R_1 R_L}{C_1 R_1 g_m r_o + C_1 R_1 + C_1 R_L + C_1 r_o + 2C_4 R_L g_m r_o + 2C_4 R_L}$

Qz: None

Wz: None

8.12 X-INVALID-NUMER-12 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{1}{C_4 s}, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_L g_m r_o + R_L + s(C_1 R_1 R_L g_m r_o + C_1 R_1 R_L)}{g_m r_o + s^2(2C_1 C_4 R_1 R_L g_m r_o + 2C_1 C_4 R_1 R_L + 2C_1 C_4 R_L r_o + C_1 C_L R_1 R_L g_m r_o + C_1 C_L R_1 R_L + C_1 C_L R_L r_o) + s(C_1 R_1 g_m r_o + C_1 R_1 + C_1 R_L + C_1 r_o + 2C_4 R_L g_m r_o + 2C_4 R_L + C_L R_L g_m r_o + C_L R_L) + 1}$$

Parameters:

Q: $\frac{2\sqrt{C_1} C_4 R_1 \sqrt{R_L} g_m r_o \sqrt{\frac{g_m r_o}{2C_4 R_1 g_m r_o + 2C_4 R_1 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o} + \frac{1}{2C_4 R_1 g_m r_o + 2C_4 R_1 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o}} + 2\sqrt{C_1} C_4 R_1 \sqrt{R_L} \sqrt{\frac{g_m r_o}{2C_4 R_1 g_m r_o + 2C_4 R_1 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o} + \frac{1}{2C_4 R_1 g_m r_o + 2C_4 R_1 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o}} + 2\sqrt{C_1} C_4 \sqrt{R_L} r_o \sqrt{\frac{g_m r_o}{2C_4 R_1 g_m r_o + 2C_4 R_1 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o} + \frac{1}{2C_4 R_1 g_m r_o + 2C_4 R_1 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o}}}{\sqrt{2C_1 C_4 R_1 R_L} \sqrt{\frac{g_m r_o + 1}{2C_1 C_4 R_1 R_L g_m r_o + 2C_1 C_4 R_1 R_L + 2C_1 C_4 R_L r_o + C_1 C_L R_1 R_L g_m r_o + C_1 C_L R_1 R_L + C_1 C_L R_L r_o}}}$

wo: $\frac{g_m r_o + 1}{2C_1 C_4 R_1 R_L g_m r_o + 2C_1 C_4 R_1 R_L + 2C_1 C_4 R_L r_o + C_1 C_L R_1 R_L g_m r_o + C_1 C_L R_1 R_L + C_1 C_L R_L r_o}$

bandwidth: $\frac{\sqrt{2C_1} C_4 R_1 \sqrt{R_L} g_m r_o \sqrt{\frac{g_m r_o}{2C_4 R_1 g_m r_o + 2C_4 R_1 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o} + \frac{1}{2C_4 R_1 g_m r_o + 2C_4 R_1 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o}} + 2\sqrt{C_1} C_4 R_1 \sqrt{R_L} \sqrt{\frac{g_m r_o}{2C_4 R_1 g_m r_o + 2C_4 R_1 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o} + \frac{1}{2C_4 R_1 g_m r_o + 2C_4 R_1 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o}} + 2\sqrt{C_1} C_4 \sqrt{R_L} r_o \sqrt{\frac{g_m r_o}{2C_4 R_1 g_m r_o + 2C_4 R_1 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o} + \frac{1}{2C_4 R_1 g_m r_o + 2C_4 R_1 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o}}}{\sqrt{2C_1 C_4 R_1 R_L} \sqrt{\frac{g_m r_o}{2C_1 C_4 R_1 R_L g_m r_o + 2C_1 C_4 R_1 R_L + 2C_1 C_4 R_L r_o + C_1 C_L R_1 R_L g_m r_o + C_1 C_L R_1 R_L + C_1 C_L R_L r_o}}}$

K-LP: R_L
K-HP: 0
K-BP: $\frac{C_1 R_1 R_L g_m r_o + C_1 R_1 R_L}{C_1 R_1 g_m r_o + C_1 R_1 + C_1 R_L + C_1 r_o + 2C_4 R_L g_m r_o + 2C_4 R_L + C_L R_L g_m r_o + C_L R_L}$
Qz: None
Wz: None

8.13 X-INVALID-NUMER-13 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, R_L \right)$

$$H(s) = \frac{R_4 R_L g_m r_o + R_4 R_L + s(C_1 R_1 R_4 R_L g_m r_o + C_1 R_1 R_4 R_L)}{R_4 g_m r_o + R_4 + 2R_L g_m r_o + 2R_L + s^2(2C_1 C_4 R_1 R_4 R_L g_m r_o + 2C_1 C_4 R_1 R_4 R_L + 2C_1 C_4 R_4 R_L r_o) + s(C_1 R_1 R_4 g_m r_o + C_1 R_1 R_4 + 2C_1 R_1 R_L g_m r_o + 2C_1 R_1 R_L + C_1 R_4 R_L + C_1 R_4 r_o + 2C_1 R_L r_o + 2C_4 R_4 R_L g_m r_o + 2C_4 R_4 R_L)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_4}\sqrt{R_L}g_m r_o\sqrt{\frac{R_4 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_4}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_L g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_L}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_4}\sqrt{R_L}\sqrt{\frac{R_4 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_4}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_L g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_L}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_4}\sqrt{R_L}r_o\sqrt{\frac{R_4 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_4}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_L g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_L}{R_1 g_m r_o + R_1 + r_o}}}{C_1 R_1 R_4 g_m r_o + C_1 R_1 R_4 + 2C_1 R_1 R_L g_m r_o + 2C_1 R_1 R_L + C_1 R_4 R_L + C_1 R_4 r_o + 2C_1 R_L r_o + 2C_4 R_4 R_L g_m r_o + 2C_4 R_4 R_L}$

wo: $\frac{R_4 g_m r_o + R_4 + 2R_L g_m r_o + 2R_L}{2C_1 C_4 R_1 R_4 R_L g_m r_o + 2C_1 C_4 R_1 R_4 R_L + 2C_1 C_4 R_4 R_L r_o}$

bandwidth: $\frac{\sqrt{\frac{R_4 g_m r_o + R_4 + 2R_L g_m r_o + 2R_L}{2C_1 C_4 R_1 R_4 R_L g_m r_o + 2C_1 C_4 R_1 R_4 R_L + 2C_1 C_4 R_4 R_L r_o}} (C_1 R_1 R_4 g_m r_o + C_1 R_1 R_4 + 2C_1 R_1 R_L g_m r_o + 2C_1 R_1 R_L + C_1 R_4 R_L + C_1 R_4 r_o + 2C_1 R_L r_o + 2C_4 R_4 R_L g_m r_o + 2C_4 R_4 R_L)}{\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_4}\sqrt{R_L}g_m r_o\sqrt{\frac{R_4 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_4}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_L g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_L}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_4}\sqrt{R_L}\sqrt{\frac{R_4 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_4}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_L g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_L}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_4}\sqrt{R_L}r_o\sqrt{\frac{R_4 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_4}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_L g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{2R_L}{R_1 g_m r_o + R_1 + r_o}}}$

K-LP: $\frac{R_4 R_L}{R_4 + 2R_L}$
K-HP: 0
K-BP: $\frac{C_1 R_1 R_4 R_L g_m r_o + C_1 R_1 R_4 R_L}{C_1 R_1 R_4 g_m r_o + C_1 R_1 R_4 + 2C_1 R_1 R_L g_m r_o + 2C_1 R_1 R_L + C_1 R_4 R_L + C_1 R_4 r_o + 2C_1 R_L r_o + 2C_4 R_4 R_L g_m r_o + 2C_4 R_4 R_L}$
Qz: None
Wz: None

8.14 X-INVALID-NUMER-14 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_4 g_m r_o + R_4 + s(C_1 R_1 R_4 g_m r_o + C_1 R_1 R_4)}{2g_m r_o + s^2(2C_1 C_4 R_1 R_4 g_m r_o + 2C_1 C_4 R_1 R_4 + 2C_1 C_4 R_4 r_o + C_1 C_L R_1 R_4 g_m r_o + C_1 C_L R_1 R_4 + C_1 C_L R_4 r_o) + s(2C_1 R_1 g_m r_o + 2C_1 R_1 + C_1 R_4 + 2C_1 r_o + 2C_4 R_4 g_m r_o + 2C_4 R_4 + C_L R_4 g_m r_o + C_L R_4) + 2}$$

Parameters:

Q: $\frac{2\sqrt{2}\sqrt{C_1}C_4R_1\sqrt{R_4}g_m r_o\sqrt{\frac{g_m r_o}{2C_4R_1g_m r_o+2C_4R_1+2C_4r_o+C_LR_1g_m r_o+C_LR_1+C_Lr_o}+\frac{1}{2C_4R_1g_m r_o+2C_4R_1+2C_4r_o+C_LR_1g_m r_o+C_LR_1+C_Lr_o}}+2\sqrt{2}\sqrt{C_1}C_4R_1\sqrt{R_4}\sqrt{\frac{g_m r_o}{2C_4R_1g_m r_o+2C_4R_1+2C_4r_o+C_LR_1g_m r_o+C_LR_1+C_Lr_o}+\frac{1}{2C_4R_1g_m r_o+2C_4R_1+2C_4r_o+C_LR_1g_m r_o+C_LR_1+C_Lr_o}}+2\sqrt{2}\sqrt{C_1}C_4\sqrt{R_4}r_o\sqrt{\frac{g_m r_o}{2C_4R_1g_m r_o+2C_4R_1+2C_4r_o+C_LR_1g_m r_o+C_LR_1+C_Lr_o}+\frac{1}{2C_4R_1g_m r_o+2C_4R_1+2C_4r_o+C_LR_1g_m r_o+C_LR_1+C_Lr_o}}}{\sqrt{2}\sqrt{C_1}C_4R_1R_4g_m r_o+2C_1C_4R_1R_4+2C_1C_4R_4r_o+C_1C_LR_1R_4g_m r_o+C_1C_LR_1R_4+C_1C_LR_4r_o}}$

wo: $\frac{2g_m r_o + 2}{2C_1 C_4 R_1 R_4 g_m r_o + 2C_1 C_4 R_1 R_4 + 2C_1 C_4 R_4 r_o + C_1 C_L R_1 R_4 g_m r_o + C_1 C_L R_1 R_4 + C_1 C_L R_4 r_o}$

bandwidth: $\frac{\sqrt{2}\sqrt{C_1}C_4R_1\sqrt{R_4}g_m r_o\sqrt{\frac{g_m r_o}{2C_4R_1g_m r_o+2C_4R_1+2C_4r_o+C_LR_1g_m r_o+C_LR_1+C_Lr_o}+\frac{1}{2C_4R_1g_m r_o+2C_4R_1+2C_4r_o+C_LR_1g_m r_o+C_LR_1+C_Lr_o}}+2\sqrt{2}\sqrt{C_1}C_4R_1\sqrt{R_4}\sqrt{\frac{g_m r_o}{2C_4R_1g_m r_o+2C_4R_1+2C_4r_o+C_LR_1g_m r_o+C_LR_1+C_Lr_o}+\frac{1}{2C_4R_1g_m r_o+2C_4R_1+2C_4r_o+C_LR_1g_m r_o+C_LR_1+C_Lr_o}}+2\sqrt{2}\sqrt{C_1}C_4\sqrt{R_4}r_o\sqrt{\frac{g_m r_o}{2C_4R_1g_m r_o+2C_4R_1+2C_4r_o+C_LR_1g_m r_o+C_LR_1+C_Lr_o}+\frac{1}{2C_4R_1g_m r_o+2C_4R_1+2C_4r_o+C_LR_1g_m r_o+C_LR_1+C_Lr_o}}}{\sqrt{2}\sqrt{C_1}C_4R_1R_4g_m r_o+2C_1C_4R_1R_4+2C_1C_4R_4r_o+C_1C_LR_1R_4g_m r_o+C_1C_LR_1R_4+C_1C_LR_4r_o}}$

K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $\frac{C_1 R_1 R_4 g_m r_o + C_1 R_1 R_4}{2C_1 R_1 g_m r_o + 2C_1 R_1 + C_1 R_4 + 2C_1 r_o + 2C_4 R_4 g_m r_o + 2C_4 R_4 + C_L R_4 g_m r_o + C_L R_4}$
Qz: None
Wz: None

8.15 X-INVALID-NUMER-15 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_4 R_L g_m r_o + R_4 R_L + s (C_1 R_1 R_4 R_L g_m r_o + C_1 R_1 R_4 R_L)}{R_4 g_m r_o + R_4 + 2 R_L g_m r_o + 2 R_L + s^2 (2 C_1 C_4 R_1 R_4 R_L g_m r_o + 2 C_1 C_4 R_1 R_4 R_L + 2 C_1 C_4 R_4 R_L r_o + C_1 C_L R_1 R_4 R_L g_m r_o + C_1 C_L R_1 R_4 R_L + C_1 C_L R_4 R_L r_o) + s (C_1 R_1 R_4 g_m r_o + C_1 R_1 R_4 + 2 C_1 R_1 R_L g_m r_o + 2 C_1 R_1 R_L + C_1 R_4 R_L + C_1 R_4 r_o + 2 C_1 R_L r_o + 2 C_4 R_4 R_L g_m r_o)}$$

Parameters:

Q: $\frac{2\sqrt{C_1}C_4R_1\sqrt{R_4}\sqrt{R_L}g_m r_o\sqrt{\frac{R_4g_m r_o}{2C_4R_1g_m r_o+2C_4R_1+2C_4r_o+C_LR_1g_m r_o+C_LR_1+C_Lr_o}+\frac{R_4}{2C_4R_1g_m r_o+2C_4R_1+2C_4r_o+C_LR_1g_m r_o+C_LR_1+C_Lr_o}+\frac{2R_Lg_m r_o}{2C_4R_1g_m r_o+2C_4R_1+2C_4r_o+C_LR_1g_m r_o+C_LR_1+C_Lr_o}+\frac{2R_L}{2C_4R_1g_m r_o+2C_4R_1+2C_4r_o+C_LR_1g_m r_o+C_LR_1+C_Lr_o}}+2\sqrt{C_1}C_4R_1\sqrt{R_4}\sqrt{R_L}\sqrt{\frac{R_4g_m r_o}{2C_4R_1g_m r_o+2C_4R_1+2C_4r_o+C_LR_1g_m r_o+C_LR_1+C_Lr_o}+\frac{R_4}{2C_4R_1g_m r_o+2C_4R_1+2C_4r_o+C_LR_1g_m r_o+C_LR_1+C_Lr_o}+\frac{2R_Lg_m r_o}{2C_4R_1g_m r_o+2C_4R_1+2C_4r_o+C_LR_1g_m r_o+C_LR_1+C_Lr_o}+\frac{2R_L}{2C_4R_1g_m r_o+2C_4R_1+2C_4r_o+C_LR_1g_m r_o+C_LR_1+C_Lr_o}}}$

wo: $\sqrt{\frac{R_4g_m r_o+R_4+2R_Lg_m r_o+2R_L}{2C_1C_4R_1R_4R_Lg_m r_o+2C_1C_4R_1R_4R_L+2C_1C_4R_4R_Lr_o+C_1C_LR_1R_4R_Lg_m r_o+C_1C_LR_1R_4R_L+C_1C_LR_4R_Lr_o}}$

bandwidth: $\frac{2\sqrt{C_1}C_4R_1\sqrt{R_4}\sqrt{R_L}g_m r_o\sqrt{\frac{R_4g_m r_o}{2C_4R_1g_m r_o+2C_4R_1+2C_4r_o+C_LR_1g_m r_o+C_LR_1+C_Lr_o}+\frac{R_4}{2C_4R_1g_m r_o+2C_4R_1+2C_4r_o+C_LR_1g_m r_o+C_LR_1+C_Lr_o}+\frac{2R_Lg_m r_o}{2C_4R_1g_m r_o+2C_4R_1+2C_4r_o+C_LR_1g_m r_o+C_LR_1+C_Lr_o}+\frac{2R_L}{2C_4R_1g_m r_o+2C_4R_1+2C_4r_o+C_LR_1g_m r_o+C_LR_1+C_Lr_o}}+2\sqrt{C_1}C_4R_1\sqrt{R_4}\sqrt{R_L}\sqrt{\frac{R_4g_m r_o}{2C_4R_1g_m r_o+2C_4R_1+2C_4r_o+C_LR_1g_m r_o+C_LR_1+C_Lr_o}+\frac{R_4}{2C_4R_1g_m r_o+2C_4R_1+2C_4r_o+C_LR_1g_m r_o+C_LR_1+C_Lr_o}+\frac{2R_Lg_m r_o}{2C_4R_1g_m r_o+2C_4R_1+2C_4r_o+C_LR_1g_m r_o+C_LR_1+C_Lr_o}+\frac{2R_L}{2C_4R_1g_m r_o+2C_4R_1+2C_4r_o+C_LR_1g_m r_o+C_LR_1+C_Lr_o}}}$

K-LP: $\frac{R_4R_L}{R_4+2R_L}$

K-HP: 0

K-BP: $\frac{C_1R_1R_4R_Lg_m r_o+C_1R_1R_4R_L}{C_1R_1R_4g_m r_o+C_1R_1R_4+2C_1R_1R_Lg_m r_o+2C_1R_1R_L+C_1R_4R_L+C_1R_4r_o+2C_1R_Lr_o+2C_4R_4R_Lg_m r_o+2C_4R_4R_L+C_LR_4R_Lg_m r_o+C_LR_4R_L}$

Qz: None

Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (R_1, \infty, \infty, R_4, \infty, R_L)$

$$H(s) = \frac{R_1 R_4 R_L g_m r_o + R_1 R_4 R_L}{R_1 R_4 g_m r_o + R_1 R_4 + 2 R_1 R_L g_m r_o + 2 R_1 R_L + R_4 R_L + R_4 r_o + 2 R_L r_o}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(R_1, \infty, \infty, R_4, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 R_4 g_m r_o + R_1 R_4}{2 R_1 g_m r_o + 2 R_1 + R_4 + 2 r_o + s (C_L R_1 R_4 g_m r_o + C_L R_1 R_4 + C_L R_4 r_o)}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(R_1, \infty, \infty, R_4, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_1 R_4 R_L g_m r_o + R_1 R_4 R_L}{R_1 R_4 g_m r_o + R_1 R_4 + 2 R_1 R_L g_m r_o + 2 R_1 R_L + R_4 R_L + R_4 r_o + 2 R_L r_o + s (C_L R_1 R_4 R_L g_m r_o + C_L R_1 R_4 R_L + C_L R_4 R_L r_o)}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(R_1, \infty, \infty, R_4, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 R_4 g_m r_o + R_1 R_4 + s (C_L R_1 R_4 R_L g_m r_o + C_L R_1 R_4 R_L)}{2 R_1 g_m r_o + 2 R_1 + R_4 + 2 r_o + s (C_L R_1 R_4 g_m r_o + C_L R_1 R_4 + 2 C_L R_1 R_L g_m r_o + 2 C_L R_1 R_L + C_L R_4 R_L + C_L R_4 r_o + 2 C_L R_L r_o)}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(R_1, \infty, \infty, \frac{1}{C_4 s}, \infty, R_L \right)$

$$H(s) = \frac{R_1 R_L g_m r_o + R_1 R_L}{R_1 g_m r_o + R_1 + R_L + r_o + s (2 C_4 R_1 R_L g_m r_o + 2 C_4 R_1 R_L + 2 C_4 R_L r_o)}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(R_1, \infty, \infty, \frac{1}{C_4 s}, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 g_m r_o + R_1}{s (2 C_4 R_1 g_m r_o + 2 C_4 R_1 + 2 C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o) + 1}$$

9.7 X-INVALID-ORDER-7 $Z(s) = \left(R_1, \infty, \infty, \frac{1}{C_4 s}, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_1 R_L g_m r_o + R_1 R_L}{R_1 g_m r_o + R_1 + R_L + r_o + s(2C_4 R_1 R_L g_m r_o + 2C_4 R_1 R_L + 2C_4 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o)}$$

9.8 X-INVALID-ORDER-8 $Z(s) = \left(R_1, \infty, \infty, \frac{1}{C_4 s}, \infty, L_L s + \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s^2 (C_L L_L R_1 g_m r_o + C_L L_L R_1)}{C_L L_L s^2 + s^3 (2C_4 C_L L_L R_1 g_m r_o + 2C_4 C_L L_L R_1 + 2C_4 C_L L_L r_o) + s(2C_4 R_1 g_m r_o + 2C_4 R_1 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o) + 1}$$

9.9 X-INVALID-ORDER-9 $Z(s) = \left(R_1, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, R_L \right)$

$$H(s) = \frac{R_1 R_4 R_L g_m r_o + R_1 R_4 R_L}{R_1 R_4 g_m r_o + R_1 R_4 + 2R_1 R_L g_m r_o + 2R_1 R_L + R_4 R_L + R_4 r_o + 2R_L r_o + s(2C_4 R_1 R_4 R_L g_m r_o + 2C_4 R_1 R_4 R_L + 2C_4 R_4 R_L r_o)}$$

9.10 X-INVALID-ORDER-10 $Z(s) = \left(R_1, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 R_4 g_m r_o + R_1 R_4}{2R_1 g_m r_o + 2R_1 + R_4 + 2r_o + s(2C_4 R_1 R_4 g_m r_o + 2C_4 R_1 R_4 + 2C_4 R_4 r_o + C_L R_1 R_4 g_m r_o + C_L R_1 R_4 + C_L R_4 r_o)}$$

9.11 X-INVALID-ORDER-11 $Z(s) = \left(R_1, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_1 R_4 R_L g_m r_o + R_1 R_4 R_L}{R_1 R_4 g_m r_o + R_1 R_4 + 2R_1 R_L g_m r_o + 2R_1 R_L + R_4 R_L + R_4 r_o + 2R_L r_o + s(2C_4 R_1 R_4 R_L g_m r_o + 2C_4 R_1 R_4 R_L + 2C_4 R_4 R_L r_o + C_L R_1 R_4 R_L g_m r_o + C_L R_1 R_4 R_L + C_L R_4 R_L r_o)}$$

9.12 X-INVALID-ORDER-12 $Z(s) = \left(R_1, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, L_L s + \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 R_4 g_m r_o + R_1 R_4 + s^2 (C_L L_L R_1 R_4 g_m r_o + C_L L_L R_1 R_4)}{2R_1 g_m r_o + 2R_1 + R_4 + 2r_o + s^3 (2C_4 C_L L_L R_1 R_4 g_m r_o + 2C_4 C_L L_L R_1 R_4 + 2C_4 C_L L_L R_4 r_o) + s^2 (2C_L L_L R_1 g_m r_o + 2C_L L_L R_1 + C_L L_L R_4 + 2C_L L_L r_o) + s(2C_4 R_1 R_4 g_m r_o + 2C_4 R_1 R_4 + 2C_4 R_4 r_o + C_L R_1 R_4 g_m r_o + C_L R_1 R_4 + C_L R_4 r_o)}$$

9.13 X-INVALID-ORDER-13 $Z(s) = \left(R_1, \infty, \infty, R_4 + \frac{1}{C_4 s}, \infty, R_L \right)$

$$H(s) = \frac{R_1 R_L g_m r_o + R_1 R_L + s(C_4 R_1 R_4 R_L g_m r_o + C_4 R_1 R_4 R_L)}{R_1 g_m r_o + R_1 + R_L + r_o + s(C_4 R_1 R_4 g_m r_o + C_4 R_1 R_4 + 2C_4 R_1 R_L g_m r_o + 2C_4 R_1 R_L + C_4 R_4 R_L + C_4 R_4 r_o + 2C_4 R_L r_o)}$$

9.14 X-INVALID-ORDER-14 $Z(s) = \left(R_1, \infty, \infty, R_4 + \frac{1}{C_4 s}, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$

$$H(s) = \frac{s^2 (C_4 L_L R_1 R_4 g_m r_o + C_4 L_L R_1 R_4) + s(L_L R_1 g_m r_o + L_L R_1)}{R_1 g_m r_o + R_1 + r_o + s^3 (C_4 C_L L_L R_1 R_4 g_m r_o + C_4 C_L L_L R_1 R_4 + C_4 C_L L_L R_4 r_o) + s^2 (2C_4 L_L R_1 g_m r_o + 2C_4 L_L R_1 + C_4 L_L R_4 + 2C_4 L_L r_o + C_L L_L R_1 g_m r_o + C_L L_L R_1 + C_L L_L r_o) + s(C_4 R_1 R_4 g_m r_o + C_4 R_1 R_4 + C_4 R_4 r_o + L_L)}$$

9.15 X-INVALID-ORDER-15 $Z(s) = \left(R_1, \infty, \infty, R_4 + \frac{1}{C_4 s}, \infty, L_L s + \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s^3 (C_4 C_L L_L R_1 R_4 g_m r_o + C_4 C_L L_L R_1 R_4) + s^2 (C_L L_L R_1 g_m r_o + C_L L_L R_1) + s(C_4 R_1 R_4 g_m r_o + C_4 R_1 R_4)}{s^3 (2C_4 C_L L_L R_1 g_m r_o + 2C_4 C_L L_L R_1 + C_4 C_L L_L R_4 + 2C_4 C_L L_L r_o) + s^2 (C_4 C_L R_1 R_4 g_m r_o + C_4 C_L R_1 R_4 + C_4 C_L R_4 r_o + C_L L_L) + s(2C_4 R_1 g_m r_o + 2C_4 R_1 + C_4 R_4 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o) + 1}$$

9.16 X-INVALID-ORDER-16 $Z(s) = \left(R_1, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{s^2 (C_L L_4 R_1 R_L g_m r_o + C_L L_4 R_1 R_L) + s(L_4 R_1 g_m r_o + L_4 R_1)}{2R_1 g_m r_o + 2R_1 + 2r_o + s^3 (2C_4 C_L L_4 R_1 R_L g_m r_o + 2C_4 C_L L_4 R_1 R_L + 2C_4 C_L L_4 R_L r_o) + s^2 (2C_4 L_4 R_1 g_m r_o + 2C_4 L_4 R_1 + 2C_4 L_4 r_o + C_L L_4 R_1 g_m r_o + C_L L_4 R_1 + C_L L_4 R_L + C_L L_4 r_o) + s(2C_L R_1 R_L g_m r_o + 2C_L R_1 R_L + 2C_L R_L r_o + L_4)}$$

$$\mathbf{9.17 \quad X-INVALID-ORDER-17} \quad Z(s) = \left(R_1, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s^3 (C_L L_4 L_L R_1 g_m r_o + C_L L_4 L_L R_1) + s (L_4 R_1 g_m r_o + L_4 R_1)}{C_L L_4 L_L s^3 + L_4 s + 2 R_1 g_m r_o + 2 R_1 + 2 r_o + s^4 (2 C_4 C_L L_4 L_L R_1 g_m r_o + 2 C_4 C_L L_4 L_L R_1 + 2 C_4 C_L L_4 L_L r_o) + s^2 (2 C_4 L_4 R_1 g_m r_o + 2 C_4 L_4 R_1 + 2 C_4 L_4 r_o + C_L L_4 R_1 g_m r_o + C_L L_4 R_1 + C_L L_4 r_o + 2 C_L L_L R_1 g_m r_o + 2 C_L L_L R_1 + 2 C_L L_L r_o)}$$

$$\mathbf{9.18 \quad X-INVALID-ORDER-18} \quad Z(s) = \left(R_1, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \infty, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s^2 (C_4 L_4 R_1 g_m r_o + C_4 L_4 R_1)}{C_4 L_4 s^2 + s^3 (C_4 C_L L_4 R_1 g_m r_o + C_4 C_L L_4 R_1 + C_4 C_L L_4 r_o) + s (2 C_4 R_1 g_m r_o + 2 C_4 R_1 + 2 C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o) + 1}$$

$$\mathbf{9.19 \quad X-INVALID-ORDER-19} \quad Z(s) = \left(R_1, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \infty, \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{R_1 R_L g_m r_o + R_1 R_L + s^2 (C_4 L_4 R_1 R_L g_m r_o + C_4 L_4 R_1 R_L)}{R_1 g_m r_o + R_1 + R_L + r_o + s^3 (C_4 C_L L_4 R_1 R_L g_m r_o + C_4 C_L L_4 R_1 R_L + C_4 C_L L_4 R_L r_o) + s^2 (C_4 L_4 R_1 g_m r_o + C_4 L_4 R_1 + C_4 L_4 R_L + C_4 L_4 r_o) + s (2 C_4 R_1 R_L g_m r_o + 2 C_4 R_1 R_L + 2 C_4 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o)}$$

$$\mathbf{9.20 \quad X-INVALID-ORDER-20} \quad Z(s) = \left(R_1, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \infty, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s^3 (C_4 C_L L_4 R_1 R_L g_m r_o + C_4 C_L L_4 R_1 R_L) + s^2 (C_4 L_4 R_1 g_m r_o + C_4 L_4 R_1) + s (C_L R_1 R_L g_m r_o + C_L R_1 R_L)}{s^3 (C_4 C_L L_4 R_1 g_m r_o + C_4 C_L L_4 R_1 + C_4 C_L L_4 R_L + C_4 C_L L_4 r_o) + s^2 (2 C_4 C_L R_1 R_L g_m r_o + 2 C_4 C_L R_1 R_L + 2 C_4 C_L R_L r_o + C_4 L_4) + s (2 C_4 R_1 g_m r_o + 2 C_4 R_1 + 2 C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L R_L + C_L r_o) + 1}$$

$$\mathbf{9.21 \quad X-INVALID-ORDER-21} \quad Z(s) = \left(R_1, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s^3 (C_4 L_4 L_L R_1 g_m r_o + C_4 L_4 L_L R_1) + s (L_L R_1 g_m r_o + L_L R_1)}{C_4 L_4 L_L s^3 + L_L s + R_1 g_m r_o + R_1 + r_o + s^4 (C_4 C_L L_4 L_L R_1 g_m r_o + C_4 C_L L_4 L_L R_1 + C_4 C_L L_4 L_L r_o) + s^2 (C_4 L_4 R_1 g_m r_o + C_4 L_4 R_1 + C_4 L_4 r_o + 2 C_4 L_L R_1 g_m r_o + 2 C_4 L_L R_1 + 2 C_4 L_L r_o + C_L L_L R_1 g_m r_o + C_L L_L R_1 + C_L L_L r_o)}$$

$$\mathbf{9.22 \quad X-INVALID-ORDER-22} \quad Z(s) = \left(R_1, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \infty, L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s^4 (C_4 C_L L_4 L_L R_1 g_m r_o + C_4 C_L L_4 L_L R_1) + s^2 (C_4 L_4 R_1 g_m r_o + C_4 L_4 R_1 + C_L L_L R_1 g_m r_o + C_L L_L R_1)}{C_4 C_L L_4 L_L s^4 + s^3 (C_4 C_L L_4 R_1 g_m r_o + C_4 C_L L_4 R_1 + C_4 C_L L_4 r_o + 2 C_4 C_L L_L R_1 g_m r_o + 2 C_4 C_L L_L R_1 + 2 C_4 C_L L_L r_o) + s^2 (C_4 L_4 + C_L L_L) + s (2 C_4 R_1 g_m r_o + 2 C_4 R_1 + 2 C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o) + 1}$$

$$\mathbf{9.23 \quad X-INVALID-ORDER-23} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, R_4, \infty, R_L \right)$$

$$H(s) = \frac{R_4 R_L g_m r_o + R_4 R_L}{R_4 g_m r_o + R_4 + 2 R_L g_m r_o + 2 R_L + s (C_1 R_4 R_L + C_1 R_4 r_o + 2 C_1 R_L r_o)}$$

$$\mathbf{9.24 \quad X-INVALID-ORDER-24} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, R_4, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s (L_L R_4 g_m r_o + L_L R_4)}{C_1 C_L L_L R_4 r_o s^3 + R_4 g_m r_o + R_4 + s^2 (C_1 L_L R_4 + 2 C_1 L_L r_o + C_L L_L R_4 g_m r_o + C_L L_L R_4) + s (C_1 R_4 r_o + 2 L_L g_m r_o + 2 L_L)}$$

$$\mathbf{9.25 \quad X-INVALID-ORDER-25} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, R_4, \infty, L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_4 g_m r_o + R_4 + s^2 (C_L L_L R_4 g_m r_o + C_L L_L R_4)}{2 g_m r_o + s^3 (C_1 C_L L_L R_4 + 2 C_1 C_L L_L r_o) + s^2 (C_1 C_L R_4 r_o + 2 C_L L_L g_m r_o + 2 C_L L_L) + s (C_1 R_4 + 2 C_1 r_o + C_L R_4 g_m r_o + C_L R_4) + 2}$$

$$\mathbf{9.26 \quad X-INVALID-ORDER-26} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{1}{C_4 s}, \infty, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m r_o + 1}{s^2 (2 C_1 C_4 r_o + C_1 C_L r_o) + s (C_1 + 2 C_4 g_m r_o + 2 C_4 + C_L g_m r_o + C_L)}$$

9.27 X-INVALID-ORDER-27 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{1}{C_4 s}, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{g_m r_o + s(C_L R_L g_m r_o + C_L R_L) + 1}{2C_1 C_4 C_L R_L r_o s^3 + s^2(2C_1 C_4 r_o + C_1 C_L R_L + C_1 C_L r_o + 2C_4 C_L R_L g_m r_o + 2C_4 C_L R_L) + s(C_1 + 2C_4 g_m r_o + 2C_4 + C_L g_m r_o + C_L)}$$

9.28 X-INVALID-ORDER-28 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{1}{C_4 s}, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$

$$H(s) = \frac{s(L_L g_m r_o + L_L)}{C_1 r_o s + g_m r_o + s^3(2C_1 C_4 L_L r_o + C_1 C_L L_L r_o) + s^2(C_1 L_L + 2C_4 L_L g_m r_o + 2C_4 L_L + C_L L_L g_m r_o + C_L L_L) + 1}$$

9.29 X-INVALID-ORDER-29 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{1}{C_4 s}, \infty, L_L s + \frac{1}{C_L s} \right)$

$$H(s) = \frac{g_m r_o + s^2(C_L L_L g_m r_o + C_L L_L) + 1}{2C_1 C_4 C_L L_L r_o s^4 + s^3(C_1 C_L L_L + 2C_4 C_L L_L g_m r_o + 2C_4 C_L L_L) + s^2(2C_1 C_4 r_o + C_1 C_L r_o) + s(C_1 + 2C_4 g_m r_o + 2C_4 + C_L g_m r_o + C_L)}$$

9.30 X-INVALID-ORDER-30 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_4 g_m r_o + R_4 + s(C_L R_4 R_L g_m r_o + C_L R_4 R_L)}{2C_1 C_4 C_L R_4 R_L r_o s^3 + 2g_m r_o + s^2(2C_1 C_4 R_4 r_o + C_1 C_L R_4 R_L + C_1 C_L R_4 r_o + 2C_1 C_L R_L r_o + 2C_4 C_L R_4 R_L g_m r_o + 2C_4 C_L R_4 R_L) + s(C_1 R_4 + 2C_1 r_o + 2C_4 R_4 g_m r_o + 2C_4 R_4 + C_L R_4 g_m r_o + C_L R_4 + 2C_L R_L g_m r_o + 2C_L R_L) + 2}$$

9.31 X-INVALID-ORDER-31 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$

$$H(s) = \frac{s(L_L R_4 g_m r_o + L_L R_4)}{R_4 g_m r_o + R_4 + s^3(2C_1 C_4 L_L R_4 r_o + C_1 C_L L_L R_4 r_o) + s^2(C_1 L_L R_4 + 2C_1 L_L r_o + 2C_4 L_L R_4 g_m r_o + 2C_4 L_L R_4 + C_L L_L R_4 g_m r_o + C_L L_L R_4) + s(C_1 R_4 r_o + 2L_L g_m r_o + 2L_L)}$$

9.32 X-INVALID-ORDER-32 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, L_L s + \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_4 g_m r_o + R_4 + s^2(C_L L_L R_4 g_m r_o + C_L L_L R_4)}{2C_1 C_4 C_L L_L R_4 r_o s^4 + 2g_m r_o + s^3(C_1 C_L L_L R_4 + 2C_1 C_L L_L r_o + 2C_4 C_L L_L R_4 g_m r_o + 2C_4 C_L L_L R_4) + s^2(2C_1 C_4 R_4 r_o + C_1 C_L R_4 r_o + 2C_L L_L g_m r_o + 2C_L L_L) + s(C_1 R_4 + 2C_1 r_o + 2C_4 R_4 g_m r_o + 2C_4 R_4 + C_L R_4 g_m r_o + C_L R_4) + 2}$$

9.33 X-INVALID-ORDER-33 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{g_m r_o + s(C_4 R_4 g_m r_o + C_4 R_4) + 1}{C_1 C_4 C_L R_4 r_o s^3 + s^2(C_1 C_4 R_4 + 2C_1 C_4 r_o + C_1 C_L r_o + C_4 C_L R_4 g_m r_o + C_4 C_L R_4) + s(C_1 + 2C_4 g_m r_o + 2C_4 + C_L g_m r_o + C_L)}$$

9.34 X-INVALID-ORDER-34 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_L g_m r_o + R_L + s(C_4 R_4 R_L g_m r_o + C_4 R_4 R_L)}{C_1 C_4 C_L R_4 R_L r_o s^3 + g_m r_o + s^2(C_1 C_4 R_4 R_L + C_1 C_4 R_4 r_o + 2C_1 C_4 R_L r_o + C_1 C_L R_L r_o + C_4 C_L R_4 R_L g_m r_o + C_4 C_L R_4 R_L) + s(C_1 R_L + C_1 r_o + C_4 R_4 g_m r_o + C_4 R_4 + 2C_4 R_L g_m r_o + 2C_4 R_L + C_L R_L g_m r_o + C_L R_L) + 1}$$

9.35 X-INVALID-ORDER-35 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{g_m r_o + s^2(C_4 C_L R_4 R_L g_m r_o + C_4 C_L R_4 R_L) + s(C_4 R_4 g_m r_o + C_4 R_4 + C_L R_L g_m r_o + C_L R_L) + 1}{s^3(C_1 C_4 C_L R_4 R_L + C_1 C_4 C_L R_4 r_o + 2C_1 C_4 C_L R_L r_o) + s^2(C_1 C_4 R_4 + 2C_1 C_4 r_o + C_1 C_L R_L + C_1 C_L r_o + C_4 C_L R_4 g_m r_o + C_4 C_L R_4 + 2C_4 C_L R_L g_m r_o + 2C_4 C_L R_L) + s(C_1 + 2C_4 g_m r_o + 2C_4 + C_L g_m r_o + C_L)}$$

9.36 X-INVALID-ORDER-36 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$

$$H(s) = \frac{s^2(C_4 L_L R_4 g_m r_o + C_4 L_L R_4) + s(L_L g_m r_o + L_L)}{C_1 C_4 C_L L_L R_4 r_o s^4 + g_m r_o + s^3(C_1 C_4 L_L R_4 + 2C_1 C_4 L_L r_o + C_1 C_L L_L r_o + C_4 C_L L_L R_4 g_m r_o + C_4 C_L L_L R_4) + s^2(C_1 C_4 R_4 r_o + C_1 L_L + 2C_4 L_L g_m r_o + 2C_4 L_L + C_L L_L g_m r_o + C_L L_L) + s(C_1 r_o + C_4 R_4 g_m r_o + C_4 R_4) + 1}$$

9.37 X-INVALID-ORDER-37 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \infty, L_L s + \frac{1}{C_L s} \right)$

$$H(s) = \frac{g_m r_o + s^3 (C_4 C_L L_L R_4 g_m r_o + C_4 C_L L_L R_4) + s^2 (C_L L_L g_m r_o + C_L L_L) + s (C_4 R_4 g_m r_o + C_4 R_4) + 1}{s^4 (C_1 C_4 C_L L_L R_4 + 2 C_1 C_4 C_L L_L r_o) + s^3 (C_1 C_4 C_L R_4 r_o + C_1 C_L L_L + 2 C_4 C_L L_L g_m r_o + 2 C_4 C_L L_L) + s^2 (C_1 C_4 R_4 + 2 C_1 C_4 r_o + C_1 C_L r_o + C_4 C_L R_4 g_m r_o + C_4 C_L R_4) + s (C_1 + 2 C_4 g_m r_o + 2 C_4 + C_L g_m r_o + C_L)}$$

9.38 X-INVALID-ORDER-38 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, R_L \right)$

$$H(s) = \frac{s (L_4 R_L g_m r_o + L_4 R_L)}{2 C_1 C_4 L_4 R_L r_o s^3 + 2 R_L g_m r_o + 2 R_L + s^2 (C_1 L_4 R_L + C_1 L_4 r_o + 2 C_4 L_4 R_L g_m r_o + 2 C_4 L_4 R_L) + s (2 C_1 R_L r_o + L_4 g_m r_o + L_4)}$$

9.39 X-INVALID-ORDER-39 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{s (L_4 g_m r_o + L_4)}{2 C_1 r_o s + 2 g_m r_o + s^3 (2 C_1 C_4 L_4 r_o + C_1 C_L L_4 r_o) + s^2 (C_1 L_4 + 2 C_4 L_4 g_m r_o + 2 C_4 L_4 + C_L L_4 g_m r_o + C_L L_4) + 2}$$

9.40 X-INVALID-ORDER-40 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{s (L_4 R_L g_m r_o + L_4 R_L)}{2 R_L g_m r_o + 2 R_L + s^3 (2 C_1 C_4 L_4 R_L r_o + C_1 C_L L_4 R_L r_o) + s^2 (C_1 L_4 R_L + C_1 L_4 r_o + 2 C_4 L_4 R_L g_m r_o + 2 C_4 L_4 R_L + C_L L_4 R_L g_m r_o + C_L L_4 R_L) + s (2 C_1 R_L r_o + L_4 g_m r_o + L_4)}$$

9.41 X-INVALID-ORDER-41 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{s^2 (C_L L_4 R_L g_m r_o + C_L L_4 R_L) + s (L_4 g_m r_o + L_4)}{2 C_1 C_4 C_L L_4 R_L r_o s^4 + 2 g_m r_o + s^3 (2 C_1 C_4 L_4 r_o + C_1 C_L L_4 R_L + C_1 C_L L_4 r_o + 2 C_4 C_L L_4 R_L g_m r_o + 2 C_4 C_L L_4 R_L) + s^2 (2 C_1 C_L R_L r_o + C_1 L_4 + 2 C_4 L_4 g_m r_o + 2 C_4 L_4 + C_L L_4 g_m r_o + C_L L_4) + s (2 C_1 r_o + 2 C_L R_L g_m r_o + 2 C_L R_L) + 2}$$

9.42 X-INVALID-ORDER-42 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$

$$H(s) = \frac{s (L_4 L_L g_m r_o + L_4 L_L)}{L_4 g_m r_o + L_4 + 2 L_L g_m r_o + 2 L_L + s^3 (2 C_1 C_4 L_4 L_L r_o + C_1 C_L L_4 L_L r_o) + s^2 (C_1 L_4 L_L + 2 C_4 L_4 L_L g_m r_o + 2 C_4 L_4 L_L + C_L L_4 L_L g_m r_o + C_L L_4 L_L) + s (C_1 L_4 r_o + 2 C_1 L_L r_o)}$$

9.43 X-INVALID-ORDER-43 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, L_L s + \frac{1}{C_L s} \right)$

$$H(s) = \frac{s^3 (C_L L_4 L_L g_m r_o + C_L L_4 L_L) + s (L_4 g_m r_o + L_4)}{2 C_1 C_4 C_L L_4 L_L r_o s^5 + 2 C_1 r_o s + 2 g_m r_o + s^4 (C_1 C_L L_4 L_L + 2 C_4 C_L L_4 L_L g_m r_o + 2 C_4 C_L L_4 L_L) + s^3 (2 C_1 C_4 L_4 r_o + C_1 C_L L_4 r_o + 2 C_1 C_L L_L r_o) + s^2 (C_1 L_4 + 2 C_4 L_4 g_m r_o + 2 C_4 L_4 + C_L L_4 g_m r_o + C_L L_4 + 2 C_L L_L g_m r_o + 2 C_L L_L) + 2}$$

9.44 X-INVALID-ORDER-44 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \infty, R_L \right)$

$$H(s) = \frac{R_L g_m r_o + R_L + s^2 (C_4 L_4 R_L g_m r_o + C_4 L_4 R_L)}{g_m r_o + s^3 (C_1 C_4 L_4 R_L + C_1 C_4 L_4 r_o) + s^2 (2 C_1 C_4 R_L r_o + C_4 L_4 g_m r_o + C_4 L_4) + s (C_1 R_L + C_1 r_o + 2 C_4 R_L g_m r_o + 2 C_4 R_L) + 1}$$

9.45 X-INVALID-ORDER-45 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{g_m r_o + s^2 (C_4 L_4 g_m r_o + C_4 L_4) + 1}{C_1 C_4 C_L L_4 r_o s^4 + s^3 (C_1 C_4 L_4 + C_4 C_L L_4 g_m r_o + C_4 C_L L_4) + s^2 (2 C_1 C_4 r_o + C_1 C_L r_o) + s (C_1 + 2 C_4 g_m r_o + 2 C_4 + C_L g_m r_o + C_L)}$$

9.46 X-INVALID-ORDER-46 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_L g_m r_o + R_L + s^2 (C_4 L_4 R_L g_m r_o + C_4 L_4 R_L)}{C_1 C_4 C_L L_4 R_L r_o s^4 + g_m r_o + s^3 (C_1 C_4 L_4 R_L + C_1 C_4 L_4 r_o + C_4 C_L L_4 R_L g_m r_o + C_4 C_L L_4 R_L) + s^2 (2 C_1 C_4 R_L r_o + C_1 C_L R_L r_o + C_4 L_4 g_m r_o + C_4 L_4) + s (C_1 R_L + C_1 r_o + 2 C_4 R_L g_m r_o + 2 C_4 R_L + C_L R_L g_m r_o + C_L R_L) + 1}$$

9.47 X-INVALID-ORDER-47 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{g_m r_o + s^3 (C_4 C_L L_4 R_L g_m r_o + C_4 C_L L_4 R_L) + s^2 (C_4 L_4 g_m r_o + C_4 L_4) + s (C_L R_L g_m r_o + C_L R_L) + 1}{s^4 (C_1 C_4 C_L L_4 R_L + C_1 C_4 C_L L_4 r_o) + s^3 (2C_1 C_4 C_L R_L r_o + C_1 C_4 L_4 + C_4 C_L L_4 g_m r_o + C_4 C_L L_4) + s^2 (2C_1 C_4 r_o + C_1 C_L R_L + C_1 C_L r_o + 2C_4 C_L R_L g_m r_o + 2C_4 C_L R_L) + s (C_1 + 2C_4 g_m r_o + 2C_4 + C_L g_m r_o + C_L)}$$

9.48 X-INVALID-ORDER-48 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$

$$H(s) = \frac{s^3 (C_4 L_4 L_L g_m r_o + C_4 L_4 L_L) + s (L_L g_m r_o + L_L)}{C_1 C_4 C_L L_4 L_L r_o s^5 + C_1 r_o s + g_m r_o + s^4 (C_1 C_4 L_4 L_L + C_4 C_L L_4 L_L g_m r_o + C_4 C_L L_4 L_L) + s^3 (C_1 C_4 L_4 r_o + 2C_1 C_4 L_L r_o + C_1 C_L L_L r_o) + s^2 (C_1 L_L + C_4 L_4 g_m r_o + C_4 L_4 + 2C_4 L_L g_m r_o + 2C_4 L_L + C_L L_L g_m r_o + C_L L_L) + 1}$$

9.49 X-INVALID-ORDER-49 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \infty, L_L s + \frac{1}{C_L s} \right)$

$$H(s) = \frac{g_m r_o + s^4 (C_4 C_L L_4 L_L g_m r_o + C_4 C_L L_4 L_L) + s^2 (C_4 L_4 g_m r_o + C_4 L_4 + C_L L_L g_m r_o + C_L L_L) + 1}{C_1 C_4 C_L L_4 L_L s^5 + s^4 (C_1 C_4 C_L L_4 r_o + 2C_1 C_4 C_L L_L r_o) + s^3 (C_1 C_4 L_4 + C_1 C_L L_L + C_4 C_L L_4 g_m r_o + C_4 C_L L_4 + 2C_4 C_L L_L g_m r_o + 2C_4 C_L L_L) + s^2 (2C_1 C_4 r_o + C_1 C_L r_o) + s (C_1 + 2C_4 g_m r_o + 2C_4 + C_L g_m r_o + C_L)}$$

9.50 X-INVALID-ORDER-50 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4, \infty, R_L \right)$

$$H(s) = \frac{R_1 R_4 R_L g_m r_o + R_1 R_4 R_L}{R_1 R_4 g_m r_o + R_1 R_4 + 2R_1 R_L g_m r_o + 2R_1 R_L + R_4 R_L + R_4 r_o + 2R_L r_o + s (C_1 R_1 R_4 R_L + C_1 R_1 R_4 r_o + 2C_1 R_1 R_L r_o)}$$

9.51 X-INVALID-ORDER-51 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$

$$H(s) = \frac{s (L_L R_1 R_4 g_m r_o + L_L R_1 R_4)}{C_1 C_L L_L R_1 R_4 r_o s^3 + R_1 R_4 g_m r_o + R_1 R_4 + R_4 r_o + s^2 (C_1 L_L R_1 R_4 + 2C_1 L_L R_1 r_o + C_L L_L R_1 R_4 g_m r_o + C_L L_L R_1 R_4 + C_L L_L R_4 r_o) + s (C_1 R_1 R_4 r_o + 2L_L R_1 g_m r_o + 2L_L R_1 + L_L R_4 + 2L_L r_o)}$$

9.52 X-INVALID-ORDER-52 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4, \infty, L_L s + \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 R_4 g_m r_o + R_1 R_4 + s^2 (C_L L_L R_1 R_4 g_m r_o + C_L L_L R_1 R_4)}{2R_1 g_m r_o + 2R_1 + R_4 + 2r_o + s^3 (C_1 C_L L_L R_1 R_4 + 2C_1 C_L L_L R_1 r_o) + s^2 (C_1 C_L R_1 R_4 r_o + 2C_L L_L R_1 g_m r_o + 2C_L L_L R_1 + C_L L_L R_4 + 2C_L L_L r_o) + s (C_1 R_1 R_4 + 2C_1 R_1 r_o + C_L R_1 R_4 g_m r_o + C_L R_1 R_4 + C_L R_4 r_o)}$$

9.53 X-INVALID-ORDER-53 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{1}{C_4 s}, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s (C_L R_1 R_L g_m r_o + C_L R_1 R_L)}{2C_1 C_4 C_L R_1 R_L r_o s^3 + s^2 (2C_1 C_4 R_1 r_o + C_1 C_L R_1 R_L + C_1 C_L R_1 r_o + 2C_4 C_L R_1 R_L g_m r_o + 2C_4 C_L R_1 R_L + 2C_4 C_L R_L r_o) + s (C_1 R_1 + 2C_4 R_1 g_m r_o + 2C_4 R_1 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L R_L + C_L r_o) + 1}$$

9.54 X-INVALID-ORDER-54 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{1}{C_4 s}, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$

$$H(s) = \frac{s (L_L R_1 g_m r_o + L_L R_1)}{R_1 g_m r_o + R_1 + r_o + s^3 (2C_1 C_4 L_L R_1 r_o + C_1 C_L L_L R_1 r_o) + s^2 (C_1 L_L R_1 + 2C_4 L_L R_1 g_m r_o + 2C_4 L_L R_1 + 2C_4 L_L r_o + C_L L_L R_1 g_m r_o + C_L L_L R_1 + C_L L_L r_o) + s (C_1 R_1 r_o + L_L)}$$

9.55 X-INVALID-ORDER-55 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{1}{C_4 s}, \infty, L_L s + \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s^2 (C_L L_L R_1 g_m r_o + C_L L_L R_1)}{2C_1 C_4 C_L L_L R_1 r_o s^4 + s^3 (C_1 C_L L_L R_1 + 2C_4 C_L L_L R_1 g_m r_o + 2C_4 C_L L_L R_1 + 2C_4 C_L L_L r_o) + s^2 (2C_1 C_4 R_1 r_o + C_1 C_L R_1 r_o + C_L L_L) + s (C_1 R_1 + 2C_4 R_1 g_m r_o + 2C_4 R_1 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o) + 1}$$

9.56 X-INVALID-ORDER-56 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 R_4 g_m r_o + R_1 R_4 + s (C_L R_1 R_4 R_L g_m r_o + C_L R_1 R_4 R_L)}{2C_1 C_4 C_L R_1 R_4 R_L r_o s^3 + 2R_1 g_m r_o + 2R_1 + R_4 + 2r_o + s^2 (2C_1 C_4 R_1 R_4 r_o + C_1 C_L R_1 R_4 R_L + C_1 C_L R_1 R_4 r_o + 2C_1 C_L R_1 R_L r_o + 2C_4 C_L R_1 R_4 R_L g_m r_o + 2C_4 C_L R_1 R_4 R_L + 2C_4 C_L R_4 R_L r_o) + s (C_1 R_1 R_4 + 2C_1 R_1 r_o + 2C_4 R_1 R_4 g_m r_o + 2C_4 R_1 R_4 + 2C_4 R_4 r_o + C_L R_1)}$$

$$\mathbf{9.57 \quad X-INVALID-ORDER-57} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s (L_L R_1 R_4 g_m r_o + L_L R_1 R_4)}{R_1 R_4 g_m r_o + R_1 R_4 + R_4 r_o + s^3 (2 C_1 C_4 L_L R_1 R_4 r_o + C_1 C_L L_L R_1 R_4 r_o) + s^2 (C_1 L_L R_1 R_4 + 2 C_1 L_L R_1 r_o + 2 C_4 L_L R_1 R_4 g_m r_o + 2 C_4 L_L R_1 R_4 + 2 C_4 L_L R_4 r_o + C_L L_L R_1 R_4 g_m r_o + C_L L_L R_1 R_4 + C_L L_L R_4 r_o) + s (C_1 R_1 R_4 r_o + 2 L_L R_1 g_m r_o + 2 L_L R_1 + L_L R_4 + 2 L_L r_o)}$$

$$\mathbf{9.58 \quad X-INVALID-ORDER-58} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 R_4 g_m r_o + R_1 R_4 + s^2 (C_L L_L R_1 R_4 g_m r_o + C_L L_L R_1 R_4)}{2 C_1 C_4 C_L L_L R_1 R_4 r_o s^4 + 2 R_1 g_m r_o + 2 R_1 + R_4 + 2 r_o + s^3 (C_1 C_L L_L R_1 R_4 + 2 C_1 C_L L_L R_1 r_o + 2 C_4 C_L L_L R_1 R_4 g_m r_o + 2 C_4 C_L L_L R_1 R_4 + 2 C_4 C_L L_L R_4 r_o) + s^2 (2 C_1 C_4 R_1 R_4 r_o + C_1 C_L R_1 R_4 r_o + 2 C_L L_L R_1 g_m r_o + 2 C_L L_L R_1 + C_L L_L R_4 + 2 C_L L_L r_o) + s (C_1 R_1 R_4 + 2 L_L R_1 g_m r_o + 2 L_L R_1 + L_L R_4 + 2 L_L r_o)}$$

$$\mathbf{9.59 \quad X-INVALID-ORDER-59} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \infty, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s (C_4 R_1 R_4 g_m r_o + C_4 R_1 R_4)}{C_1 C_4 C_L R_1 R_4 r_o s^3 + s^2 (C_1 C_4 R_1 R_4 + 2 C_1 C_4 R_1 r_o + C_1 C_L R_1 r_o + C_4 C_L R_1 R_4 g_m r_o + C_4 C_L R_1 R_4 + C_4 C_L R_4 r_o) + s (C_1 R_1 + 2 C_4 R_1 g_m r_o + 2 C_4 R_1 + C_4 R_4 + 2 C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o) + 1}$$

$$\mathbf{9.60 \quad X-INVALID-ORDER-60} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \infty, \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{R_1 R_L g_m r_o + R_1 R_L + s (C_4 R_1 R_4 R_L g_m r_o + C_4 R_1 R_4 R_L)}{C_1 C_4 C_L R_1 R_4 R_L r_o s^3 + R_1 g_m r_o + R_1 + R_L + r_o + s^2 (C_1 C_4 R_1 R_4 R_L + C_1 C_4 R_1 R_4 r_o + 2 C_1 C_4 R_1 R_L r_o + C_1 C_L R_1 R_L r_o + C_4 C_L R_1 R_4 R_L g_m r_o + C_4 C_L R_1 R_4 R_L + C_4 C_L R_4 R_L r_o) + s (C_1 R_1 R_L + C_1 R_1 r_o + C_4 R_1 R_4 g_m r_o + C_4 R_1 R_4 + 2 C_4 R_1 R_L g_m r_o + 2 C_4 R_1 R_L + 2 C_4 R_1 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o)}$$

$$\mathbf{9.61 \quad X-INVALID-ORDER-61} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \infty, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s^2 (C_4 C_L R_1 R_4 R_L g_m r_o + C_4 C_L R_1 R_4 R_L) + s (C_4 R_1 R_4 g_m r_o + C_4 R_1 R_4 + C_L R_1 R_L g_m r_o + C_L R_1 R_L)}{s^3 (C_1 C_4 C_L R_1 R_4 R_L + C_1 C_4 C_L R_1 R_4 r_o + 2 C_1 C_4 C_L R_1 R_L r_o) + s^2 (C_1 C_4 R_1 R_4 + 2 C_1 C_4 R_1 r_o + C_1 C_L R_1 R_L + C_1 C_L R_1 r_o + C_4 C_L R_1 R_4 g_m r_o + C_4 C_L R_1 R_4 + 2 C_4 C_L R_1 R_L g_m r_o + 2 C_4 C_L R_1 R_L + C_4 C_L R_4 R_L + C_4 C_L R_4 r_o + 2 C_4 C_L R_L r_o) + s (C_1 R_1 + 2 C_4 R_1 g_m r_o + 2 C_4 R_1 + C_4 R_4 + 2 C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o)}$$

$$\mathbf{9.62 \quad X-INVALID-ORDER-62} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s^2 (C_4 L_L R_1 R_4 g_m r_o + C_4 L_L R_1 R_4) + s (L_L R_1 g_m r_o + L_L R_1)}{C_1 C_4 C_L L_L R_1 R_4 r_o s^4 + R_1 g_m r_o + R_1 + r_o + s^3 (C_1 C_4 L_L R_1 R_4 + 2 C_1 C_4 L_L R_1 r_o + C_1 C_L L_L R_1 r_o + C_4 C_L L_L R_1 R_4 g_m r_o + C_4 C_L L_L R_1 R_4 + C_4 C_L L_L R_4 r_o) + s^2 (C_1 C_4 R_1 R_4 r_o + C_1 L_L R_1 + 2 C_4 L_L R_1 g_m r_o + 2 C_4 L_L R_1 + C_4 L_L R_4 + 2 C_4 L_L r_o + C_L L_L R_1 g_m r_o + C_L L_L R_1 + C_L L_L r_o) + s (C_1 R_1 + 2 C_4 R_1 g_m r_o + 2 C_4 R_1 + C_4 R_4 + 2 C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o)}$$

$$\mathbf{9.63 \quad X-INVALID-ORDER-63} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \infty, L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s^3 (C_4 C_L L_L R_1 R_4 g_m r_o + C_4 C_L L_L R_1 R_4) + s^2 (C_L L_L R_1 g_m r_o + C_L L_L R_1) + s (C_4 R_1 R_4 g_m r_o + C_4 R_1 R_4)}{s^4 (C_1 C_4 C_L L_L R_1 R_4 + 2 C_1 C_4 C_L L_L R_1 r_o) + s^3 (C_1 C_4 C_L R_1 R_4 r_o + C_1 C_L L_L R_1 + 2 C_4 C_L L_L R_1 g_m r_o + 2 C_4 C_L L_L R_1 + C_4 C_L L_L R_4 + 2 C_4 C_L L_L r_o) + s^2 (C_1 C_4 R_1 R_4 + 2 C_1 C_4 R_1 r_o + C_1 C_L R_1 r_o + C_4 C_L R_1 R_4 g_m r_o + C_4 C_L R_1 R_4 + C_4 C_L R_4 r_o + C_L L_L) + s (C_1 R_1 + 2 C_4 R_1 g_m r_o + 2 C_4 R_1 + C_4 R_4 + 2 C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o)}$$

$$\mathbf{9.64 \quad X-INVALID-ORDER-64} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, R_L \right)$$

$$H(s) = \frac{s (L_4 R_1 R_L g_m r_o + L_4 R_1 R_L)}{2 C_1 C_4 L_4 R_1 R_L r_o s^3 + 2 R_1 R_L g_m r_o + 2 R_1 R_L + 2 R_L r_o + s^2 (C_1 L_4 R_1 R_L + C_1 L_4 R_1 r_o + 2 C_4 L_4 R_1 R_L g_m r_o + 2 C_4 L_4 R_1 R_L + 2 C_4 L_4 R_L r_o) + s (2 C_1 R_1 R_L r_o + L_4 R_1 g_m r_o + L_4 R_1 + L_4 R_L + L_4 r_o)}$$

$$\mathbf{9.65 \quad X-INVALID-ORDER-65} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s (L_4 R_1 g_m r_o + L_4 R_1)}{2 R_1 g_m r_o + 2 R_1 + 2 r_o + s^3 (2 C_1 C_4 L_4 R_1 r_o + C_1 C_L L_4 R_1 r_o) + s^2 (C_1 L_4 R_1 + 2 C_4 L_4 R_1 g_m r_o + 2 C_4 L_4 R_1 + 2 C_4 L_4 r_o + C_L L_4 R_1 g_m r_o + C_L L_4 R_1 + C_L L_4 r_o) + s (2 C_1 R_1 r_o + L_4)}$$

$$\mathbf{9.66 \quad X-INVALID-ORDER-66} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{s (L_4 R_1 R_L g_m r_o + L_4 R_1 R_L)}{2 R_1 R_L g_m r_o + 2 R_1 R_L + 2 R_L r_o + s^3 (2 C_1 C_4 L_4 R_1 R_L r_o + C_1 C_L L_4 R_1 R_L r_o) + s^2 (C_1 L_4 R_1 R_L + C_1 L_4 R_1 r_o + 2 C_4 L_4 R_1 R_L g_m r_o + 2 C_4 L_4 R_1 R_L + 2 C_4 L_4 R_L r_o + C_L L_4 R_1 R_L g_m r_o + C_L L_4 R_1 R_L + C_L L_4 R_L r_o) + s (2 C_1 R_1 R_L r_o + L_4 R_1 g_m r_o + L_4 R_1 + L_4 R_L + L_4 r_o)}$$

$$\mathbf{9.67 \quad X-INVALID-ORDER-67} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s^2 (C_L L_4 R_1 R_L g_m r_o + C_L L_4 R_1 R_L) + s (L_4 R_1 g_m r_o + L_4 R_1)}{2C_1 C_4 C_L L_4 R_1 R_L r_o s^4 + 2R_1 g_m r_o + 2R_1 + 2r_o + s^3 (2C_1 C_4 L_4 R_1 r_o + C_1 C_L L_4 R_1 R_L + C_1 C_L L_4 R_1 r_o + 2C_4 C_L L_4 R_1 R_L g_m r_o + 2C_4 C_L L_4 R_1 R_L + 2C_4 C_L L_4 R_L r_o) + s^2 (2C_1 C_L R_1 R_L r_o + C_1 L_4 R_1 + 2C_4 L_4 R_1 g_m r_o + 2C_4 L_4 R_1 + 2C_4 L_4 r_o + C_L L_4 R_1 g_m r_o + C_L L_4 R_1)}$$

$$\mathbf{9.68 \quad X-INVALID-ORDER-68} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s (L_4 L_L R_1 g_m r_o + L_4 L_L R_1)}{L_4 R_1 g_m r_o + L_4 R_1 + L_4 r_o + 2L_L R_1 g_m r_o + 2L_L R_1 + 2L_L r_o + s^3 (2C_1 C_4 L_4 L_L R_1 r_o + C_1 C_L L_4 L_L R_1 r_o) + s^2 (C_1 L_4 L_L R_1 + 2C_4 L_4 L_L R_1 g_m r_o + 2C_4 L_4 L_L R_1 + 2C_4 L_4 L_L r_o + C_L L_4 L_L R_1 g_m r_o + C_L L_4 L_L R_1 + C_L L_4 L_L r_o) + s (C_1 L_4 R_1 r_o + 2C_1 L_L R_1 r_o + L_4 L_L)}$$

$$\mathbf{9.69 \quad X-INVALID-ORDER-69} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s^3 (C_L L_4 L_L R_1 g_m r_o + C_L L_4 L_L R_1) + s (L_4 R_1 g_m r_o + L_4 R_1)}{2C_1 C_4 C_L L_4 L_L R_1 r_o s^5 + 2R_1 g_m r_o + 2R_1 + 2r_o + s^4 (C_1 C_L L_4 L_L R_1 + 2C_4 C_L L_4 L_L R_1 g_m r_o + 2C_4 C_L L_4 L_L R_1 + 2C_4 C_L L_4 L_L r_o) + s^3 (2C_1 C_4 L_4 R_1 r_o + C_1 C_L L_4 R_1 r_o + 2C_1 C_L L_L R_1 r_o + C_L L_4 L_L) + s^2 (C_1 L_4 R_1 + 2C_4 L_4 R_1 g_m r_o + 2C_4 L_4 R_1 + 2C_4 L_4 r_o + C_L L_4 R_1 g_m r_o + C_L L_4 R_1)}$$

$$\mathbf{9.70 \quad X-INVALID-ORDER-70} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad R_L \right)$$

$$H(s) = \frac{R_1 R_L g_m r_o + R_1 R_L + s^2 (C_4 L_4 R_1 R_L g_m r_o + C_4 L_4 R_1 R_L)}{R_1 g_m r_o + R_1 + R_L + r_o + s^3 (C_1 C_4 L_4 R_1 R_L + C_1 C_4 L_4 R_1 r_o) + s^2 (2C_1 C_4 R_1 R_L r_o + C_4 L_4 R_1 g_m r_o + C_4 L_4 R_1 + C_4 L_4 R_L + C_4 L_4 r_o) + s (C_1 R_1 R_L + C_1 R_1 r_o + 2C_4 R_1 R_L g_m r_o + 2C_4 R_1 R_L + 2C_4 R_L r_o)}$$

$$\mathbf{9.71 \quad X-INVALID-ORDER-71} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s^2 (C_4 L_4 R_1 g_m r_o + C_4 L_4 R_1)}{C_1 C_4 C_L L_4 R_1 r_o s^4 + s^3 (C_1 C_4 L_4 R_1 + C_4 C_L L_4 R_1 g_m r_o + C_4 C_L L_4 R_1 + C_4 C_L L_4 r_o) + s^2 (2C_1 C_4 R_1 r_o + C_1 C_L R_1 r_o + C_4 L_4) + s (C_1 R_1 + 2C_4 R_1 g_m r_o + 2C_4 R_1 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o) + 1}$$

$$\mathbf{9.72 \quad X-INVALID-ORDER-72} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{R_1 R_L g_m r_o + R_1 R_L + s^2 (C_4 L_4 R_1 R_L g_m r_o + C_4 L_4 R_1 R_L)}{C_1 C_4 C_L L_4 R_1 R_L r_o s^4 + R_1 g_m r_o + R_1 + R_L + r_o + s^3 (C_1 C_4 L_4 R_1 R_L + C_1 C_4 L_4 R_1 r_o + C_4 C_L L_4 R_1 R_L g_m r_o + C_4 C_L L_4 R_1 R_L + C_4 C_L L_4 R_L r_o) + s^2 (2C_1 C_4 R_1 R_L r_o + C_1 C_L R_1 R_L r_o + C_4 L_4 R_1 g_m r_o + C_4 L_4 R_1 + C_4 L_4 R_L + C_4 L_4 r_o) + s (C_1 R_1 R_L + C_1 R_1 r_o + 2C_4 R_1 R_L g_m r_o + 2C_4 R_1 R_L + 2C_4 R_L r_o)}$$

$$\mathbf{9.73 \quad X-INVALID-ORDER-73} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s^3 (C_4 C_L L_4 R_1 R_L g_m r_o + C_4 C_L L_4 R_1 R_L) + s^2 (C_4 L_4 R_1 g_m r_o + C_4 L_4 R_1) + s (C_L R_1 R_L g_m r_o + C_L R_1 R_L)}{s^4 (C_1 C_4 C_L L_4 R_1 R_L + C_1 C_4 C_L L_4 R_1 r_o) + s^3 (2C_1 C_4 C_L R_1 R_L r_o + C_1 C_4 L_4 R_1 + C_4 C_L L_4 R_1 g_m r_o + C_4 C_L L_4 R_1 + C_4 C_L L_4 R_L + C_4 C_L L_4 r_o) + s^2 (2C_1 C_4 R_1 r_o + C_1 C_L R_1 R_L + C_1 C_L R_1 r_o + 2C_4 C_L R_1 R_L g_m r_o + 2C_4 C_L R_1 R_L + 2C_4 C_L R_L r_o + C_4 L_4) + s (C_1 R_1 + 2C_4 R_1 g_m r_o + 2C_4 R_1 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o) + 1}$$

$$\mathbf{9.74 \quad X-INVALID-ORDER-74} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s^3 (C_4 L_4 L_L R_1 g_m r_o + C_4 L_4 L_L R_1) + s (L_L R_1 g_m r_o + L_L R_1)}{C_1 C_4 C_L L_4 L_L R_1 r_o s^5 + R_1 g_m r_o + R_1 + r_o + s^4 (C_1 C_4 L_4 L_L R_1 + C_4 C_L L_4 L_L R_1 g_m r_o + C_4 C_L L_4 L_L R_1 + C_4 C_L L_4 L_L r_o) + s^3 (C_1 C_4 L_4 R_1 r_o + 2C_1 C_4 L_L R_1 r_o + C_1 C_L L_L R_1 r_o + C_4 L_4 L_L) + s^2 (C_1 L_L R_1 + C_4 L_4 R_1 g_m r_o + C_4 L_4 R_1 + C_4 L_4 r_o + 2C_4 L_L R_1 g_m r_o + 2C_4 L_L R_1)}$$

$$\mathbf{9.75 \quad X-INVALID-ORDER-75} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s^4 (C_4 C_L L_4 L_L R_1 g_m r_o + C_4 C_L L_4 L_L R_1) + s^2 (C_4 L_4 R_1 g_m r_o + C_4 L_4 R_1 + C_L L_L R_1 g_m r_o + C_L L_L R_1)}{C_1 C_4 C_L L_4 L_L R_1 s^5 + s^4 (C_1 C_4 C_L L_4 R_1 r_o + 2C_1 C_4 C_L L_L R_1 r_o + C_4 C_L L_4 L_L) + s^3 (C_1 C_4 L_4 R_1 + C_1 C_L L_L R_1 + C_4 C_L L_4 R_1 g_m r_o + C_4 C_L L_4 R_1 + C_4 C_L L_4 r_o + 2C_4 C_L L_L R_1 g_m r_o + 2C_4 C_L L_L R_1 + 2C_4 C_L L_L r_o) + s^2 (2C_1 C_4 R_1 r_o + C_1 C_L R_1 r_o + C_4 L_4 + C_L L_L) + s (C_1 R_1 + 2C_4 R_1 g_m r_o + 2C_4 R_1 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o) + 1}$$

$$\mathbf{9.76 \quad X-INVALID-ORDER-76} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \infty, \quad R_L \right)$$

$$H(s) = \frac{R_4 R_L g_m r_o + R_4 R_L + s (C_1 R_1 R_4 R_L g_m r_o + C_1 R_1 R_4 R_L)}{R_4 g_m r_o + R_4 + 2R_L g_m r_o + 2R_L + s (C_1 R_1 R_4 g_m r_o + C_1 R_1 R_4 + 2C_1 R_1 R_L g_m r_o + 2C_1 R_1 R_L + C_1 R_4 R_L + C_1 R_4 r_o + 2C_1 R_L r_o)}$$

$$\mathbf{9.77 \quad X-INVALID-ORDER-77} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \infty, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s^2 (C_1 L_L R_1 R_4 g_m r_o + C_1 L_L R_1 R_4) + s (L_L R_4 g_m r_o + L_L R_4)}{R_4 g_m r_o + R_4 + s^3 (C_1 C_L L_L R_1 R_4 g_m r_o + C_1 C_L L_L R_1 R_4 + C_1 C_L L_L R_4 r_o) + s^2 (2C_1 L_L R_1 g_m r_o + 2C_1 L_L R_1 + C_1 L_L R_4 + 2C_1 L_L r_o + C_L L_L R_4 g_m r_o + C_L L_L R_4) + s (C_1 R_1 R_4 g_m r_o + C_1 R_1 R_4 + C_1 R_4 r_o + 2L_L g_m r_o + 2L_L)}$$

$$\mathbf{9.78 \quad X-INVALID-ORDER-78} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \infty, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_4 g_m r_o + R_4 + s^3 (C_1 C_L L_L R_1 R_4 g_m r_o + C_1 C_L L_L R_1 R_4) + s^2 (C_L L_L R_4 g_m r_o + C_L L_L R_4) + s (C_1 R_1 R_4 g_m r_o + C_1 R_1 R_4)}{2g_m r_o + s^3 (2C_1 C_L L_L R_1 g_m r_o + 2C_1 C_L L_L R_1 + C_1 C_L L_L R_4 + 2C_1 C_L L_L r_o) + s^2 (C_1 C_L R_1 R_4 g_m r_o + C_1 C_L R_1 R_4 + C_1 C_L R_4 r_o + 2C_L L_L g_m r_o + 2C_L L_L) + s (2C_1 R_1 g_m r_o + 2C_1 R_1 + C_1 R_4 + 2C_1 r_o + C_L R_4 g_m r_o + C_L R_4) + 2}$$

$$\mathbf{9.79 \quad X-INVALID-ORDER-79} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \infty, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m r_o + s (C_1 R_1 g_m r_o + C_1 R_1) + 1}{s^2 (2C_1 C_4 R_1 g_m r_o + 2C_1 C_4 R_1 + 2C_1 C_4 r_o + C_1 C_L R_1 g_m r_o + C_1 C_L R_1 + C_1 C_L r_o) + s (C_1 + 2C_4 g_m r_o + 2C_4 + C_L g_m r_o + C_L)}$$

$$\mathbf{9.80 \quad X-INVALID-ORDER-80} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \infty, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m r_o + s^2 (C_1 C_L R_1 R_L g_m r_o + C_1 C_L R_1 R_L) + s (C_1 R_1 g_m r_o + C_1 R_1 + C_L R_L g_m r_o + C_L R_L) + 1}{s^3 (2C_1 C_4 C_L R_1 R_L g_m r_o + 2C_1 C_4 C_L R_1 R_L + 2C_1 C_4 C_L R_L r_o) + s^2 (2C_1 C_4 R_1 g_m r_o + 2C_1 C_4 R_1 + 2C_1 C_4 r_o + C_1 C_L R_1 g_m r_o + C_1 C_L R_1 + C_1 C_L R_L + C_1 C_L r_o + 2C_4 C_L R_L g_m r_o + 2C_4 C_L R_L) + s (C_1 + 2C_4 g_m r_o + 2C_4 + C_L g_m r_o + C_L)}$$

$$\mathbf{9.81 \quad X-INVALID-ORDER-81} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \infty, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s^2 (C_1 L_L R_1 g_m r_o + C_1 L_L R_1) + s (L_L g_m r_o + L_L)}{g_m r_o + s^3 (2C_1 C_4 L_L R_1 g_m r_o + 2C_1 C_4 L_L R_1 + 2C_1 C_4 L_L r_o + C_1 C_L L_L R_1 g_m r_o + C_1 C_L L_L R_1 + C_1 C_L L_L r_o) + s^2 (C_1 L_L + 2C_4 L_L g_m r_o + 2C_4 L_L + C_L L_L g_m r_o + C_L L_L) + s (C_1 R_1 g_m r_o + C_1 R_1 + C_1 r_o) + 1}$$

$$\mathbf{9.82 \quad X-INVALID-ORDER-82} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \infty, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m r_o + s^3 (C_1 C_L L_L R_1 g_m r_o + C_1 C_L L_L R_1) + s^2 (C_L L_L g_m r_o + C_L L_L) + s (C_1 R_1 g_m r_o + C_1 R_1) + 1}{s^4 (2C_1 C_4 C_L L_L R_1 g_m r_o + 2C_1 C_4 C_L L_L R_1 + 2C_1 C_4 C_L L_L r_o) + s^3 (C_1 C_L L_L + 2C_4 C_L L_L g_m r_o + 2C_4 C_L L_L) + s^2 (2C_1 C_4 R_1 g_m r_o + 2C_1 C_4 R_1 + 2C_1 C_4 r_o + C_1 C_L R_1 g_m r_o + C_1 C_L R_1 + C_1 C_L r_o) + s (C_1 + 2C_4 g_m r_o + 2C_4 + C_L g_m r_o + C_L)}$$

$$\mathbf{9.83 \quad X-INVALID-ORDER-83} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_4 g_m r_o + R_4 + s^2 (C_1 C_L R_1 R_4 R_L g_m r_o + C_1 C_L R_1 R_4 R_L) + s (C_1 R_1 R_4 g_m r_o + C_1 R_1 R_4 + C_L R_4 R_L g_m r_o + C_L R_4 R_L)}{2g_m r_o + s^3 (2C_1 C_4 C_L R_1 R_4 R_L g_m r_o + 2C_1 C_4 C_L R_1 R_4 R_L + 2C_1 C_4 C_L R_4 R_L r_o) + s^2 (2C_1 C_4 R_1 R_4 g_m r_o + 2C_1 C_4 R_1 R_4 + 2C_1 C_4 R_4 r_o + C_1 C_L R_1 R_4 g_m r_o + C_1 C_L R_1 R_4 + 2C_1 C_L R_1 R_L g_m r_o + 2C_1 C_L R_1 R_L + C_1 C_L R_4 R_L + C_1 C_L R_4 r_o + 2C_1 C_L R_L r_o + 2C_4 C_L R_4 R_L g_m r_o + 2C_4 C_L R_4 R_L) + s (C_1 R_1 R_4 g_m r_o + C_1 R_1 R_4 + C_L R_4 R_L g_m r_o + C_L R_4 R_L) + 1}$$

$$\mathbf{9.84 \quad X-INVALID-ORDER-84} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s^2 (C_1 L_L R_1 R_4 g_m r_o + C_1 L_L R_1 R_4) + s (L_L R_4 g_m r_o + L_L R_4)}{R_4 g_m r_o + R_4 + s^3 (2C_1 C_4 L_L R_1 R_4 g_m r_o + 2C_1 C_4 L_L R_1 R_4 + 2C_1 C_4 L_L R_4 r_o + C_1 C_L L_L R_1 R_4 g_m r_o + C_1 C_L L_L R_1 R_4 + C_1 C_L L_L R_4 r_o) + s^2 (2C_1 L_L R_1 g_m r_o + 2C_1 L_L R_1 + C_1 L_L R_4 + 2C_1 L_L r_o + 2C_4 L_L R_4 g_m r_o + 2C_4 L_L R_4 + C_L L_L R_4 g_m r_o + C_L L_L R_4) + s (C_1 R_1 R_4 g_m r_o + C_1 R_1 R_4 + C_L R_4 g_m r_o + C_L R_4) + 1}$$

$$\mathbf{9.85 \quad X-INVALID-ORDER-85} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_4 g_m r_o + R_4 + s^3 (C_1 C_L L_L R_1 R_4 g_m r_o + C_1 C_L L_L R_1 R_4) + s^2 (C_L L_L R_4 g_m r_o + C_L L_L R_4) + s (C_1 R_1 R_4 g_m r_o + C_1 R_1 R_4)}{2g_m r_o + s^4 (2C_1 C_4 C_L L_L R_1 R_4 g_m r_o + 2C_1 C_4 C_L L_L R_1 R_4 + 2C_1 C_4 C_L L_L R_4 r_o) + s^3 (2C_1 C_L L_L R_1 g_m r_o + 2C_1 C_L L_L R_1 + C_1 C_L L_L R_4 + 2C_1 C_L L_L r_o + 2C_4 C_L L_L R_4 g_m r_o + 2C_4 C_L L_L R_4) + s^2 (2C_1 C_4 R_1 R_4 g_m r_o + 2C_1 C_4 R_1 R_4 + 2C_1 C_4 R_4 r_o + C_1 C_L R_1 R_4 g_m r_o + C_1 C_L R_1 R_4 + C_1 C_L R_4 r_o) + s (C_1 R_1 R_4 g_m r_o + C_1 R_1 R_4 + C_L R_4 g_m r_o + C_L R_4) + 1}$$

$$\mathbf{9.86 \quad X-INVALID-ORDER-86} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m r_o + s^2 (C_1 C_4 R_1 R_4 g_m r_o + C_1 C_4 R_1 R_4) + s (C_1 R_1 g_m r_o + C_1 R_1 + C_4 R_4 g_m r_o + C_4 R_4) + 1}{s^3 (C_1 C_4 C_L R_1 R_4 g_m r_o + C_1 C_4 C_L R_1 R_4 + C_1 C_4 C_L R_4 r_o) + s^2 (2C_1 C_4 R_1 g_m r_o + 2C_1 C_4 R_1 + C_1 C_4 R_4 + 2C_1 C_4 r_o + C_1 C_L R_1 g_m r_o + C_1 C_L R_1 + C_1 C_L r_o + C_4 C_L R_4 g_m r_o + C_4 C_L R_4) + s (C_1 + 2C_4 g_m r_o + 2C_4 + C_L g_m r_o + C_L)}$$

$$\mathbf{9.87 \quad X-INVALID-ORDER-87} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{R_L g_m r_o + R_L + s^2 (C_1 C_4 R_1 R_4 R_L g_m r_o + C_1 C_4 R_1 R_4 R_L) + s (C_1 R_1 R_L g_m r_o + C_1 R_1 R_L + C_4 R_4 R_L g_m r_o + C_4 R_4 R_L)}{g_m r_o + s^3 (C_1 C_4 C_L R_1 R_4 R_L g_m r_o + C_1 C_4 C_L R_1 R_4 R_L + C_1 C_4 C_L R_4 R_L r_o) + s^2 (C_1 C_4 R_1 R_4 g_m r_o + C_1 C_4 R_1 R_4 + 2C_1 C_4 R_1 R_L g_m r_o + 2C_1 C_4 R_1 R_L + C_1 C_4 R_4 R_L + C_1 C_4 R_4 r_o + 2C_1 C_4 R_L r_o + C_1 C_L R_1 R_L g_m r_o + C_1 C_L R_1 R_L + C_1 C_L R_L r_o + C_4 C_L R_4 R_L g_m r_o + C_4 C_L R_4 R_L) + s (C_1 C_4 C_L R_1 R_4 g_m r_o + C_1 C_4 C_L R_1 R_4 + 2C_1 C_4 C_L R_1 R_L g_m r_o + 2C_1 C_4 C_L R_1 R_L + C_1 C_4 C_L R_4 R_L + C_1 C_4 C_L R_4 r_o + 2C_1 C_4 C_L R_L r_o) + s^2 (2C_1 C_4 R_1 g_m r_o + 2C_1 C_4 R_1 + C_1 C_4 R_4 + 2C_1 C_4 r_o + C_1 C_L R_1 g_m r_o + C_1 C_L R_1 + C_1 C_L R_L + C_1 C_L r_o + C_4 C_L R_4 g_m r_o + C_4 C_L R_4 + 2C_1 C_L R_L g_m r_o + 2C_1 C_L R_L + C_1 C_L r_o + L_L g_m r_o + L_L)}$$

$$\mathbf{9.88 \quad X-INVALID-ORDER-88} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m r_o + s^3 (C_1 C_4 C_L R_1 R_4 R_L g_m r_o + C_1 C_4 C_L R_1 R_4 R_L) + s^2 (C_1 C_4 R_1 R_4 g_m r_o + C_1 C_4 R_1 R_4 + C_1 C_L R_1 R_L g_m r_o + C_1 C_L R_1 R_L + C_4 C_L R_4 R_L g_m r_o + C_4 C_L R_4 R_L) + s (C_1 R_1 g_m r_o + C_1 R_1 + C_4 R_4 g_m r_o + C_4 R_4)}{s^3 (C_1 C_4 C_L R_1 R_4 g_m r_o + C_1 C_4 C_L R_1 R_4 + 2C_1 C_4 C_L R_1 R_L g_m r_o + 2C_1 C_4 C_L R_1 R_L + C_1 C_4 C_L R_4 R_L + C_1 C_4 C_L R_4 r_o + 2C_1 C_4 C_L R_L r_o) + s^2 (2C_1 C_4 R_1 g_m r_o + 2C_1 C_4 R_1 + C_1 C_4 R_4 + 2C_1 C_4 r_o + C_1 C_L R_1 g_m r_o + C_1 C_L R_1 + C_1 C_L R_L + C_1 C_L r_o + C_4 C_L R_4 g_m r_o + C_4 C_L R_4 + 2C_1 C_L R_L g_m r_o + 2C_1 C_L R_L + C_1 C_L r_o + L_L g_m r_o + L_L)}$$

$$\mathbf{9.89 \quad X-INVALID-ORDER-89} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s^3 (C_1 C_4 L_L R_1 R_4 g_m r_o + C_1 C_4 L_L R_1 R_4) + s^2 (C_1 L_L R_1 g_m r_o + C_1 L_L R_1 + C_4 L_L R_4 g_m r_o + C_4 L_L R_4) + s (L_L g_m r_o + L_L)}{g_m r_o + s^4 (C_1 C_4 C_L L_L R_1 R_4 g_m r_o + C_1 C_4 C_L L_L R_1 R_4 + C_1 C_4 C_L L_L R_4 r_o) + s^3 (2C_1 C_4 L_L R_1 g_m r_o + 2C_1 C_4 L_L R_1 + C_1 C_4 L_L R_4 + 2C_1 C_4 L_L r_o + C_1 C_L L_L R_1 g_m r_o + C_1 C_L L_L R_1 + C_1 C_L L_L r_o + C_4 C_L L_L R_4 g_m r_o + C_4 C_L L_L R_4) + s^2 (C_1 C_4 R_1 R_4 g_m r_o + C_1 C_4 R_1 R_4 + C_1 C_L R_1 g_m r_o + C_1 C_L R_1 + C_1 C_L R_L + C_1 C_L r_o + C_4 C_L R_4 g_m r_o + C_4 C_L R_4 + 2C_1 C_L R_L g_m r_o + 2C_1 C_L R_L + C_1 C_L r_o + L_L g_m r_o + L_L)}$$

$$\mathbf{9.90 \quad X-INVALID-ORDER-90} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m r_o + s^4 (C_1 C_4 C_L L_L R_1 R_4 g_m r_o + C_1 C_4 C_L L_L R_1 R_4) + s^3 (C_1 C_L L_L R_1 g_m r_o + C_1 C_L L_L R_1 + C_4 C_L L_L R_4 g_m r_o + C_4 C_L L_L R_4) + s^2 (C_1 C_4 R_1 R_4 g_m r_o + C_1 C_4 R_1 R_4 + C_L L_L g_m r_o + C_L L_L) + s (C_1 R_1 g_m r_o + C_1 R_1 + C_4 R_4 g_m r_o + C_4 R_4)}{s^4 (2C_1 C_4 C_L L_L R_1 g_m r_o + 2C_1 C_4 C_L L_L R_1 + C_1 C_4 C_L L_L R_4 + 2C_1 C_4 C_L L_L r_o) + s^3 (C_1 C_4 C_L R_1 R_4 g_m r_o + C_1 C_4 C_L R_1 R_4 + C_1 C_4 C_L R_4 r_o + C_1 C_L L_L + 2C_4 C_L L_L g_m r_o + 2C_4 C_L L_L) + s^2 (2C_1 C_4 R_1 g_m r_o + 2C_1 C_4 R_1 + C_1 C_4 R_4 + 2C_1 C_4 r_o + C_1 C_L R_1 g_m r_o + C_1 C_L R_1 + C_1 C_L R_L + C_1 C_L r_o + C_4 C_L R_4 g_m r_o + C_4 C_L R_4 + 2C_1 C_L R_L g_m r_o + 2C_1 C_L R_L + C_1 C_L r_o + L_L g_m r_o + L_L)}$$

$$\mathbf{9.91 \quad X-INVALID-ORDER-91} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad R_L \right)$$

$$H(s) = \frac{s^2 (C_1 L_4 R_1 R_L g_m r_o + C_1 L_4 R_1 R_L) + s (L_4 R_L g_m r_o + L_4 R_L)}{2R_L g_m r_o + 2R_L + s^3 (2C_1 C_4 L_4 R_1 R_L g_m r_o + 2C_1 C_4 L_4 R_1 R_L + 2C_1 C_4 L_4 R_L r_o) + s^2 (C_1 L_4 R_1 g_m r_o + C_1 L_4 R_1 + C_1 L_4 R_L + C_1 L_4 r_o + 2C_4 L_4 R_L g_m r_o + 2C_4 L_4 R_L) + s (2C_1 R_1 R_L g_m r_o + 2C_1 R_1 R_L + 2C_1 R_L r_o + L_4 g_m r_o + L_4)}$$

$$\mathbf{9.92 \quad X-INVALID-ORDER-92} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s^2 (C_1 L_4 R_1 g_m r_o + C_1 L_4 R_1) + s (L_4 g_m r_o + L_4)}{2g_m r_o + s^3 (2C_1 C_4 L_4 R_1 g_m r_o + 2C_1 C_4 L_4 R_1 + 2C_1 C_4 L_4 r_o + C_1 C_L L_4 R_1 g_m r_o + C_1 C_L L_4 R_1 + C_1 C_L L_4 r_o) + s^2 (C_1 L_4 + 2C_4 L_4 g_m r_o + 2C_4 L_4 + C_L L_4 g_m r_o + C_L L_4) + s (2C_1 R_1 g_m r_o + 2C_1 R_1 + 2C_1 r_o) + 2}$$

$$\mathbf{9.93 \quad X-INVALID-ORDER-93} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{s^2 (C_1 L_4 R_1 R_L g_m r_o + C_1 L_4 R_1 R_L) + s (L_4 R_L g_m r_o + L_4 R_L)}{2R_L g_m r_o + 2R_L + s^3 (2C_1 C_4 L_4 R_1 R_L g_m r_o + 2C_1 C_4 L_4 R_1 R_L + 2C_1 C_4 L_4 R_L r_o + C_1 C_L L_4 R_1 R_L g_m r_o + C_1 C_L L_4 R_1 R_L + C_1 C_L L_4 R_L r_o) + s^2 (C_1 L_4 R_1 g_m r_o + C_1 L_4 R_1 + C_1 L_4 R_L + C_1 L_4 r_o + 2C_4 L_4 R_L g_m r_o + 2C_4 L_4 R_L + C_L L_4 R_L g_m r_o + C_L L_4 R_L) + s (2C_1 R_1 R_L g_m r_o + 2C_1 R_1 R_L + 2C_1 R_L r_o + L_4 g_m r_o + L_4)}$$

$$\mathbf{9.94 \quad X-INVALID-ORDER-94} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s^3 (C_1 C_L L_4 R_1 R_L g_m r_o + C_1 C_L L_4 R_1 R_L) + s^2 (C_1 L_4 R_1 g_m r_o + C_1 L_4 R_1 + C_L L_4 R_L g_m r_o + C_L L_4 R_L) + s (L_4 g_m r_o + L_4)}{2g_m r_o + s^4 (2C_1 C_4 C_L L_4 R_1 R_L g_m r_o + 2C_1 C_4 C_L L_4 R_1 R_L + 2C_1 C_4 C_L L_4 R_L r_o) + s^3 (2C_1 C_4 L_4 R_1 g_m r_o + 2C_1 C_4 L_4 R_1 + 2C_1 C_4 L_4 r_o + C_1 C_L L_4 R_1 g_m r_o + C_1 C_L L_4 R_1 + C_1 C_L L_4 R_L + C_1 C_L L_4 r_o + 2C_4 C_L L_4 R_L g_m r_o + 2C_4 C_L L_4 R_L) + s^2 (2C_1 C_L R_1 R_L g_m r_o + 2C_1 C_L R_1 R_L + 2C_1 C_L R_L r_o + L_4 g_m r_o + L_4)}$$

$$\mathbf{9.95 \quad X-INVALID-ORDER-95} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s^2 (C_1 L_4 L_L R_1 g_m r_o + C_1 L_4 L_L R_1) + s (L_4 L_L g_m r_o + L_4 L_L)}{L_4 g_m r_o + L_4 + 2L_L g_m r_o + 2L_L + s^3 (2C_1 C_4 L_4 L_L R_1 g_m r_o + 2C_1 C_4 L_4 L_L R_1 + 2C_1 C_4 L_4 L_L r_o + C_1 C_L L_4 L_L R_1 g_m r_o + C_1 C_L L_4 L_L R_1 + C_1 C_L L_4 L_L r_o) + s^2 (C_1 L_4 L_L + 2C_4 L_4 L_L g_m r_o + 2C_4 L_4 L_L + C_L L_4 L_L g_m r_o + C_L L_4 L_L) + s (C_1 L_4 R_1 g_m r_o + C_1 L_4 R_1 + C_1 L_4 r_o + L_4 g_m r_o + L_4)}$$

$$\mathbf{9.96 \quad X-INVALID-ORDER-96} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s^4 (C_1 C_L L_4 L_L R_1 g_m r_o + C_1 C_L L_4 L_L R_1) + s^3 (C_L L_4 L_L g_m r_o + C_L L_4 L_L) + s^2 (C_1 L_4 R_1 g_m r_o + C_1 L_4 R_1) + s (L_4 g_m r_o + L_4)}{2g_m r_o + s^5 (2C_1 C_4 C_L L_4 L_L R_1 g_m r_o + 2C_1 C_4 C_L L_4 L_L R_1 + 2C_1 C_4 C_L L_4 L_L r_o) + s^4 (C_1 C_L L_4 L_L + 2C_4 C_L L_4 L_L g_m r_o + 2C_4 C_L L_4 L_L) + s^3 (2C_1 C_4 L_4 R_1 g_m r_o + 2C_1 C_4 L_4 R_1 + 2C_1 C_4 L_4 r_o + C_1 C_L L_4 R_1 g_m r_o + C_1 C_L L_4 R_1 + C_1 C_L L_4 r_o + 2C_1 C_L L_L R_1 g_m r_o + 2C_1 C_L L_L R_1 + 2C_1 C_L L_L r_o + L_4 g_m r_o + L_4)}$$

$$\mathbf{9.97 \quad X-INVALID-ORDER-97} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad R_L \right)$$

$$H(s) = \frac{R_L g_m r_o + R_L + s^3 (C_1 C_4 L_4 R_1 R_L g_m r_o + C_1 C_4 L_4 R_1 R_L) + s^2 (C_4 L_4 R_L g_m r_o + C_4 L_4 R_L) + s (C_1 R_1 R_L g_m r_o + C_1 R_1 R_L)}{g_m r_o + s^3 (C_1 C_4 L_4 R_1 g_m r_o + C_1 C_4 L_4 R_1 + C_1 C_4 L_4 R_L + C_1 C_4 L_4 r_o) + s^2 (2C_1 C_4 R_1 R_L g_m r_o + 2C_1 C_4 R_1 R_L + 2C_1 C_4 R_L r_o + C_4 L_4 g_m r_o + C_4 L_4) + s (C_1 R_1 g_m r_o + C_1 R_1 + C_1 R_L + C_1 r_o + 2C_4 R_L g_m r_o + 2C_4 R_L) + 1}$$

$$\mathbf{9.98 \quad X-INVALID-ORDER-98} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m r_o + s^3 (C_1 C_4 L_4 R_1 g_m r_o + C_1 C_4 L_4 R_1) + s^2 (C_4 L_4 g_m r_o + C_4 L_4) + s (C_1 R_1 g_m r_o + C_1 R_1) + 1}{s^4 (C_1 C_4 C_L L_4 R_1 g_m r_o + C_1 C_4 C_L L_4 R_1 + C_1 C_4 C_L L_4 r_o) + s^3 (C_1 C_4 L_4 + C_4 C_L L_4 g_m r_o + C_4 C_L L_4) + s^2 (2C_1 C_4 R_1 g_m r_o + 2C_1 C_4 R_1 + 2C_1 C_4 r_o + C_1 C_L R_1 g_m r_o + C_1 C_L R_1 + C_1 C_L r_o) + s (C_1 + 2C_4 g_m r_o + 2C_4 + C_L g_m r_o + C_L)}$$

$$\mathbf{9.99 \quad X-INVALID-ORDER-99} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{R_L g_m r_o + R_L + s^3 (C_1 C_4 L_4 R_1 R_L g_m r_o + C_1 C_4 L_4 R_1 R_L) + s^2 (C_4 L_4 R_L g_m r_o + C_4 L_4 R_L) + s (C_1 R_1 R_L g_m r_o + C_1 R_1 R_L)}{g_m r_o + s^4 (C_1 C_4 C_L L_4 R_1 R_L g_m r_o + C_1 C_4 C_L L_4 R_1 R_L + C_1 C_4 C_L L_4 R_L r_o) + s^3 (C_1 C_4 L_4 R_1 g_m r_o + C_1 C_4 L_4 R_1 + C_1 C_4 L_4 R_L + C_1 C_4 L_4 r_o + C_4 C_L L_4 R_L g_m r_o + C_4 C_L L_4 R_L) + s^2 (2C_1 C_4 R_1 R_L g_m r_o + 2C_1 C_4 R_1 R_L + 2C_1 C_4 R_L r_o + C_1 C_L R_1 R_L g_m r_o + C_1 C_L R_1 R_L + C_1 C_L R_L r_o) + s (C_1 + 2C_4 g_m r_o + 2C_4 + C_L g_m r_o + C_L)}$$

$$\mathbf{9.100 \quad X-INVALID-ORDER-100} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m r_o + s^4 (C_1 C_4 C_L L_4 R_1 R_L g_m r_o + C_1 C_4 C_L L_4 R_1 R_L) + s^3 (C_1 C_4 L_4 R_1 g_m r_o + C_1 C_4 L_4 R_1 + C_4 C_L L_4 R_L g_m r_o + C_4 C_L L_4 R_L) + s^2 (C_1 C_L R_1 R_L g_m r_o + C_1 C_L R_1 R_L + C_4 L_4 g_m r_o + C_4 L_4) + s (C_1 R_1 g_m r_o + C_1 R_1 + C_1 R_L + C_1 r_o + 2C_4 R_L g_m r_o + 2C_4 R_L) + 1}{s^4 (C_1 C_4 C_L L_4 R_1 g_m r_o + C_1 C_4 C_L L_4 R_1 + C_1 C_4 C_L L_4 R_L + C_1 C_4 C_L L_4 r_o) + s^3 (2C_1 C_4 C_L R_1 R_L g_m r_o + 2C_1 C_4 C_L R_1 R_L + 2C_1 C_4 C_L R_L r_o + C_1 C_4 L_4 + C_4 C_L L_4 g_m r_o + C_4 C_L L_4) + s^2 (2C_1 C_4 R_1 g_m r_o + 2C_1 C_4 R_1 + 2C_1 C_4 r_o + C_1 C_L R_1 g_m r_o + C_1 C_L R_1 + C_1 C_L R_L r_o) + s (C_1 + 2C_4 g_m r_o + 2C_4 + C_L g_m r_o + C_L)}$$

$$\mathbf{9.101 \quad X-INVALID-ORDER-101} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s^4 (C_1 C_4 L_4 L_L R_1 g_m r_o + C_1 C_4 L_4 L_L R_1) + s^3 (C_4 L_4 L_L g_m r_o + C_4 L_4 L_L) + s^2 (C_1 L_L R_1 g_m r_o + C_1 L_L R_1) + s (L_L g_m r_o + L_L)}{g_m r_o + s^5 (C_1 C_4 C_L L_4 L_L R_1 g_m r_o + C_1 C_4 C_L L_4 L_L R_1 + C_1 C_4 C_L L_4 L_L r_o) + s^4 (C_1 C_4 L_4 L_L + C_4 C_L L_4 L_L g_m r_o + C_4 C_L L_4 L_L) + s^3 (C_1 C_4 L_4 R_1 g_m r_o + C_1 C_4 L_4 R_1 + C_1 C_4 L_4 r_o + 2C_1 C_4 L_L R_1 g_m r_o + 2C_1 C_4 L_L R_1 + 2C_1 C_4 L_L r_o + C_1 C_L L_L R_1 g_m r_o + C_1 C_L L_L R_1 + C_1 C_L L_L r_o) + s^2 (2C_1 C_4 R_1 g_m r_o + 2C_1 C_4 R_1 + 2C_1 C_4 r_o + C_1 C_L R_1 g_m r_o + C_1 C_L R_1 + C_1 C_L R_L r_o) + s (C_1 + 2C_4 g_m r_o + 2C_4 + C_L g_m r_o + C_L)}$$

$$\mathbf{9.102 \quad X-INVALID-ORDER-102} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m r_o + s^5 (C_1 C_4 C_L L_4 L_L R_1 g_m r_o + C_1 C_4 C_L L_4 L_L R_1) + s^4 (C_4 C_L L_4 L_L g_m r_o + C_4 C_L L_4 L_L) + s^3 (C_1 C_4 L_4 R_1 g_m r_o + C_1 C_4 L_4 R_1 + C_1 C_L L_L R_1 g_m r_o + C_1 C_L L_L R_1) + s^2 (C_4 L_4 g_m r_o + C_4 L_4 + C_L L_L g_m r_o + C_L L_L) + s (C_1 R_1 g_m r_o + C_1 R_1 + C_1 R_L + C_1 r_o + 2C_4 R_L g_m r_o + 2C_4 R_L) + 1}{C_1 C_4 C_L L_4 L_L s^5 + s^4 (C_1 C_4 C_L L_4 R_1 g_m r_o + C_1 C_4 C_L L_4 R_1 + C_1 C_4 C_L L_4 r_o + 2C_1 C_4 C_L L_L R_1 g_m r_o + 2C_1 C_4 C_L L_L R_1 + 2C_1 C_4 C_L L_L r_o) + s^3 (C_1 C_4 L_4 + C_1 C_L L_L + C_4 C_L L_4 g_m r_o + C_4 C_L L_4 + 2C_4 C_L L_L g_m r_o + 2C_4 C_L L_L) + s^2 (2C_1 C_4 R_1 g_m r_o + 2C_1 C_4 R_1 + 2C_1 C_4 r_o + C_1 C_L R_1 g_m r_o + C_1 C_L R_1 + C_1 C_L R_L r_o) + s (C_1 + 2C_4 g_m r_o + 2C_4 + C_L g_m r_o + C_L)}$$

$$\mathbf{9.103 \quad X-INVALID-ORDER-103} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4, \quad \infty, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s (L_1 R_4 g_m r_o + L_1 R_4)}{C_1 C_L L_1 R_4 r_o s^3 + R_4 + 2r_o + s^2 (C_1 L_1 R_4 + 2C_1 L_1 r_o + C_L L_1 R_4 g_m r_o + C_L L_1 R_4) + s (C_L R_4 r_o + 2L_1 g_m r_o + 2L_1)}$$

$$\mathbf{9.104 \quad X-INVALID-ORDER-104} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4, \quad \infty, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{s (L_1 R_4 R_L g_m r_o + L_1 R_4 R_L)}{C_1 C_L L_1 R_4 R_L r_o s^3 + R_4 R_L + R_4 r_o + 2R_L r_o + s^2 (C_1 L_1 R_4 R_L + C_1 L_1 R_4 r_o + 2C_1 L_1 R_L r_o + C_L L_1 R_4 R_L g_m r_o + C_L L_1 R_4 R_L) + s (C_L R_4 R_L r_o + L_1 R_4 g_m r_o + L_1 R_4 + 2L_1 R_L g_m r_o + 2L_1 R_L)}$$

$$\mathbf{9.105 \quad X-INVALID-ORDER-105} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4, \quad \infty, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s^2 (C_L L_1 R_4 R_L g_m r_o + C_L L_1 R_4 R_L) + s (L_1 R_4 g_m r_o + L_1 R_4)}{R_4 + 2r_o + s^3 (C_1 C_L L_1 R_4 R_L + C_1 C_L L_1 R_4 r_o + 2C_1 C_L L_1 R_L r_o) + s^2 (C_1 L_1 R_4 + 2C_1 L_1 r_o + C_L L_1 R_4 g_m r_o + C_L L_1 R_4 + 2C_L L_1 R_L g_m r_o + 2C_L L_1 R_L) + s (C_L R_4 R_L + C_L R_4 r_o + 2C_L R_L r_o + 2L_1 g_m r_o + 2L_1)}$$

$$\mathbf{9.106 \quad X-INVALID-ORDER-106} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4, \quad \infty, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s^2 (L_1 L_L R_4 g_m r_o + L_1 L_L R_4)}{C_1 C_L L_1 L_L R_4 r_o s^4 + R_4 r_o + s^3 (C_1 L_1 L_L R_4 + 2C_1 L_1 L_L r_o + C_L L_1 L_L R_4 g_m r_o + C_L L_1 L_L R_4) + s^2 (C_1 L_1 R_4 r_o + C_L L_L R_4 r_o + 2L_1 L_L g_m r_o + 2L_1 L_L) + s (L_1 R_4 g_m r_o + L_1 R_4 + L_L R_4 + 2L_L r_o)}$$

$$\mathbf{9.107 \quad X-INVALID-ORDER-107} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4, \quad \infty, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s^3 (C_L L_1 L_L R_4 g_m r_o + C_L L_1 L_L R_4) + s (L_1 R_4 g_m r_o + L_1 R_4)}{R_4 + 2r_o + s^4 (C_1 C_L L_1 L_L R_4 + 2C_1 C_L L_1 L_L r_o) + s^3 (C_1 C_L L_1 R_4 r_o + 2C_L L_1 L_L g_m r_o + 2C_L L_1 L_L) + s^2 (C_1 L_1 R_4 + 2C_1 L_1 r_o + C_L L_1 R_4 g_m r_o + C_L L_1 R_4 + C_L L_L R_4 + 2C_L L_L r_o) + s (C_L R_4 r_o + 2L_1 g_m r_o + 2L_1)}$$

$$\mathbf{9.108 \quad X-INVALID-ORDER-108} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \infty, \quad R_L \right)$$

$$H(s) = \frac{s (L_1 R_L g_m r_o + L_1 R_L)}{2C_1 C_4 L_1 R_L r_o s^3 + R_L + r_o + s^2 (C_1 L_1 R_L + C_1 L_1 r_o + 2C_4 L_1 R_L g_m r_o + 2C_4 L_1 R_L) + s (2C_4 R_L r_o + L_1 g_m r_o + L_1)}$$

$$\mathbf{9.109 \quad X-INVALID-ORDER-109} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \infty, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s (L_1 g_m r_o + L_1)}{s^3 (2C_1 C_4 L_1 r_o + C_1 C_L L_1 r_o) + s^2 (C_1 L_1 + 2C_4 L_1 g_m r_o + 2C_4 L_1 + C_L L_1 g_m r_o + C_L L_1) + s (2C_4 r_o + C_L r_o) + 1}$$

$$\mathbf{9.110 \quad X-INVALID-ORDER-110} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \infty, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{s (L_1 R_L g_m r_o + L_1 R_L)}{R_L + r_o + s^3 (2C_1 C_4 L_1 R_L r_o + C_1 C_L L_1 R_L r_o) + s^2 (C_1 L_1 R_L + C_1 L_1 r_o + 2C_4 L_1 R_L g_m r_o + 2C_4 L_1 R_L + C_L L_1 R_L g_m r_o + C_L L_1 R_L) + s (2C_4 R_L r_o + C_L R_L r_o + L_1 g_m r_o + L_1)}$$

$$\mathbf{9.111 \quad X-INVALID-ORDER-111} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \infty, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s^2 (C_L L_1 R_L g_m r_o + C_L L_1 R_L) + s (L_1 g_m r_o + L_1)}{2C_1 C_4 C_L L_1 R_L r_o s^4 + s^3 (2C_1 C_4 L_1 r_o + C_1 C_L L_1 R_L + C_1 C_L L_1 r_o + 2C_4 C_L L_1 R_L g_m r_o + 2C_4 C_L L_1 R_L) + s^2 (C_1 L_1 + 2C_4 C_L R_L r_o + 2C_4 L_1 g_m r_o + 2C_4 L_1 + C_L L_1 g_m r_o + C_L L_1) + s (2C_4 r_o + C_L R_L + C_L r_o) + 1}$$

$$\mathbf{9.112 \quad X-INVALID-ORDER-112} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \infty, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s^2 (L_1 L_L g_m r_o + L_1 L_L)}{r_o + s^4 (2C_1 C_4 L_1 L_L r_o + C_1 C_L L_1 L_L r_o) + s^3 (C_1 L_1 L_L + 2C_4 L_1 L_L g_m r_o + 2C_4 L_1 L_L + C_L L_1 L_L g_m r_o + C_L L_1 L_L) + s^2 (C_1 L_1 r_o + 2C_4 L_L r_o + C_L L_L r_o) + s (L_1 g_m r_o + L_1 + L_L)}$$

$$\mathbf{9.113 \quad X-INVALID-ORDER-113} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \infty, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s^3 (C_L L_1 L_L g_m r_o + C_L L_1 L_L) + s (L_1 g_m r_o + L_1)}{2C_1 C_4 C_L L_1 L_L r_o s^5 + s^4 (C_1 C_L L_1 L_L + 2C_4 C_L L_1 L_L g_m r_o + 2C_4 C_L L_1 L_L) + s^3 (2C_1 C_4 L_1 r_o + C_1 C_L L_1 r_o + 2C_4 C_L L_L r_o) + s^2 (C_1 L_1 + 2C_4 L_1 g_m r_o + 2C_4 L_1 + C_L L_1 g_m r_o + C_L L_1 + C_L L_L) + s (2C_4 r_o + C_L r_o) + 1}$$

$$\mathbf{9.114 \quad X-INVALID-ORDER-114} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad R_L \right)$$

$$H(s) = \frac{s (L_1 R_4 R_L g_m r_o + L_1 R_4 R_L)}{2C_1 C_4 L_1 R_4 R_L r_o s^3 + R_4 R_L + R_4 r_o + 2R_L r_o + s^2 (C_1 L_1 R_4 R_L + C_1 L_1 R_4 r_o + 2C_1 L_1 R_L r_o + 2C_4 L_1 R_4 R_L g_m r_o + 2C_4 L_1 R_4 R_L) + s (2C_4 R_4 R_L r_o + L_1 R_4 g_m r_o + L_1 R_4 + 2L_1 R_L g_m r_o + 2L_1 R_L)}$$

$$\mathbf{9.115 \quad X-INVALID-ORDER-115} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s (L_1 R_4 g_m r_o + L_1 R_4)}{R_4 + 2r_o + s^3 (2C_1 C_4 L_1 R_4 r_o + C_1 C_L L_1 R_4 r_o) + s^2 (C_1 L_1 R_4 + 2C_1 L_1 r_o + 2C_4 L_1 R_4 g_m r_o + 2C_4 L_1 R_4 + C_L L_1 R_4 g_m r_o + C_L L_1 R_4) + s (2C_4 R_4 r_o + C_L R_4 r_o + 2L_1 g_m r_o + 2L_1)}$$

$$\mathbf{9.116 \quad X-INVALID-ORDER-116} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{s (L_1 R_4 R_L g_m r_o + L_1 R_4 R_L)}{R_4 R_L + R_4 r_o + 2R_L r_o + s^3 (2C_1 C_4 L_1 R_4 R_L r_o + C_1 C_L L_1 R_4 R_L r_o) + s^2 (C_1 L_1 R_4 R_L + C_1 L_1 R_4 r_o + 2C_1 L_1 R_L r_o + 2C_4 L_1 R_4 R_L g_m r_o + 2C_4 L_1 R_4 R_L + C_L L_1 R_4 R_L g_m r_o + C_L L_1 R_4 R_L) + s (2C_4 R_4 R_L r_o + C_L R_4 R_L r_o + L_1 R_4 g_m r_o + L_1 R_4 + 2L_1 R_L g_m r_o + 2L_1 R_L)}$$

$$\mathbf{9.117 \quad X-INVALID-ORDER-117} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s^2 (C_L L_1 R_4 R_L g_m r_o + C_L L_1 R_4 R_L) + s (L_1 R_4 g_m r_o + L_1 R_4)}{2C_1 C_4 C_L L_1 R_4 R_L r_o s^4 + R_4 + 2r_o + s^3 (2C_1 C_4 L_1 R_4 r_o + C_1 C_L L_1 R_4 R_L + C_1 C_L L_1 R_4 r_o + 2C_1 C_L L_1 R_L r_o + 2C_4 C_L L_1 R_4 R_L g_m r_o + 2C_4 C_L L_1 R_4 R_L) + s^2 (C_1 L_1 R_4 + 2C_1 L_1 r_o + 2C_4 C_L R_4 R_L r_o + 2C_4 L_1 R_4 g_m r_o + 2C_4 L_1 R_4 + C_L L_1 R_4 g_m r_o + C_L L_1 R_4 + 2C_L L_1 R_L r_o)}$$

$$\mathbf{9.118 \quad X-INVALID-ORDER-118} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s^2 (L_1 L_L R_4 g_m r_o + L_1 L_L R_4)}{R_4 r_o + s^4 (2C_1 C_4 L_1 L_L R_4 r_o + C_1 C_L L_1 L_L R_4 r_o) + s^3 (C_1 L_1 L_L R_4 + 2C_1 L_1 L_L r_o + 2C_4 L_1 L_L R_4 g_m r_o + 2C_4 L_1 L_L R_4 + C_L L_1 L_L R_4 g_m r_o + C_L L_1 L_L R_4) + s^2 (C_1 L_1 R_4 r_o + 2C_4 L_L R_4 r_o + C_L L_L R_4 r_o + 2L_1 L_L g_m r_o + 2L_1 L_L) + s (L_1 R_4 g_m r_o + L_1 R_4 + L_L R_4 + 2L_L r_o)}$$

$$\mathbf{9.119 \quad X-INVALID-ORDER-119} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s^3 (C_L L_1 L_L R_4 g_m r_o + C_L L_1 L_L R_4) + s (L_1 R_4 g_m r_o + L_1 R_4)}{2C_1 C_4 C_L L_1 L_L R_4 r_o s^5 + R_4 + 2r_o + s^4 (C_1 C_L L_1 L_L R_4 + 2C_1 C_L L_1 L_L r_o + 2C_4 C_L L_1 L_L R_4 g_m r_o + 2C_4 C_L L_1 L_L R_4) + s^3 (2C_1 C_4 L_1 R_4 r_o + C_1 C_L L_1 R_4 r_o + 2C_4 C_L L_L R_4 r_o + 2C_L L_1 L_L g_m r_o + 2C_L L_1 L_L) + s^2 (C_1 L_1 R_4 + 2C_1 L_1 r_o + 2C_4 L_1 R_4 g_m r_o + 2C_4 L_1 R_4 + C_L L_1 R_4 g_m r_o + C_L L_1 R_4 + 2C_L L_1 R_L r_o)}$$

$$\mathbf{9.120 \quad X-INVALID-ORDER-120} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad R_L \right)$$

$$H(s) = \frac{s^2 (C_4 L_1 R_4 R_L g_m r_o + C_4 L_1 R_4 R_L) + s (L_1 R_L g_m r_o + L_1 R_L)}{R_L + r_o + s^3 (C_1 C_4 L_1 R_4 R_L + C_1 C_4 L_1 R_4 r_o + 2C_1 C_4 L_1 R_L r_o) + s^2 (C_1 L_1 R_L + C_1 L_1 r_o + C_4 L_1 R_4 g_m r_o + C_4 L_1 R_4 + 2C_4 L_1 R_L g_m r_o + 2C_4 L_1 R_L) + s (C_4 R_4 R_L + C_4 R_4 r_o + 2C_4 R_L r_o + L_1 g_m r_o + L_1)}$$

$$\mathbf{9.121 \quad X-INVALID-ORDER-121} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s^2 (C_4 L_1 R_4 g_m r_o + C_4 L_1 R_4) + s (L_1 g_m r_o + L_1)}{C_1 C_4 C_L L_1 R_4 r_o s^4 + s^3 (C_1 C_4 L_1 R_4 + 2C_1 C_4 L_1 r_o + C_1 C_L L_1 r_o + C_4 C_L L_1 R_4 g_m r_o + C_4 C_L L_1 R_4) + s^2 (C_1 L_1 + C_4 C_L R_4 r_o + 2C_4 L_1 g_m r_o + 2C_4 L_1 + C_L L_1 g_m r_o + C_L L_1) + s (C_4 R_4 + 2C_4 r_o + C_L r_o) + 1}$$

$$\mathbf{9.122 \quad X-INVALID-ORDER-122} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{s^2 (C_4 L_1 R_4 R_L g_m r_o + C_4 L_1 R_4 R_L) + s (L_1 R_L g_m r_o + L_1 R_L)}{C_1 C_4 C_L L_1 R_4 R_L r_o s^4 + R_L + r_o + s^3 (C_1 C_4 L_1 R_4 R_L + C_1 C_4 L_1 R_4 r_o + 2C_1 C_4 L_1 R_L r_o + C_1 C_L L_1 R_L r_o + C_4 C_L L_1 R_4 R_L g_m r_o + C_4 C_L L_1 R_4 R_L) + s^2 (C_1 L_1 R_L + C_1 L_1 r_o + C_4 C_L R_4 R_L r_o + C_4 L_1 R_4 g_m r_o + C_4 L_1 R_4 + 2C_4 L_1 R_L g_m r_o + 2C_4 L_1 R_L + C_L L_1 R_L g_m r_o + C_L L_1 R_L r_o)}$$

$$\mathbf{9.123 \quad X-INVALID-ORDER-123} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s^3 (C_4 C_L L_1 R_4 R_L g_m r_o + C_4 C_L L_1 R_4 R_L) + s^2 (C_4 L_1 R_4 g_m r_o + C_4 L_1 R_4 + C_L L_1 R_L g_m r_o + C_L L_1 R_L) + s (L_1 g_m r_o + L_1)}{s^4 (C_1 C_4 C_L L_1 R_4 R_L + C_1 C_4 C_L L_1 R_4 r_o + 2C_1 C_4 C_L L_1 R_L r_o) + s^3 (C_1 C_4 L_1 R_4 + 2C_1 C_4 L_1 r_o + C_1 C_L L_1 R_L + C_1 C_L L_1 r_o + C_4 C_L L_1 R_4 g_m r_o + C_4 C_L L_1 R_4 + 2C_4 C_L L_1 R_L g_m r_o + 2C_4 C_L L_1 R_L) + s^2 (C_1 L_1 + C_4 C_L R_4 R_L + C_4 C_L R_4 r_o + 2C_4 C_L R_L r_o + 2C_4 L_1 g_m r_o + C_L L_1 g_m r_o + C_L L_1 r_o)}$$

$$\mathbf{9.124 \quad X-INVALID-ORDER-124} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s^3 (C_4 L_1 L_L R_4 g_m r_o + C_4 L_1 L_L R_4) + s^2 (L_1 L_L g_m r_o + L_1 L_L)}{C_1 C_4 C_L L_1 L_L R_4 r_o s^5 + r_o + s^4 (C_1 C_4 L_1 L_L R_4 + 2C_1 C_4 L_1 L_L r_o + C_1 C_L L_1 L_L r_o + C_4 C_L L_1 L_L R_4 g_m r_o + C_4 C_L L_1 L_L R_4) + s^3 (C_1 C_4 L_1 R_4 r_o + C_1 L_1 L_L + C_4 C_L L_L R_4 r_o + 2C_4 L_1 L_L g_m r_o + 2C_4 L_1 L_L + C_L L_1 L_L g_m r_o + C_L L_1 L_L) + s^2 (C_1 L_1 r_o + C_4 L_1 R_4 g_m r_o + C_4 L_1 R_4 r_o)}$$

$$\mathbf{9.125 \quad X-INVALID-ORDER-125} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s^4 (C_4 C_L L_1 L_L R_4 g_m r_o + C_4 C_L L_1 L_L R_4) + s^3 (C_L L_1 L_L g_m r_o + C_L L_1 L_L) + s^2 (C_4 L_1 R_4 g_m r_o + C_4 L_1 R_4) + s (L_1 g_m r_o + L_1)}{s^5 (C_1 C_4 C_L L_1 L_L R_4 + 2C_1 C_4 C_L L_1 L_L r_o) + s^4 (C_1 C_4 C_L L_1 R_4 r_o + C_1 C_L L_1 L_L + 2C_4 C_L L_1 L_L g_m r_o + 2C_4 C_L L_1 L_L) + s^3 (C_1 C_4 L_1 R_4 + 2C_1 C_4 L_1 r_o + C_1 C_L L_1 r_o + C_4 C_L L_1 R_4 g_m r_o + C_4 C_L L_1 R_4 + C_4 C_L L_L R_4 + 2C_4 C_L L_L r_o) + s^2 (C_1 L_1 + C_4 C_L R_4 r_o + 2C_4 L_1 g_m r_o + C_L L_1 g_m r_o + C_L L_1 r_o)}$$

$$\mathbf{9.126 \quad X-INVALID-ORDER-126} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad R_L \right)$$

$$H(s) = \frac{s^2 (L_1 L_4 R_L g_m r_o + L_1 L_4 R_L)}{2C_1 C_4 L_1 L_4 R_L r_o s^4 + 2R_L r_o + s^3 (C_1 L_1 L_4 R_L + C_1 L_1 L_4 r_o + 2C_4 L_1 L_4 R_L g_m r_o + 2C_4 L_1 L_4 R_L) + s^2 (2C_1 L_1 R_L r_o + 2C_4 L_4 R_L r_o + L_1 L_4 g_m r_o + L_1 L_4) + s (2L_1 R_L g_m r_o + 2L_1 R_L + L_4 R_L + L_4 r_o)}$$

$$\mathbf{9.127 \quad X-INVALID-ORDER-127} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s^2 (L_1 L_4 g_m r_o + L_1 L_4)}{2r_o + s^4 (2C_1 C_4 L_1 L_4 r_o + C_1 C_L L_1 L_4 r_o) + s^3 (C_1 L_1 L_4 + 2C_4 L_1 L_4 g_m r_o + 2C_4 L_1 L_4 + C_L L_1 L_4 g_m r_o + C_L L_1 L_4) + s^2 (2C_1 L_1 r_o + 2C_4 L_4 r_o + C_L L_4 r_o) + s (2L_1 g_m r_o + 2L_1 + L_4)}$$

$$\mathbf{9.128 \quad X-INVALID-ORDER-128} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{s^2 (L_1 L_4 R_L g_m r_o + L_1 L_4 R_L)}{2R_L r_o + s^4 (2C_1 C_4 L_1 L_4 R_L r_o + C_1 C_L L_1 L_4 R_L r_o) + s^3 (C_1 L_1 L_4 R_L + C_1 L_1 L_4 r_o + 2C_4 L_1 L_4 R_L g_m r_o + 2C_4 L_1 L_4 R_L + C_L L_1 L_4 R_L g_m r_o + C_L L_1 L_4 R_L) + s^2 (2C_1 L_1 R_L r_o + 2C_4 L_4 R_L r_o + C_L L_4 R_L r_o + L_1 L_4 g_m r_o + L_1 L_4) + s (2L_1 R_L g_m r_o + 2L_1 R_L + L_4 R_L + L_4 r_o)}$$

$$\mathbf{9.129 \quad X-INVALID-ORDER-129} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s^3 (C_L L_1 L_4 R_L g_m r_o + C_L L_1 L_4 R_L) + s^2 (L_1 L_4 g_m r_o + L_1 L_4)}{2C_1 C_4 C_L L_1 L_4 R_L r_o s^5 + 2r_o + s^4 (2C_1 C_4 L_1 L_4 r_o + C_1 C_L L_1 L_4 R_L + C_1 C_L L_1 L_4 r_o + 2C_4 C_L L_1 L_4 R_L g_m r_o + 2C_4 C_L L_1 L_4 R_L) + s^3 (2C_1 C_L L_1 R_L r_o + C_1 L_1 L_4 + 2C_4 C_L L_4 R_L r_o + 2C_4 L_1 L_4 g_m r_o + 2C_4 L_1 L_4 + C_L L_1 L_4 g_m r_o + C_L L_1 L_4) + s^2 (2C_1 L_1 r_o + 2C_4 L_4 r_o + 2C_L L_4 r_o)}$$

$$\mathbf{9.130 \quad X-INVALID-ORDER-130} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s^2 (L_1 L_4 L_L g_m r_o + L_1 L_4 L_L)}{L_4 r_o + 2L_L r_o + s^4 (2C_1 C_4 L_1 L_4 L_L r_o + C_1 C_L L_1 L_4 L_L r_o) + s^3 (C_1 L_1 L_4 L_L + 2C_4 L_1 L_4 L_L g_m r_o + 2C_4 L_1 L_4 L_L + C_L L_1 L_4 L_L g_m r_o + C_L L_1 L_4 L_L) + s^2 (C_1 L_1 L_4 r_o + 2C_1 L_1 L_L r_o + 2C_4 L_4 L_L r_o + C_L L_4 L_L r_o) + s (L_1 L_4 g_m r_o + L_1 L_4 + 2L_1 L_L g_m r_o + 2L_1 L_L + L_4 L_L)}$$

$$\mathbf{9.131 \quad X-INVALID-ORDER-131} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \infty, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s^4 (C_L L_1 L_4 L_L g_m r_o + C_L L_1 L_4 L_L) + s^2 (L_1 L_4 g_m r_o + L_1 L_4)}{2C_1 C_4 C_L L_1 L_4 L_L r_o s^6 + 2r_o + s^5 (C_1 C_L L_1 L_4 L_L + 2C_4 C_L L_1 L_4 L_L g_m r_o + 2C_4 C_L L_1 L_4 L_L) + s^4 (2C_1 C_4 L_1 L_4 r_o + C_1 C_L L_1 L_4 r_o + 2C_1 C_L L_1 L_L r_o + 2C_4 C_L L_4 L_L r_o) + s^3 (C_1 L_1 L_4 + 2C_4 L_1 L_4 g_m r_o + 2C_4 L_1 L_4 + C_L L_1 L_4 g_m r_o + C_L L_1 L_4 + 2C_L L_1 L_L g_m r_o + 2C_L L_1 L_L)}$$

$$\mathbf{9.132 \quad X-INVALID-ORDER-132} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad R_L \right)$$

$$H(s) = \frac{s^3 (C_4 L_1 L_4 R_L g_m r_o + C_4 L_1 L_4 R_L) + s (L_1 R_L g_m r_o + L_1 R_L)}{R_L + r_o + s^4 (C_1 C_4 L_1 L_4 R_L + C_1 C_4 L_1 L_4 r_o) + s^3 (2C_1 C_4 L_1 R_L r_o + C_4 L_1 L_4 g_m r_o + C_4 L_1 L_4) + s^2 (C_1 L_1 R_L + C_1 L_1 r_o + 2C_4 L_1 R_L g_m r_o + 2C_4 L_1 R_L + C_4 L_4 R_L + C_4 L_4 r_o) + s (2C_4 R_L r_o + L_1 g_m r_o + L_1)}$$

$$\mathbf{9.133 \quad X-INVALID-ORDER-133} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s^3 (C_4 L_1 L_4 g_m r_o + C_4 L_1 L_4) + s (L_1 g_m r_o + L_1)}{C_1 C_4 C_L L_1 L_4 r_o s^5 + s^4 (C_1 C_4 L_1 L_4 + C_4 C_L L_1 L_4 g_m r_o + C_4 C_L L_1 L_4) + s^3 (2C_1 C_4 L_1 r_o + C_1 C_L L_1 r_o + C_4 C_L L_4 r_o) + s^2 (C_1 L_1 + 2C_4 L_1 g_m r_o + 2C_4 L_1 + C_4 L_4 + C_L L_1 g_m r_o + C_L L_1) + s (2C_4 r_o + C_L r_o) + 1}$$

$$\mathbf{9.134 \quad X-INVALID-ORDER-134} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{s^3 (C_4 L_1 L_4 R_L g_m r_o + C_4 L_1 L_4 R_L) + s (L_1 R_L g_m r_o + L_1 R_L)}{C_1 C_4 C_L L_1 L_4 R_L r_o s^5 + R_L + r_o + s^4 (C_1 C_4 L_1 L_4 R_L + C_1 C_4 L_1 L_4 r_o + C_4 C_L L_1 L_4 R_L g_m r_o + C_4 C_L L_1 L_4 R_L) + s^3 (2C_1 C_4 L_1 R_L r_o + C_1 C_L L_1 R_L r_o + C_4 C_L L_4 R_L r_o + C_4 L_1 L_4 g_m r_o + C_4 L_1 L_4) + s^2 (C_1 L_1 R_L + C_1 L_1 r_o + 2C_4 L_1 R_L g_m r_o + 2C_4 L_1 R_L + C_4 L_4 R_L + C_4 L_4 r_o)}$$

$$\mathbf{9.135 \quad X-INVALID-ORDER-135} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s^4 (C_4 C_L L_1 L_4 R_L g_m r_o + C_4 C_L L_1 L_4 R_L) + s^3 (C_4 L_1 L_4 g_m r_o + C_4 L_1 L_4) + s^2 (C_L L_1 R_L g_m r_o + C_L L_1 R_L) + s (L_1 g_m r_o + L_1)}{s^5 (C_1 C_4 C_L L_1 L_4 R_L + C_1 C_4 C_L L_1 L_4 r_o) + s^4 (2C_1 C_4 C_L L_1 R_L r_o + C_1 C_4 L_1 L_4 + C_4 C_L L_1 L_4 g_m r_o + C_4 C_L L_1 L_4) + s^3 (2C_1 C_4 L_1 r_o + C_1 C_L L_1 R_L + C_1 C_L L_1 r_o + 2C_4 C_L L_1 R_L g_m r_o + 2C_4 C_L L_1 R_L + C_4 C_L L_4 R_L + C_4 C_L L_4 r_o) + s^2 (C_1 L_1 + 2C_4 C_L R_L r_o + 2C_4 L_1 g_m r_o)}$$

$$\mathbf{9.136 \quad X-INVALID-ORDER-136} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s^4 (C_4 L_1 L_4 L_L g_m r_o + C_4 L_1 L_4 L_L) + s^2 (L_1 L_L g_m r_o + L_1 L_L)}{C_1 C_4 C_L L_1 L_4 L_L r_o s^6 + r_o + s^5 (C_1 C_4 L_1 L_4 L_L + C_4 C_L L_1 L_4 L_L g_m r_o + C_4 C_L L_1 L_4 L_L) + s^4 (C_1 C_4 L_1 L_4 r_o + 2C_1 C_4 L_1 L_L r_o + C_1 C_L L_1 L_L r_o + C_4 C_L L_4 L_L r_o) + s^3 (C_1 L_1 L_L + C_4 L_1 L_4 g_m r_o + C_4 L_1 L_4 + 2C_4 L_1 L_L g_m r_o + 2C_4 L_1 L_L + C_4 L_4 L_L + C_L L_1 L_L g_m r_o + C_L L_1 L_L)}$$

$$\mathbf{9.137 \quad X-INVALID-ORDER-137} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \infty, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s^5 (C_4 C_L L_1 L_4 L_L g_m r_o + C_4 C_L L_1 L_4 L_L) + s^3 (C_4 L_1 L_4 g_m r_o + C_4 L_1 L_4 + C_L L_1 L_L g_m r_o + C_L L_1 L_L) + s (L_1 g_m r_o + L_1)}{C_1 C_4 C_L L_1 L_4 L_L s^6 + s^5 (C_1 C_4 C_L L_1 L_4 r_o + 2 C_1 C_4 C_L L_1 L_L r_o) + s^4 (C_1 C_4 L_1 L_4 + C_1 C_L L_1 L_L + C_4 C_L L_1 L_4 g_m r_o + C_4 C_L L_1 L_4 + 2 C_4 C_L L_1 L_L g_m r_o + 2 C_4 C_L L_1 L_L + C_4 C_L L_4 L_L) + s^3 (2 C_1 C_4 L_1 r_o + C_1 C_L L_1 r_o + C_4 C_L L_4 r_o + 2 C_4 C_L L_L r_o) + s^2 (C_1 L_1 + 2 C_4 L_1 g_m r_o)}$$

$$\mathbf{9.138 \quad X-INVALID-ORDER-138} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \infty, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_4 g_m r_o + R_4 + s^2 (C_1 L_1 R_4 g_m r_o + C_1 L_1 R_4)}{2 g_m r_o + s^3 (C_1 C_L L_1 R_4 g_m r_o + C_1 C_L L_1 R_4) + s^2 (C_1 C_L R_4 r_o + 2 C_1 L_1 g_m r_o + 2 C_1 L_1) + s (C_1 R_4 + 2 C_1 r_o + C_L R_4 g_m r_o + C_L R_4) + 2}$$

$$\mathbf{9.139 \quad X-INVALID-ORDER-139} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \infty, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{R_4 R_L g_m r_o + R_4 R_L + s^2 (C_1 L_1 R_4 R_L g_m r_o + C_1 L_1 R_4 R_L)}{R_4 g_m r_o + R_4 + 2 R_L g_m r_o + 2 R_L + s^3 (C_1 C_L L_1 R_4 R_L g_m r_o + C_1 C_L L_1 R_4 R_L) + s^2 (C_1 C_L R_4 R_L r_o + C_1 L_1 R_4 g_m r_o + C_1 L_1 R_4 + 2 C_1 L_1 R_L g_m r_o + 2 C_1 L_1 R_L) + s (C_1 R_4 R_L + C_1 R_4 r_o + 2 C_1 R_L r_o + C_L R_4 R_L g_m r_o + C_L R_4 R_L)}$$

$$\mathbf{9.140 \quad X-INVALID-ORDER-140} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \infty, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_4 g_m r_o + R_4 + s^3 (C_1 C_L L_1 R_4 R_L g_m r_o + C_1 C_L L_1 R_4 R_L) + s^2 (C_1 L_1 R_4 g_m r_o + C_1 L_1 R_4) + s (C_L R_4 R_L g_m r_o + C_L R_4 R_L)}{2 g_m r_o + s^3 (C_1 C_L L_1 R_4 g_m r_o + C_1 C_L L_1 R_4 + 2 C_1 C_L L_1 R_L g_m r_o + 2 C_1 C_L L_1 R_L) + s^2 (C_1 C_L R_4 R_L + C_1 C_L R_4 r_o + 2 C_1 C_L R_L r_o + 2 C_1 L_1 g_m r_o + 2 C_1 L_1) + s (C_1 R_4 + 2 C_1 r_o + C_L R_4 g_m r_o + C_L R_4 + 2 C_L R_L g_m r_o + 2 C_L R_L) + 2}$$

$$\mathbf{9.141 \quad X-INVALID-ORDER-141} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \infty, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s^3 (C_1 L_1 L_L R_4 g_m r_o + C_1 L_1 L_L R_4) + s (L_L R_4 g_m r_o + L_L R_4)}{R_4 g_m r_o + R_4 + s^4 (C_1 C_L L_1 L_L R_4 g_m r_o + C_1 C_L L_1 L_L R_4) + s^3 (C_1 C_L L_L R_4 r_o + 2 C_1 L_1 L_L g_m r_o + 2 C_1 L_1 L_L) + s^2 (C_1 L_1 R_4 g_m r_o + C_1 L_1 R_4 + C_1 L_L R_4 + 2 C_1 L_L r_o + C_L L_L R_4 g_m r_o + C_L L_L R_4) + s (C_1 R_4 r_o + 2 L_L g_m r_o + 2 L_L)}$$

$$\mathbf{9.142 \quad X-INVALID-ORDER-142} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \infty, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_4 g_m r_o + R_4 + s^4 (C_1 C_L L_1 L_L R_4 g_m r_o + C_1 C_L L_1 L_L R_4) + s^2 (C_1 L_1 R_4 g_m r_o + C_1 L_1 R_4 + C_L L_L R_4 g_m r_o + C_L L_L R_4)}{2 g_m r_o + s^4 (2 C_1 C_L L_1 L_L g_m r_o + 2 C_1 C_L L_1 L_L) + s^3 (C_1 C_L L_1 R_4 g_m r_o + C_1 C_L L_1 R_4 + C_1 C_L L_L R_4 + 2 C_1 C_L L_L r_o) + s^2 (C_1 C_L R_4 r_o + 2 C_1 L_1 g_m r_o + 2 C_1 L_1 + 2 C_L L_L g_m r_o + 2 C_L L_L) + s (C_1 R_4 + 2 C_1 r_o + C_L R_4 g_m r_o + C_L R_4) + 2}$$

$$\mathbf{9.143 \quad X-INVALID-ORDER-143} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \infty, \quad R_L \right)$$

$$H(s) = \frac{R_L g_m r_o + R_L + s^2 (C_1 L_1 R_L g_m r_o + C_1 L_1 R_L)}{g_m r_o + s^3 (2 C_1 C_4 L_1 R_L g_m r_o + 2 C_1 C_4 L_1 R_L) + s^2 (2 C_1 C_4 R_L r_o + C_1 L_1 g_m r_o + C_1 L_1) + s (C_1 R_L + C_1 r_o + 2 C_4 R_L g_m r_o + 2 C_4 R_L) + 1}$$

$$\mathbf{9.144 \quad X-INVALID-ORDER-144} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \infty, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m r_o + s^2 (C_1 L_1 g_m r_o + C_1 L_1) + 1}{s^3 (2 C_1 C_4 L_1 g_m r_o + 2 C_1 C_4 L_1 + C_1 C_L L_1 g_m r_o + C_1 C_L L_1) + s^2 (2 C_1 C_4 r_o + C_1 C_L r_o) + s (C_1 + 2 C_4 g_m r_o + 2 C_4 + C_L g_m r_o + C_L)}$$

$$\mathbf{9.145 \quad X-INVALID-ORDER-145} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \infty, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{R_L g_m r_o + R_L + s^2 (C_1 L_1 R_L g_m r_o + C_1 L_1 R_L)}{g_m r_o + s^3 (2 C_1 C_4 L_1 R_L g_m r_o + 2 C_1 C_4 L_1 R_L + C_1 C_L L_1 R_L g_m r_o + C_1 C_L L_1 R_L) + s^2 (2 C_1 C_4 R_L r_o + C_1 C_L R_L r_o + C_1 L_1 g_m r_o + C_1 L_1) + s (C_1 R_L + C_1 r_o + 2 C_4 R_L g_m r_o + 2 C_4 R_L + C_L R_L g_m r_o + C_L R_L) + 1}$$

$$\mathbf{9.146 \quad X-INVALID-ORDER-146} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \infty, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m r_o + s^3 (C_1 C_L L_1 R_L g_m r_o + C_1 C_L L_1 R_L) + s^2 (C_1 L_1 g_m r_o + C_1 L_1) + s (C_L R_L g_m r_o + C_L R_L) + 1}{s^4 (2 C_1 C_4 C_L L_1 R_L g_m r_o + 2 C_1 C_4 C_L L_1 R_L) + s^3 (2 C_1 C_4 C_L R_L r_o + 2 C_1 C_4 L_1 g_m r_o + 2 C_1 C_4 L_1 + C_1 C_L L_1 g_m r_o + C_1 C_L L_1) + s^2 (2 C_1 C_4 r_o + C_1 C_L R_L + C_1 C_L r_o + 2 C_4 C_L R_L g_m r_o + 2 C_4 C_L R_L) + s (C_1 + 2 C_4 g_m r_o + 2 C_4 + C_L g_m r_o + C_L)}$$

$$\mathbf{9.147 \quad X-INVALID-ORDER-147} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \infty, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s^3 (C_1 L_1 L_L g_m r_o + C_1 L_1 L_L) + s (L_L g_m r_o + L_L)}{C_1 r_o s + g_m r_o + s^4 (2C_1 C_4 L_1 L_L g_m r_o + 2C_1 C_4 L_1 L_L + C_1 C_L L_1 L_L g_m r_o + C_1 C_L L_1 L_L) + s^3 (2C_1 C_4 L_L r_o + C_1 C_L L_L r_o) + s^2 (C_1 L_1 g_m r_o + C_1 L_1 + C_1 L_L + 2C_4 L_L g_m r_o + 2C_4 L_L + C_L L_L g_m r_o + C_L L_L) + 1}$$

$$\mathbf{9.148 \quad X-INVALID-ORDER-148} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \infty, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m r_o + s^4 (C_1 C_L L_1 L_L g_m r_o + C_1 C_L L_1 L_L) + s^2 (C_1 L_1 g_m r_o + C_1 L_1 + C_L L_L g_m r_o + C_L L_L) + 1}{2C_1 C_4 C_L L_L r_o s^4 + s^5 (2C_1 C_4 C_L L_1 L_L g_m r_o + 2C_1 C_4 C_L L_1 L_L) + s^3 (2C_1 C_4 L_1 g_m r_o + 2C_1 C_4 L_1 + C_1 C_L L_1 g_m r_o + C_1 C_L L_1 + C_1 C_L L_L + 2C_4 C_L L_L g_m r_o + 2C_4 C_L L_L) + s^2 (2C_1 C_4 r_o + C_1 C_L r_o) + s (C_1 + 2C_4 g_m r_o + 2C_4 + C_L g_m r_o + C_L)}$$

$$\mathbf{9.149 \quad X-INVALID-ORDER-149} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad R_L \right)$$

$$H(s) = \frac{R_4 R_L g_m r_o + R_4 R_L + s^2 (C_1 L_1 R_4 R_L g_m r_o + C_1 L_1 R_4 R_L)}{R_4 g_m r_o + R_4 + 2R_L g_m r_o + 2R_L + s^3 (2C_1 C_4 L_1 R_4 R_L g_m r_o + 2C_1 C_4 L_1 R_4 R_L) + s^2 (2C_1 C_4 R_4 R_L r_o + C_1 L_1 R_4 g_m r_o + C_1 L_1 R_4 + 2C_1 L_1 R_L g_m r_o + 2C_1 L_1 R_L) + s (C_1 R_4 R_L + C_1 R_4 r_o + 2C_1 R_L r_o + 2C_4 R_4 R_L g_m r_o + 2C_4 R_4 R_L)}$$

$$\mathbf{9.150 \quad X-INVALID-ORDER-150} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_4 g_m r_o + R_4 + s^2 (C_1 L_1 R_4 g_m r_o + C_1 L_1 R_4)}{2g_m r_o + s^3 (2C_1 C_4 L_1 R_4 g_m r_o + 2C_1 C_4 L_1 R_4 + C_1 C_L L_1 R_4 g_m r_o + C_1 C_L L_1 R_4) + s^2 (2C_1 C_4 R_4 r_o + C_1 C_L R_4 r_o + 2C_1 L_1 g_m r_o + 2C_1 L_1) + s (C_1 R_4 + 2C_1 r_o + 2C_4 R_4 g_m r_o + 2C_4 R_4 + C_L R_4 g_m r_o + C_L R_4) + 2}$$

$$\mathbf{9.151 \quad X-INVALID-ORDER-151} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{R_4 R_L g_m r_o + R_4 R_L + s^2 (C_1 L_1 R_4 R_L g_m r_o + C_1 L_1 R_4 R_L)}{R_4 g_m r_o + R_4 + 2R_L g_m r_o + 2R_L + s^3 (2C_1 C_4 L_1 R_4 R_L g_m r_o + 2C_1 C_4 L_1 R_4 R_L + C_1 C_L L_1 R_4 R_L g_m r_o + C_1 C_L L_1 R_4 R_L) + s^2 (2C_1 C_4 R_4 R_L r_o + C_1 C_L R_4 R_L r_o + C_1 L_1 R_4 g_m r_o + C_1 L_1 R_4 + 2C_1 L_1 R_L g_m r_o + 2C_1 L_1 R_L) + s (C_1 R_4 R_L + C_1 R_4 r_o + 2C_1 R_L r_o + 2C_4 R_4 R_L)}$$

$$\mathbf{9.152 \quad X-INVALID-ORDER-152} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_4 g_m r_o + R_4 + s^3 (C_1 C_L L_1 R_4 R_L g_m r_o + C_1 C_L L_1 R_4 R_L) + s^2 (C_1 L_1 R_4 g_m r_o + C_1 L_1 R_4) + s (C_L R_4 R_L g_m r_o + C_L R_4 R_L)}{2g_m r_o + s^4 (2C_1 C_4 C_L L_1 R_4 R_L g_m r_o + 2C_1 C_4 C_L L_1 R_4 R_L) + s^3 (2C_1 C_4 C_L R_4 R_L r_o + 2C_1 C_4 L_1 R_4 g_m r_o + 2C_1 C_4 L_1 R_4 + C_1 C_L L_1 R_4 g_m r_o + C_1 C_L L_1 R_4 + 2C_1 C_L L_1 R_L g_m r_o + 2C_1 C_L L_1 R_L) + s^2 (2C_1 C_4 R_4 r_o + C_1 C_L R_4 R_L + C_1 C_L R_4 r_o + 2C_1 C_L R_L r_o + 2C_1 L_1 g_m r_o)}$$

$$\mathbf{9.153 \quad X-INVALID-ORDER-153} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s^3 (C_1 L_1 L_L R_4 g_m r_o + C_1 L_1 L_L R_4) + s (L_L R_4 g_m r_o + L_L R_4)}{R_4 g_m r_o + R_4 + s^4 (2C_1 C_4 L_1 L_L R_4 g_m r_o + 2C_1 C_4 L_1 L_L R_4 + C_1 C_L L_1 L_L R_4 g_m r_o + C_1 C_L L_1 L_L R_4) + s^3 (2C_1 C_4 L_L R_4 r_o + C_1 C_L L_L R_4 r_o + 2C_1 L_1 L_L g_m r_o + 2C_1 L_1 L_L) + s^2 (C_1 L_1 R_4 g_m r_o + C_1 L_1 R_4 + C_1 L_L R_4 + 2C_1 L_L r_o + 2C_4 L_L R_4 g_m r_o + 2C_4 L_L R_4 + C_L L_L R_4 g_m r_o)}$$

$$\mathbf{9.154 \quad X-INVALID-ORDER-154} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_4 g_m r_o + R_4 + s^4 (C_1 C_L L_1 L_L R_4 g_m r_o + C_1 C_L L_1 L_L R_4) + s^2 (C_1 L_1 R_4 g_m r_o + C_1 L_1 R_4 + C_L L_L R_4 g_m r_o + C_L L_L R_4)}{2g_m r_o + s^5 (2C_1 C_4 C_L L_1 L_L R_4 g_m r_o + 2C_1 C_4 C_L L_1 L_L R_4) + s^4 (2C_1 C_4 C_L L_L R_4 r_o + 2C_1 C_L L_1 L_L g_m r_o + 2C_1 C_L L_1 L_L) + s^3 (2C_1 C_4 L_1 R_4 g_m r_o + 2C_1 C_4 L_1 R_4 + C_1 C_L L_1 R_4 g_m r_o + C_1 C_L L_1 R_4 + C_1 C_L L_L R_4 + 2C_1 C_L L_L r_o + 2C_4 C_L L_L R_4 g_m r_o + 2C_4 C_L L_L R_4) + s^2 (C_1 L_1 R_4 g_m r_o + C_1 L_1 R_4 + C_1 L_L R_4 + 2C_1 L_L r_o + 2C_4 L_L R_4 g_m r_o + 2C_4 L_L R_4 + C_L L_L R_4 g_m r_o)}$$

$$\mathbf{9.155 \quad X-INVALID-ORDER-155} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad R_L \right)$$

$$H(s) = \frac{R_L g_m r_o + R_L + s^3 (C_1 C_4 L_1 R_4 R_L g_m r_o + C_1 C_4 L_1 R_4 R_L) + s^2 (C_1 L_1 R_L g_m r_o + C_1 L_1 R_L) + s (C_4 R_4 R_L g_m r_o + C_4 R_4 R_L)}{g_m r_o + s^3 (C_1 C_4 L_1 R_4 g_m r_o + C_1 C_4 L_1 R_4 + 2C_1 C_4 L_1 R_L g_m r_o + 2C_1 C_4 L_1 R_L) + s^2 (C_1 C_4 R_4 R_L + C_1 C_4 R_4 r_o + 2C_1 C_4 R_L r_o + C_1 L_1 g_m r_o + C_1 L_1) + s (C_1 R_L + C_1 r_o + C_4 R_4 g_m r_o + C_4 R_4 + 2C_4 R_L g_m r_o + 2C_4 R_L) + 1}$$

$$\mathbf{9.156 \quad X-INVALID-ORDER-156} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m r_o + s^3 (C_1 C_4 L_1 R_4 g_m r_o + C_1 C_4 L_1 R_4) + s^2 (C_1 L_1 g_m r_o + C_1 L_1) + s (C_4 R_4 g_m r_o + C_4 R_4) + 1}{s^4 (C_1 C_4 C_L L_1 R_4 g_m r_o + C_1 C_4 C_L L_1 R_4) + s^3 (C_1 C_4 C_L R_4 r_o + 2C_1 C_4 L_1 g_m r_o + 2C_1 C_4 L_1 + C_1 C_L L_1 g_m r_o + C_1 C_L L_1) + s^2 (C_1 C_4 R_4 + 2C_1 C_4 r_o + C_1 C_L r_o + C_4 C_L R_4 g_m r_o + C_4 C_L R_4) + s (C_1 + 2C_4 g_m r_o + 2C_4 + C_L g_m r_o + C_L)}$$

9.157 X-INVALID-ORDER-157 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_L g_m r_o + R_L + s^3 (C_1 C_4 L_1 R_4 R_L g_m r_o + C_1 C_4 L_1 R_4 R_L) + s^2 (C_1 L_1 R_L g_m r_o + C_1 L_1 R_L) + s (C_4 R_4 R_L g_m r_o + C_4 R_4 R_L)}{g_m r_o + s^4 (C_1 C_4 C_L L_1 R_4 R_L g_m r_o + C_1 C_4 C_L L_1 R_4 R_L) + s^3 (C_1 C_4 C_L R_4 R_L r_o + C_1 C_4 L_1 R_4 g_m r_o + C_1 C_4 L_1 R_4 + 2 C_1 C_4 L_1 R_L g_m r_o + 2 C_1 C_4 L_1 R_L + C_1 C_L L_1 R_L g_m r_o + C_1 C_L L_1 R_L) + s^2 (C_1 C_4 R_4 R_L + C_1 C_4 R_4 r_o + 2 C_1 C_4 R_L r_o + C_1 C_L R_L r_o + C_1 L_1 g_m r_o + C_1 L_1 -$$

9.158 X-INVALID-ORDER-158 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{g_m r_o + s^4 (C_1 C_4 C_L L_1 R_4 R_L g_m r_o + C_1 C_4 C_L L_1 R_4 R_L) + s^3 (C_1 C_4 L_1 R_4 g_m r_o + C_1 C_4 L_1 R_4 + C_1 C_L L_1 R_L g_m r_o + C_1 C_L L_1 R_L) + s^2 (C_1 L_1 g_m r_o + C_1 L_1 + C_4 C_L R_4 R_L g_m r_o + C_4 C_L R_4 R_L) + s (C_4 R_4 g_m r_o + C_4 R_4) + C_4}{s^4 (C_1 C_4 C_L L_1 R_4 g_m r_o + C_1 C_4 C_L L_1 R_4 + 2 C_1 C_4 C_L L_1 R_L g_m r_o + 2 C_1 C_4 C_L L_1 R_L) + s^3 (C_1 C_4 C_L R_4 R_L + C_1 C_4 C_L R_4 r_o + 2 C_1 C_4 C_L R_L r_o + 2 C_1 C_4 L_1 g_m r_o + 2 C_1 C_4 L_1 + C_1 C_L L_1 g_m r_o + C_1 C_L L_1) + s^2 (C_1 C_4 R_4 + 2 C_1 C_4 r_o + C_1 C_L R_L + C_1 C_L r_o + C_4 C_L R_4 g_m r_o + C_4 C_L R_4) + s (C_4 R_4 g_m r_o + C_4 R_4) + C_4}$$

9.159 X-INVALID-ORDER-159 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$

$$H(s) = \frac{s^4 (C_1 C_4 L_1 L_L R_4 g_m r_o + C_1 C_4 L_1 L_L R_4) + s^3 (C_1 L_1 L_L g_m r_o + C_1 L_1 L_L) + s^2 (C_4 L_L R_4 g_m r_o + C_4 L_L R_4) + s (L_L g_m r_o + L_L)}{g_m r_o + s^5 (C_1 C_4 C_L L_1 L_L R_4 g_m r_o + C_1 C_4 C_L L_1 L_L R_4) + s^4 (C_1 C_4 C_L L_L R_4 r_o + 2 C_1 C_4 L_1 L_L g_m r_o + 2 C_1 C_4 L_1 L_L + C_1 C_L L_1 L_L g_m r_o + C_1 C_L L_1 L_L) + s^3 (C_1 C_4 L_1 R_4 g_m r_o + C_1 C_4 L_1 R_4 + C_1 C_4 L_L R_4 + 2 C_1 C_4 L_L r_o + C_1 C_L L_L r_o + C_4 C_L L_L R_4 g_m r_o + C_4 C_L L_L R_4) + s^2 (C_1 C_4 L_1 L_L R_4 g_m r_o + C_1 C_4 L_1 L_L R_4) + s (C_1 C_4 L_1 L_L g_m r_o + C_1 C_4 L_1 L_L) + C_1 C_4 L_1 L_L R_4 g_m r_o + C_1 C_4 L_1 L_L R_4}.$$

9.160 X-INVALID-ORDER-160 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \infty, L_L s + \frac{1}{C_L s} \right)$

$$H(s) = \frac{g_m r_o + s^5 (C_1 C_4 C_L L_1 L_L R_4 g_m r_o + C_1 C_4 C_L L_1 L_L R_4) + s^4 (C_1 C_L L_1 L_L g_m r_o + C_1 C_L L_1 L_L) + s^3 (C_1 C_4 L_1 R_4 g_m r_o + C_1 C_4 L_1 R_4 + C_4 C_L L_L R_4 g_m r_o + C_4 C_L L_L R_4) + s^2 (C_1 L_1 g_m r_o + C_1 L_1 + C_L L_L g_m r_o + C_L)}{s^5 (2C_1 C_4 C_L L_1 L_L g_m r_o + 2C_1 C_4 C_L L_1 L_L) + s^4 (C_1 C_4 C_L L_1 R_4 g_m r_o + C_1 C_4 C_L L_1 R_4 + C_1 C_4 C_L L_L R_4 + 2C_1 C_4 C_L L_L r_o) + s^3 (C_1 C_4 C_L R_4 r_o + 2C_1 C_4 L_1 g_m r_o + 2C_1 C_4 L_1 + C_1 C_L L_1 g_m r_o + C_1 C_L L_1 + C_1 C_L L_L + 2C_4 C_L L_L g_m r_o + 2C_4 C_L L_L) + s^2 (C_1 C_4 R_4 + 2C_1 C_4 L_1 + 2C_1 C_L L_1 + C_L L_L)}.$$

9.161 X-INVALID-ORDER-161 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, R_L \right)$

$$H(s) = \frac{s^3 (C_1 L_1 L_4 R_L g_m r_o + C_1 L_1 L_4 R_L) + s (L_4 R_L g_m r_o + L_4 R_L)}{2R_L g_m r_o + 2R_L + s^4 (2C_1 C_4 L_1 L_4 R_L g_m r_o + 2C_1 C_4 L_1 L_4 R_L) + s^3 (2C_1 C_4 L_4 R_L r_o + C_1 L_1 L_4 g_m r_o + C_1 L_1 L_4) + s^2 (2C_1 L_1 R_L g_m r_o + 2C_1 L_1 R_L + C_1 L_4 R_L + C_1 L_4 r_o + 2C_4 L_4 R_L g_m r_o + 2C_4 L_4 R_L) + s (2C_1 R_L r_o + L_4 g_m r_o + L_4)}$$

9.162 X-INVALID-ORDER-162 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{s^3 (C_1 L_1 L_4 g_m r_o + C_1 L_1 L_4) + s (L_4 g_m r_o + L_4)}{2C_1 r_o s + 2g_m r_o + s^4 (2C_1 C_4 L_1 L_4 g_m r_o + 2C_1 C_4 L_1 L_4 + C_1 C_L L_1 L_4 g_m r_o + C_1 C_L L_1 L_4) + s^3 (2C_1 C_4 L_4 r_o + C_1 C_L L_4 r_o) + s^2 (2C_1 L_1 g_m r_o + 2C_1 L_1 + C_1 L_4 + 2C_4 L_4 g_m r_o + 2C_4 L_4 + C_L L_4 g_m r_o + C_L L_4) + 2}$$

9.163 X-INVALID-ORDER-163 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{s^3 (C_1 L_1 L_4 R_L g_m r_o + C_1 L_1 L_4 R_L) + s (L_4 R_L g_m r_o + L_4 R_L)}{2R_L g_m r_o + 2R_L + s^4 (2C_1 C_4 L_1 L_4 R_L g_m r_o + 2C_1 C_4 L_1 L_4 R_L + C_1 C_L L_1 L_4 R_L g_m r_o + C_1 C_L L_1 L_4 R_L) + s^3 (2C_1 C_4 L_4 R_L r_o + C_1 C_L L_4 R_L r_o + C_1 L_1 L_4 g_m r_o + C_1 L_1 L_4) + s^2 (2C_1 L_1 R_L g_m r_o + 2C_1 L_1 R_L + C_1 L_4 R_L + C_1 L_4 r_o + 2C_4 L_4 R_L g_m r_o + 2C_4 L_4 R_L + C_L L_4 R_L)}$$

9.164 X-INVALID-ORDER-164 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{s^4 (C_1 C_L L_1 L_4 R_L g_m r_o + C_1 C_L L_1 L_4 R_L) + s^3 (C_1 L_1 L_4 g_m r_o + C_1 L_1 L_4) + s^2 (C_L L_4 R_L g_m r_o + C_L L_4 R_L) + s (L_4 g_m r_o + L_4)}{2g_m r_o + s^5 (2C_1 C_4 C_L L_1 L_4 R_L g_m r_o + 2C_1 C_4 C_L L_1 L_4 R_L) + s^4 (2C_1 C_4 C_L L_4 R_L r_o + 2C_1 C_4 L_1 L_4 g_m r_o + 2C_1 C_4 L_1 L_4 + C_1 C_L L_1 L_4 g_m r_o + C_1 C_L L_1 L_4) + s^3 (2C_1 C_4 L_4 r_o + 2C_1 C_L L_1 R_L g_m r_o + 2C_1 C_L L_1 R_L + C_1 C_L L_4 R_L + C_1 C_L L_4 r_o + 2C_4 C_L L_4 R_L g_m r_o + 2C_4 C_L L_4 R_L) + s^2 (2C_1 C_4 L_4 + 2C_1 C_L L_1 R_L + C_1 C_L L_4) + s (C_1 C_L L_4 + C_1 C_L L_1) + C_1 C_L}$$

9.165 X-INVALID-ORDER-165 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$

$$H(s) = \frac{s^3(C_1 L_1 L_4 L_L g_m r_o + C_1 L_1 L_4 L_L) + s(L_4 L_L g_m r_o + L_4 L_L)}{L_4 g_m r_o + L_4 + 2L_L g_m r_o + 2L_L + s^4(2C_1 C_4 L_1 L_4 L_L g_m r_o + 2C_1 C_4 L_1 L_4 L_L + C_1 C_L L_1 L_4 L_L g_m r_o + C_1 C_L L_1 L_4 L_L) + s^3(2C_1 C_4 L_4 L_L r_o + C_1 C_L L_4 L_L r_o) + s^2(C_1 L_1 L_4 g_m r_o + C_1 L_1 L_4 + 2C_1 L_1 L_L g_m r_o + 2C_1 L_1 L_L + C_1 L_4 L_L + 2C_4 L_4 L_L g_m r_o + 2C_4 L_4 L_L + C_L L_4)}$$

9.166 X-INVALID-ORDER-166 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \infty, L_L s + \frac{1}{C_L s} \right)$

$$H(s) = \frac{s^5 (C_1 C_L L_1 L_4 L_L g_m r_o + C_1 C_L L_1 L_4 L_L) + s^3 (C_1 L_1 L_4 g_m r_o + C_1 L_1 L_4 + C_L L_4 L_L g_m r_o + C_L L_4 L_L) + s (L_4 g_m r_o + L_4)}{2C_1 C_4 C_L L_4 L_L r_o s^5 + 2C_1 r_o s + 2g_m r_o + s^6 (2C_1 C_4 C_L L_1 L_4 L_L g_m r_o + 2C_1 C_4 C_L L_1 L_4 L_L) + s^4 (2C_1 C_4 L_1 L_4 g_m r_o + 2C_1 C_4 L_1 L_4 + C_1 C_L L_1 L_4 g_m r_o + C_1 C_L L_1 L_4 + 2C_1 C_L L_1 L_L g_m r_o + 2C_1 C_L L_1 L_L + C_1 C_L L_4 L_L + 2C_4 C_L L_4 L_L g_m r_o + 2C_4 C_L L_4 L_L) + s^3 (2C_1 C_4 L_4 L_4)}$$

9.167 X-INVALID-ORDER-167 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \infty, R_L \right)$

$$H(s) = \frac{R_L g_m r_o + R_L + s^4 (C_1 C_4 L_1 L_4 R_L g_m r_o + C_1 C_4 L_1 L_4 R_L) + s^2 (C_1 L_1 R_L g_m r_o + C_1 L_1 R_L + C_4 L_4 R_L g_m r_o + C_4 L_4 R_L)}{g_m r_o + s^4 (C_1 C_4 L_1 L_4 g_m r_o + C_1 C_4 L_1 L_4) + s^3 (2C_1 C_4 L_1 R_L g_m r_o + 2C_1 C_4 L_1 R_L + C_1 C_4 L_4 R_L + C_1 C_4 L_4 r_o) + s^2 (2C_1 C_4 R_L r_o + C_1 L_1 g_m r_o + C_1 L_1 + C_4 L_4 g_m r_o + C_4 L_4) + s (C_1 R_L + C_1 r_o + 2C_4 R_L g_m r_o + 2C_4 R_L) + 1}$$

9.168 X-INVALID-ORDER-168 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{g_m r_o + s^4 (C_1 C_4 L_1 L_4 g_m r_o + C_1 C_4 L_1 L_4) + s^2 (C_1 L_1 g_m r_o + C_1 L_1 + C_4 L_4 g_m r_o + C_4 L_4) + 1}{C_1 C_4 C_L L_4 r_o s^4 + s^5 (C_1 C_4 C_L L_1 L_4 g_m r_o + C_1 C_4 C_L L_1 L_4) + s^3 (2C_1 C_4 L_1 g_m r_o + 2C_1 C_4 L_1 + C_1 C_4 L_4 + C_1 C_L L_1 g_m r_o + C_1 C_L L_1 + C_4 C_L L_4 g_m r_o + C_4 C_L L_4) + s^2 (2C_1 C_4 r_o + C_1 C_L r_o) + s (C_1 + 2C_4 g_m r_o + 2C_4 + C_L g_m r_o + C_L)}$$

9.169 X-INVALID-ORDER-169 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_L g_m r_o + R_L + s^4 (C_1 C_4 L_1 L_4 R_L g_m r_o + C_1 C_4 L_1 L_4 R_L) + s^2 (C_1 L_1 R_L g_m r_o + C_1 L_1 R_L + C_4 L_4 R_L g_m r_o + C_4 L_4 R_L)}{g_m r_o + s^5 (C_1 C_4 C_L L_1 L_4 R_L g_m r_o + C_1 C_4 C_L L_1 L_4 R_L) + s^4 (C_1 C_4 C_L L_4 R_L r_o + C_1 C_4 L_1 L_4 g_m r_o + C_1 C_4 L_1 L_4) + s^3 (2C_1 C_4 L_1 R_L g_m r_o + 2C_1 C_4 L_1 R_L + C_1 C_4 L_4 R_L + C_1 C_4 L_4 r_o + C_1 C_L L_1 R_L g_m r_o + C_1 C_L L_1 R_L + C_4 C_L L_4 R_L g_m r_o + C_4 C_L L_4 R_L) + s^2 (2C_1 C_4 R_L r_o + C_1 C_L R_L r_o) + s (C_1 + 2C_4 g_m r_o + 2C_4 + C_L g_m r_o + C_L)}$$

9.170 X-INVALID-ORDER-170 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{g_m r_o + s^5 (C_1 C_4 C_L L_1 L_4 R_L g_m r_o + C_1 C_4 C_L L_1 L_4 R_L) + s^4 (C_1 C_4 L_1 L_4 g_m r_o + C_1 C_4 L_1 L_4) + s^3 (C_1 C_L L_1 R_L g_m r_o + C_1 C_L L_1 R_L + C_4 C_L L_4 R_L g_m r_o + C_4 C_L L_4 R_L) + s^2 (C_1 L_1 g_m r_o + C_1 L_1 + C_4 L_4 g_m r_o + C_4 L_4)}{s^5 (C_1 C_4 C_L L_1 L_4 g_m r_o + C_1 C_4 C_L L_1 L_4) + s^4 (2C_1 C_4 C_L L_1 R_L g_m r_o + 2C_1 C_4 C_L L_1 R_L + C_1 C_4 C_L L_4 R_L + C_1 C_4 C_L L_4 r_o) + s^3 (2C_1 C_4 C_L R_L r_o + 2C_1 C_4 L_1 g_m r_o + 2C_1 C_4 L_1 + C_1 C_4 L_4 + C_1 C_L L_1 g_m r_o + C_1 C_L L_1 + C_4 C_L L_4 g_m r_o + C_4 C_L L_4) + s^2 (2C_1 C_4 r_o + C_1 C_L R_L r_o) + s (C_1 + 2C_4 g_m r_o + 2C_4 + C_L g_m r_o + C_L)}$$

9.171 X-INVALID-ORDER-171 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$

$$H(s) = \frac{s^5 (C_1 C_4 L_1 L_4 L_L g_m r_o + C_1 C_4 L_1 L_4 L_L) + s^3 (C_1 L_1 L_L g_m r_o + C_1 L_1 L_L + C_4 L_4 L_L g_m r_o + C_4 L_4 L_L) + s (L_L g_m r_o + L_L)}{C_1 C_4 C_L L_4 L_L r_o s^5 + C_1 r_o s + g_m r_o + s^6 (C_1 C_4 C_L L_1 L_4 L_L g_m r_o + C_1 C_4 C_L L_1 L_4 L_L) + s^4 (C_1 C_4 L_1 L_4 g_m r_o + C_1 C_4 L_1 L_4 + 2C_1 C_4 L_1 L_L g_m r_o + 2C_1 C_4 L_1 L_L + C_1 C_4 L_4 L_L + C_1 C_L L_1 L_L g_m r_o + C_1 C_L L_1 L_L + C_4 C_L L_4 L_L g_m r_o + C_4 C_L L_4 L_L) + s^3 (C_1 C_4 L_4 r_o + 2C_1 C_4 L_4)}{}$$

9.172 X-INVALID-ORDER-172 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \infty, L_L s + \frac{1}{C_L s} \right)$

$$H(s) = \frac{g_m r_o + s^6 (C_1 C_4 C_L L_1 L_4 L_L g_m r_o + C_1 C_4 C_L L_1 L_4 L_L) + s^4 (C_1 C_4 L_1 L_4 g_m r_o + C_1 C_4 L_1 L_4 + C_1 C_L L_1 L_L g_m r_o + C_1 C_L L_1 L_L + C_4 C_L L_4 L_L g_m r_o + C_4 C_L L_4 L_L) + s^2 (C_1 L_1 g_m r_o + C_1 L_1 + C_4 L_4 g_m r_o + C_4 L_4)}{s^5 (C_1 C_4 C_L L_1 L_4 g_m r_o + C_1 C_4 C_L L_1 L_4 + 2C_1 C_4 C_L L_1 L_L g_m r_o + 2C_1 C_4 C_L L_1 L_L + C_1 C_4 C_L L_4 L_L) + s^4 (C_1 C_4 C_L L_4 r_o + 2C_1 C_4 C_L L_L r_o) + s^3 (2C_1 C_4 L_1 g_m r_o + 2C_1 C_4 L_1 + C_1 C_4 L_4 + C_1 C_L L_1 g_m r_o + C_1 C_L L_1 + C_1 C_L L_L + C_4 C_L L_4 g_m r_o + C_4 C_L L_4 + 2C_4 C_L L_L g_m r_o) + s^2 (2C_1 C_4 r_o + C_1 C_L R_L r_o) + s (C_1 + 2C_4 g_m r_o + 2C_4 + C_L g_m r_o + C_L)}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(R_1, \infty, \infty, R_4 + \frac{1}{C_4 s}, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s^2 (C_4 C_L R_1 R_4 R_L g_m r_o + C_4 C_L R_1 R_4 R_L) + s (C_4 R_1 R_4 g_m r_o + C_4 R_1 R_4 + C_L R_1 R_L g_m r_o + C_L R_1 R_L)}{s^2 (C_4 C_L R_1 R_4 g_m r_o + C_4 C_L R_1 R_4 + 2C_4 C_L R_1 R_L g_m r_o + 2C_4 C_L R_1 R_L + C_4 C_L R_4 R_L + C_4 C_L R_4 r_o + 2C_4 C_L R_L r_o) + s (2C_4 R_1 g_m r_o + 2C_4 R_1 + C_4 R_4 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L R_L + C_L r_o) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_4} \sqrt{C_L} R_1 R_4 g_m r_o \sqrt{\frac{1}{R_1 R_4 g_m r_o + R_1 R_4 + 2R_1 R_L g_m r_o + 2R_1 R_L + R_4 R_L + R_4 r_o + 2R_L r_o}} + \sqrt{C_4} \sqrt{C_L} R_1 R_4 \sqrt{\frac{1}{R_1 R_4 g_m r_o + R_1 R_4 + 2R_1 R_L g_m r_o + 2R_1 R_L + R_4 R_L + R_4 r_o + 2R_L r_o}} + 2\sqrt{C_4} \sqrt{C_L} R_1 R_L g_m r_o \sqrt{\frac{1}{R_1 R_4 g_m r_o + R_1 R_4 + 2R_1 R_L g_m r_o + 2R_1 R_L + R_4 R_L + R_4 r_o + 2R_L r_o}} + 2\sqrt{C_4} \sqrt{C_L} R_1 R_L \sqrt{\frac{1}{R_1 R_4 g_m r_o + R_1 R_4 + 2R_1 R_L g_m r_o + 2R_1 R_L + R_4 R_L + R_4 r_o + 2R_L r_o}}}{2C_4 R_1 g_m r_o + 2C_4 R_1 + C_4 R_4 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L R_L + C_L r_o}$

wo: $\sqrt{\frac{1}{C_4 C_L R_1 R_4 g_m r_o + C_4 C_L R_1 R_4 + 2C_4 C_L R_1 R_L g_m r_o + 2C_4 C_L R_1 R_L + C_4 C_L R_4 R_L + C_4 C_L R_4 r_o + 2C_4 C_L R_L r_o}}$

bandwidth: $\frac{(2C_4 R_1 g_m r_o + 2C_4 R_1 + C_4 R_4 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L R_L + C_L r_o) \sqrt{\frac{1}{C_4 C_L R_1 R_4 g_m r_o + C_4 C_L R_1 R_4 + 2C_4 C_L R_1 R_L g_m r_o + 2C_4 C_L R_1 R_L + C_4 C_L R_4 R_L + C_4 C_L R_4 r_o + 2C_4 C_L R_L r_o}}}{\sqrt{C_4} \sqrt{C_L} R_1 R_4 g_m r_o \sqrt{\frac{1}{R_1 R_4 g_m r_o + R_1 R_4 + 2R_1 R_L g_m r_o + 2R_1 R_L + R_4 R_L + R_4 r_o + 2R_L r_o}} + \sqrt{C_4} \sqrt{C_L} R_1 R_4 \sqrt{\frac{1}{R_1 R_4 g_m r_o + R_1 R_4 + 2R_1 R_L g_m r_o + 2R_1 R_L + R_4 R_L + R_4 r_o + 2R_L r_o}} + 2\sqrt{C_4} \sqrt{C_L} R_1 R_L g_m r_o \sqrt{\frac{1}{R_1 R_4 g_m r_o + R_1 R_4 + 2R_1 R_L g_m r_o + 2R_1 R_L + R_4 R_L + R_4 r_o + 2R_L r_o}} + 2\sqrt{C_4} \sqrt{C_L} R_1 R_L \sqrt{\frac{1}{R_1 R_4 g_m r_o + R_1 R_4 + 2R_1 R_L g_m r_o + 2R_1 R_L + R_4 R_L + R_4 r_o + 2R_L r_o}}}$

K-LP: $R_1 g_m r_o + R_1$

K-HP: $\frac{R_1 R_4 R_L g_m r_o + R_1 R_4 R_L}{R_1 R_4 g_m r_o + R_1 R_4 + 2R_1 R_L g_m r_o + 2R_1 R_L + R_4 R_L + R_4 r_o + 2R_L r_o}$

K-BP: $\frac{C_4 R_1 R_4 g_m r_o + C_4 R_1 R_4 + C_L R_1 R_L g_m r_o + C_L R_1 R_L}{2C_4 R_1 g_m r_o + 2C_4 R_1 + C_4 R_4 + 2C_4 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L R_L + C_L r_o}$

Qz: None

Wz: $\frac{1}{\sqrt{C_4} \sqrt{C_L} \sqrt{R_4} \sqrt{R_L}}$

10.2 X-INVALID-WZ-2 $Z(s) = \left(R_1 + \frac{1}{C_{1s}}, \infty, \infty, R_4, \infty, R_L + \frac{1}{C_{Ls}} \right)$

$$H(s) = \frac{R_4 g_m r_o + R_4 + s^2 (C_1 C_L R_1 R_4 R_L g_m r_o + C_1 C_L R_1 R_4 R_L) + s (C_1 R_1 R_4 g_m r_o + C_1 R_1 R_4 + C_L R_4 R_L g_m r_o + C_L R_4 R_L)}{2 g_m r_o + s^2 (C_1 C_L R_1 R_4 g_m r_o + C_1 C_L R_1 R_4 + 2 C_1 C_L R_1 R_L g_m r_o + 2 C_1 C_L R_1 R_L + C_1 C_L R_4 R_L + C_1 C_L R_4 r_o + 2 C_1 C_L R_L r_o) + s (2 C_1 R_1 g_m r_o + 2 C_1 R_1 + C_1 R_4 + 2 C_1 r_o + C_L R_4 g_m r_o + C_L R_4 + 2 C_L R_L g_m r_o + 2 C_L R_L) + 2}$$

Parameters:

$$\text{Q: } \frac{\sqrt{2}\sqrt{C_1}\sqrt{C_L}R_1R_4g_mr_o\sqrt{\frac{gm_r_o}{R_1R_4g_mr_o+R_1R_4+2R_1R_Lg_mr_o+2R_1R_L+R_4R_L+R_4r_o+2R_Lr_o}}+\frac{1}{R_1R_4g_mr_o+R_1R_4+2R_1R_Lg_mr_o+2R_1R_L+R_4R_L+R_4r_o+2R_Lr_o}}{\sqrt{2}\sqrt{C_1}\sqrt{C_L}R_1R_4\sqrt{\frac{gm_r_o}{R_1R_4g_mr_o+R_1R_4+2R_1R_Lg_mr_o+2R_1R_L+R_4R_L+R_4r_o+2R_Lr_o}}+\frac{1}{R_1R_4g_mr_o+R_1R_4+2R_1R_Lg_mr_o+2R_1R_L+R_4R_L+R_4r_o+2R_Lr_o}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_L}R_1R_Lg_mr_o\sqrt{\frac{gm_r_o}{R_1R_4g_mr_o+R_1R_4+2R_1R_Lg_mr_o+2R_1R_L+R_4R_L+R_4r_o+2R_Lr_o}}}$$

$$\text{WO: } \sqrt{\frac{2g_m r_o + 2}{C_1 C_L R_1 R_4 g_m r_o + C_1 C_L R_1 R_4 + 2C_1 C_L R_1 R_L g_m r_o + 2C_1 C_L R_1 R_L + C_1 C_L R_4 R_L + C_1 C_L R_4 r_o + 2C_1 C_L R_L r_o}}$$

$$\text{bandwidth: } \frac{\sqrt{2}\sqrt{C_1}\sqrt{C_L}R_1R_4gmr_o}{\sqrt{\frac{gm^2r_o}{R_1R_4gmr_o+R_1R_4+2R_1R_Lgmr_o+2R_1R_L+R_4R_L+R_4r_o+2R_Lr_o} + \frac{1}{R_1R_4gmr_o+R_1R_4+2R_1R_Lgmr_o+2R_1R_L+R_4R_L+R_4r_o+2R_Lr_o}}} + \sqrt{2}\sqrt{C_1}\sqrt{C_L}R_1R_4\sqrt{\frac{gm^2r_o}{R_1R_4gmr_o+R_1R_4+2R_1R_Lgmr_o+2R_1R_L+R_4R_L+R_4r_o+2R_Lr_o} + \frac{1}{R_1R_4gmr_o+R_1R_4+2R_1R_Lgmr_o+2R_1R_L+R_4R_L+R_4r_o+2R_Lr_o}}} + 2\sqrt{2}\sqrt{C_1}\sqrt{C_L}R_1R_Lg$$

$$\text{K-LP: } \frac{R_4}{2}$$

$$\text{K-HP: } \frac{R_1 R_4 R_L g_m r_o + R_1 R_4 R_L}{R_1 R_4 g_m r_o + R_1 R_4 + 2 R_1 R_L g_m r_o + 2 R_1 R_L + R_4 R_L + R_4 r_o + 2 R_L r_o}$$

$$\text{K-BP: } \frac{R_1 R_4 g_{m'0} + R_1 R_4 + 2 R_1 R_L g_{m'0} + 2 R_1 R_L + R_4 R_L + R_4 r_0 + 2 R_L r_0}{2 C_1 R_1 R_4 g_{m'0} + C_1 R_1 R_4 + C_1 R_1 R_L + C_1 R_4 R_L g_{m'0} + C_1 R_4 R_L}$$

Qz: None

$$W_Z: \frac{1}{\sqrt{C_1}\sqrt{C_L}\sqrt{R_1}\sqrt{R_L}}$$

10.3 X-INVALID-WZ-3 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \infty, R_L \right)$

$$H(s) = \frac{R_L g_m r_o + R_L + s^2 (C_1 C_4 R_1 R_4 R_L g_m r_o + C_1 C_4 R_1 R_4 R_L) + s (C_1 R_1 R_L g_m r_o + C_1 R_1 R_L + C_4 R_4 R_L g_m r_o + C_4 R_4 R_L)}{g_m r_o + s^2 (C_1 C_4 R_1 R_4 g_m r_o + C_1 C_4 R_1 R_4 + 2 C_1 C_4 R_1 R_L g_m r_o + 2 C_1 C_4 R_1 R_L + C_1 C_4 R_4 R_L + C_1 C_4 R_4 r_o + 2 C_1 C_4 R_L r_o) + s (C_1 R_1 g_m r_o + C_1 R_1 + C_1 R_L + C_1 r_o + C_4 R_4 g_m r_o + C_4 R_4 + 2 C_4 R_L g_m r_o + 2 C_4 R_L) + 1}$$

Parameters:

$$\text{Q: } \frac{\sqrt{C_1}\sqrt{C_4}R_1R_4g_mr_o\sqrt{\frac{gm_r_o}{R_1R_4g_mr_o+R_1R_4+2R_1R_Lg_mr_o+2R_1R_L+R_4R_L+R_4r_o+2R_Lr_o}}+\frac{1}{R_1R_4g_mr_o+R_1R_4+2R_1R_Lg_mr_o+2R_1R_L+R_4R_L+R_4r_o+2R_Lr_o}}{\sqrt{C_1}\sqrt{C_4}R_1R_4\sqrt{\frac{gm_r_o}{R_1R_4g_mr_o+R_1R_4+2R_1R_Lg_mr_o+2R_1R_L+R_4R_L+R_4r_o+2R_Lr_o}}+\frac{1}{R_1R_4g_mr_o+R_1R_4+2R_1R_Lg_mr_o+2R_1R_L+R_4R_L+R_4r_o+2R_Lr_o}}+2\sqrt{C_1}\sqrt{C_4}R_1R_Lg_mr_o\sqrt{\frac{gm_r_o}{R_1R_4g_mr_o+R_1R_4+2R_1R_Lg_mr_o+2R_1R_L+R_4R_L+R_4r_o+2R_Lr_o}}}$$

$$\text{WO: } \sqrt{\frac{g_m r_o + 1}{C_1 C_4 R_1 R_4 g_m r_o + C_1 C_4 R_1 R_4 + 2 C_1 C_4 R_1 R_L g_m r_o + 2 C_1 C_4 R_1 R_L + C_1 C_4 R_4 R_L + C_1 C_4 R_4 r_o + 2 C_1 C_4 R_L r_o}}$$

$$\text{bandwidth: } \frac{\sqrt{C_1} \sqrt{C_4} R_1 R_4 g_{m r_o}}{\sqrt{\frac{g_{m r_o}}{R_1 R_4 g_{m r_o} + R_1 R_4 + 2 R_1 R_L g_{m r_o} + 2 R_1 R_L + R_4 R_L + R_4 r_o + 2 R_L r_o} + \frac{1}{R_1 R_4 g_{m r_o} + R_1 R_4 + 2 R_1 R_L g_{m r_o} + 2 R_1 R_L + R_4 R_L + R_4 r_o + 2 R_L r_o}}} + \sqrt{C_1} \sqrt{C_4} R_1 R_4 \sqrt{\frac{g_{m r_o}}{R_1 R_4 g_{m r_o} + R_1 R_4 + 2 R_1 R_L g_{m r_o} + 2 R_1 R_L + R_4 R_L + R_4 r_o + 2 R_L r_o} + \frac{1}{R_1 R_4 g_{m r_o} + R_1 R_4 + 2 R_1 R_L g_{m r_o} + 2 R_1 R_L + R_4 R_L + R_4 r_o + 2 R_L r_o}}} + 2 \sqrt{C_1} \sqrt{C_4} R_1 R_L g_{m r_o} \sqrt{\frac{g_{m r_o}}{R_1 R_4 g_{m r_o} + R_1 R_4 + 2 R_1 R_L g_{m r_o} + 2 R_1 R_L + R_4 R_L + R_4 r_o + 2 R_L r_o} + \frac{1}{R_1 R_4 g_{m r_o} + R_1 R_4 + 2 R_1 R_L g_{m r_o} + 2 R_1 R_L + R_4 R_L + R_4 r_o + 2 R_L r_o}}}$$

K-LP: R_L

$$\text{K-HP: } \frac{R_1 R_4 R_L g_m r_o + R_1 R_4 R_L}{R_1 R_4 q_m r_o + R_1 R_4 + 2 R_1 R_L q_m r_o + 2 R_1 R_L + R_4 R_L + R_4 r_o + 2 R_L r_o}$$

$$\text{K-BP: } \frac{C_1 R_1 R_L g_m r_o + C_1 R_1 R_L + C_4 R_4 g_m r_o + C_4 R_4 R_L}{C_1 R_1 g_m r_o + C_1 R_1 + C_1 R_1 + C_4 R_4 g_m r_o + C_4 R_4 + 2C_4 R_1 g_m r_o + 2C_4 R_1}$$

Qz: None

$$\text{WZ: } \frac{1}{\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_4}}$$

11 X-PolynomialError